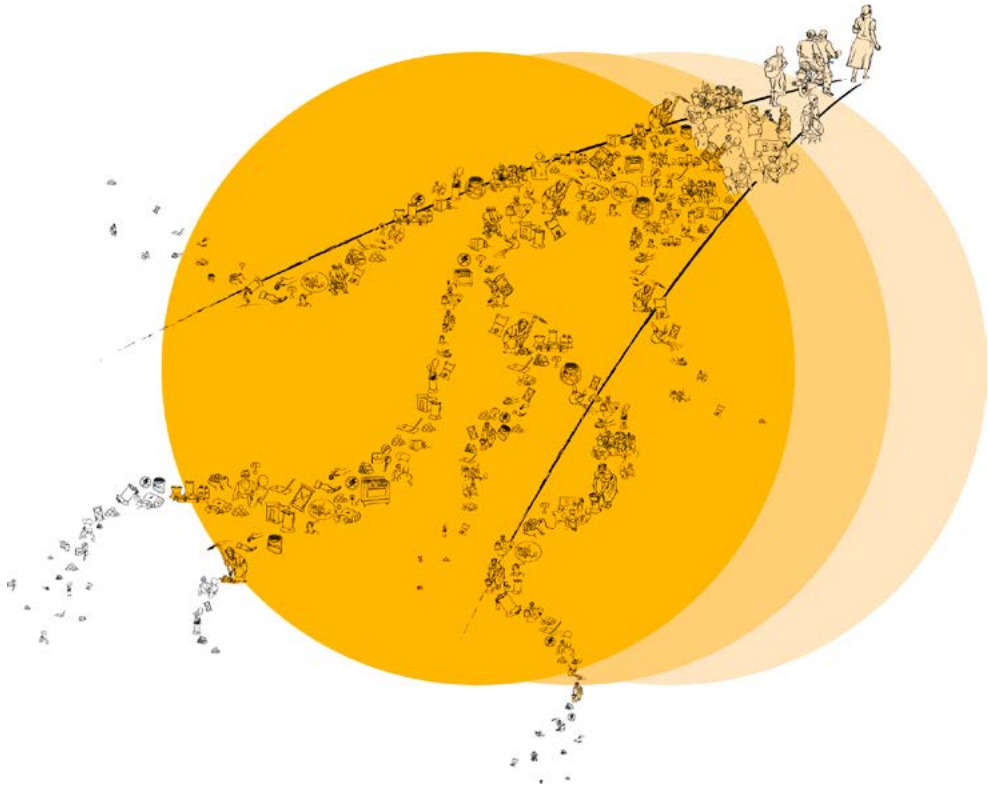
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Devising Capabilities

Service Design for Development Interventions

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Devising Capabilities: Service Design for Development Interventions

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Abstract

Despite the progress in recent decades, one in ten people globally still live in extreme poverty, and this number is set to increase in the coming years. Designing interventions to improve well-being and livelihoods is challenging because poverty is multidimensional and plays out in complex, adaptive social-ecological systems, where behaviours and practices at the local level can have unintended consequences elsewhere in the system. In such contexts, linear approaches to designing development interventions are insufficient. Service design has emerged as a human-centred, integrative approach to designing services and systems in complex settings, but knowledge gaps remain on how service design can be used to address development challenges in the Global South. Without an understanding of how service design tools and approaches function in these contexts, there is a risk they might inadvertently cause harm to research participants and local communities.

This thesis contributes new knowledge about how service design can be used to design development interventions in complex social-ecological contexts The research questions address three areas where service design could play a role: (i) how service design could be used to make sense of local level complexity and package this information for development programmers and policymakers; (ii) how service design could support local agency in the design of development projects; and (iii) how service design could be used to complement conventional methods of development research.

The research questions were addressed using three case studies of development interventions. In two of the cases, a clean cookstove intervention in Kenya and an insurance product for small-scale farmers in Uganda, a service design approach was combined with quantitative methods. In the third case study, participatory backcasting was used to inform a long-term plan for energy transition for an off-grid community in Machakos, Kenya. A conceptual framework was first developed to support the use of service design to address development challenges in complex social-ecological systems. Elements of the framework were then applied in two of the case studies: the cookstoves and insurance studies. The thesis uses service design as an approach and practice, and the capabilities approach as the main conceptual and theoretical framing.

The findings reveal that in these contexts, service design tools can become devices for understanding how value is assigned over time by users of the designed services. Archetype construction and prototyping became important devices for identifying patterns in heterogeneous needs and behaviours, while conveying key design parameters to policymakers and programmers. The research also shows that prototyping can enhance local agency by allowing research participants to challenge the core assumptions that underpin proposed interventions. The findings also demonstrate that participatory backcasting can be positioned as a device for prototyping future development pathways. It was found to facilitate individual and collective action in the short term. Combining service design devices with quantitative methods allowed triangulation of findings and a more comprehensive understanding of complex contexts.

Beyond the empirical findings on design devices, the thesis makes two important conceptual contributions. First, it positions service design as an integrative approach to conducting transdisciplinary development research in complex social-ecological contexts. Second, the thesis bridges service design and the capabilities approach and demonstrates how this can help designers anchor their work in the local context while navigating normative development objectives. The contributions are useful for service design researchers and practitioners interested in how service design can be used alongside other disciplines to support long-term development objectives. For the development community, the contributions demonstrate a radically different approach to designing interventions where complexity, messiness and non-linearity are embraced.

Keywords: Service design; capabilities approach; development interventions; Global South; complexity

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Populärvetenskaplig sammanfattning

En kollega och jag reste 2014 på landsbygden i Gujarat, Indien, för att samla in data i ett projekt med syfte att förstå hur koldioxidfinansiering kan användas för att finansiera mer effektiva vedeldade spisar till låginkomsthushåll. Vi intervjuade människor om deras erfarenheter av att använda dessa spisar som de hade köpt till reducerat pris med hjälp av en lokal aktör för mikrofinansiering. Spisarna, som hade tillverkats av ett USA-baserat företag och nu såldes som en del av ett koldioxidkompensationsprogram, var designade för att använda mindre ved, vilket är bättre för miljön, samt producera mindre rök, vilket är bättre för användarnas hälsa. Logiken i programmet var enkel. Om ett hushåll i Indien använde den effektiva vedeldade spisen i stället för sin mindre effektiva traditionella spis, skulle en viss mängd CO_2-utsläpp sparas (kompenseras).

Företag i västvärlden kunde sedan köpa denna kompensation för att täcka sina egna koldioxidutsläpp, samtidigt som de dessutom skulle bidra till en förbättrad hälsa, särskilt för kvinnor och barn som vanligtvis är mest utsatta för hushållens luftföroreningar. Intäkterna från sålda kompensationer skulle vidare användas för att subventionera kostnaderna för spisar för fattiga hushåll. På pappret såg detta ut som en win-win!

Men när vi började prata med användare av de förbättrade spisarna insåg vi snart att det verkade finnas ett glapp mellan de grundläggande antagandena i kompensationsprogrammet och användarnas verklighet. Även om spisarna hade marknadsförts som praktiska och bränslesnåla, beskrev kvinnorna vi pratade med hur de tyckte att de var svåra att tända och hur delar lätt gick sönder när de placerade sina tunga kastruller ovanpå. Att laga spisarna var knepigt eftersom det inte fanns några reservdelar eller någon lokal reparatör. Och eftersom de flesta familjerna i området är stora, skulle de ha behövt flera spisar för att klara all den dagliga matlagningen, men de flesta hade bara råd att köpa en spis.

Min kollega och jag slogs av hur detta till synes globalt lovande klimatfinansieringsprogram totalt missade målet på den lokala nivån. En linjär interventionslogik hade använts för att lösa en komplex utmaning: ett problem (dålig hälsa) hade definierats och en orsak (exponering för rök från ineffektiva spisar) hade härletts. Utifrån detta hade en lösning (effektivare spisar) identifierats. Den stökiga verkligheten med dess matlagningsbeteenden och behov passade

uppenbarligen inte in i denna interventionsmodell. Ett bristande engagemang på lokal nivå av den globala programinsatsen hade resulterat i ett misslyckat projekt för såväl spisanvändarna, som klimatet. Utöver detta hade även utvecklarna av koldioxidkompensationsprogrammet missat möjligheten att ge hushållen en röst genom att inte inkludera dem i designprocessen.

Min erfarenhet från norra Indien väckte en nyfikenhet att utforska metoder och tillvägagångssätt att utforma projekt som placerar människor i centrum för designprocessen, samtidigt som de verkar inom ett befintligt program eller policyram (som ett koldioxidfinansieringsprogram). Att ta itu med dessa typer av problem är komplicerat eftersom fattigdom är så mångdimensionellt; det upplevs olika av människor beroende på deras sammanhang och deras förmågor – det vill säga vad de kan göra och vara. Även om vi lyckas ta reda på vad som behövs på lokal nivå, och levererar en intervention utifrån detta, kan det som händer lokalt efter interventionen introducerats få oavsiktliga effekter. Interventionsarbetet kan inte ses som en punktinsats utan bör ses som ett iterativt löpande arbete. Med all denna komplexitet är det inte förvånande att en linjär, orsaks- och effektmodell, för interventionsdesign inte fungerar så bra. Och med tanke på att en av tio människor i världen för närvarande lever i extrem fattigdom, är det angeläget att hitta nya sätt att möta dessa komplexa utvecklingsutmaningar.

Kort efter fältarbetet i Gujarat föreslog en vän att tjänstedesign skulle kunna vara ett användbart tillvägagångssätt för att ta itu med dessa typer av komplexa systemutmaningar. Tjänstedesign är ett människocentrerat, integrerat tillvägagångssätt för att designa tjänster och system som framgångsrikt har använts bland annat för att designa offentliga tjänster framförallt i Europa och Nordamerika. Vi började experimentera med tjänstedesignmetoder i våra forskningsprojekt och fann bland annat att verktyg som kundresekartläggning var användbara för att förstå, koppla och illustrera individers beteende och sammanhang med utvecklingsinsatser. Jag var nyfiken på varför dessa verktyg verkade fungera så bra och undrade över hur de kunde användas för att komplettera andra discipliner och tillvägagångssätt.

Även om designtänkande växer i popularitet för att designa produkter och teknologier i ”Global South-miljöer”, är tjänstedesign i dessa sammanhang inte lika utforskat. Lite är känt om hur tjänstedesign, dess tillvägagångssätt och verktyg kan användas i denna kontext för att hantera komplexitet och ge inflytande på

lokal nivå för att utforma utvecklingsinsatser. Att förstå det här är viktigt eftersom utan att veta hur tjänstedesign (ett tillvägagångssätt som härrör från västvärlden) – fungerar i dessa sammanhang, så finns det risk att det uppstår oavsiktliga negativa effekter.

Denna avhandling undersöker hur tjänstedesign kan användas för att ta itu med den komplexitet som är förenad med att designa utvecklingsinsatser och öka handlingsfriheten för de personer som är involverade i designprocessen. Jag arbetade i tvärvetenskapliga team med lokala samarbetspartners och genomförde tre fallstudier som tittade på utformningen av tre olika utvecklingsinsatser: en förbättrad spis i Kenya, en försäkringsprodukt för småskaliga jordbrukare i Uganda och en långsiktig energiomställningsplan för ett samhälle på den kenyanska landsbygden. Vi använde servicedesignmetodik kombinerat med kvantitativa metoder i spis- och försäkringsstudierna, och för energiomställningsplanen användes ”participatory backcasting” i syfte att utforska möjliga lösningar och alternativa framtider.

Vi fann att under rätt förutsättningar, där det finns starka, förtroendebaserade samarbeten med lokala aktörer samt erfarenhet och kunskap om ämnesområdet inom teamet; kan tjänstedesignverktyg vara användbara både för att hantera komplexitet och för att säkerställa lokalt inflytande i interventionsdesignen. Till exempel så kan kartläggningen av användarresan och konstruktionen av arketyper vara redskap för att förstå hur individer värderar en tjänst över tid och för att visualisera för beslutsfattare och programdesigners, mönster i heterogeniteten kopplat till vad människor värdesätter i en tjänst. Dessa verktyg kan också användas för att identifiera viktiga förmågor för användarna av en intervention. Med andra ord kan vi, med verktygen, peka ut de grundläggande förutsättningar och agerande (görande och varanden) som användare i ett system anser vara mest värdefulla för att leva ett bra liv, och koppla dessa faktorer till designfunktioner för interventionen. Med detta kan vi säkerställa att designarbetet bidrar till det långsiktiga utvecklingsmålet kopplat till interventionen, samtidigt som det är fast förankrat i den lokala kontexten. Forskningen visade också att prototypframställning kan ge användarna utrymme att vidareutveckla, utmana eller förkasta interventionens grundantaganden. Och som ett medel för att undersöka framtider kan ”participatory backcasting” underlätta gruppers föreställningar och förståelse för alternativa utvecklingsvägar, vilket kan leda till åtgärder på gräsrotsnivåer.

Att ta itu med komplexa social-ekologiska problem kommer att kräva transdisciplinära insatser som involverar forskare från olika discipliner såväl som icke-akademiska aktörer. Denna avhandling visar hur tjänstedesign kan vara ett integrerande tillvägagångssätt för transdisciplinär forskning och praktik i komplexa utvecklingssammanhang.

Popular science summary

In 2014, a colleague and I travelled to rural Gujarat, India, to collect data for a project aimed at understanding how carbon finance can be used to fund more efficient wood-burning stoves for low-income households. We interviewed people about their experiences of using these stoves, which they had purchased at a reduced price with the help of a local microfinance company. The stoves, which had been manufactured by a US-based company and were now being sold as part of a carbon offset scheme, were designed to use less wood, which is better for the environment, and to produce less smoke, which is better for users' health. The logic of the programme was simple. If a household in India used the efficient wood burning stoves instead of their less efficient traditional stove, a certain amount of carbon dioxide (CO_2) emissions would be saved (offset). Companies in the West could then buy this offset to cover their own CO_2 emissions, while also contributing to improved health, especially for women and children who are typically most exposed to household air pollution. The proceeds from the offsets sold would then be used to subsidize the cost of efficient stoves for poor households. On paper, this looked like a win-win!

When we began talking to the users of the improved stoves, however, we soon realized that there was a gap between the basic assumptions of the compensation programme and the reality of the users. Although the stoves had been marketed as practical and fuel-efficient, the women we spoke to described how they found them difficult to light, and that parts broke easily when heavy pots were placed on them. Fixing the stoves was tricky because there were no spare parts or local repair services. Finally, since most of the households in the area were extended families, several stoves would have been needed to handle all the daily cooking, but most households could only afford one.

My colleague and I were struck by how this seemingly globally promising climate finance programme completely missed the mark at the local level. A linear intervention logic had been used to resolve a complex challenge: a problem (poor health) had been defined, a cause (exposure to smoke from inefficient stoves) had been identified and, based on this, a solution (more efficient stoves) had been found. The messy reality of cooking behaviours and needs, however, clearly did not fit this intervention model. A lack of engagement by the global programme at the local level had resulted in a failed project for both the stove

users and the climate. In addition, the developers of the carbon offset programme had missed an opportunity to give households a voice by not including them in the design process.

My experience in northern India sparked my curiosity to explore approaches to designing projects that put people at the centre of the design process while operating within an existing programme or policy framework, such as a carbon financing programme. Addressing these types of problems is complicated because poverty is multidimensional: how people experience it depends on their context and their capabilities – that is, what they can do and what they can be. Even if it is possible to identify what is needed at the local level, and to deliver an intervention based on what we have found out, what happens locally after the intervention has been introduced can have unintended consequences. Intervention work cannot be viewed as a one-off effort but should be seen as iterative and ongoing. With all this complexity, it is not surprising that a linear, cause-and-effect model of intervention design does not work very well. Given that one in ten people in the world are currently living in extreme poverty, we urgently need to find new ways to meet these complex development challenges.

Shortly after the fieldwork in Gujarat, a friend suggested that service design might be a useful approach to addressing these types of complex, systemic challenges. Service design is a human-centred, integrated approach to designing services and systems that has been used successfully, primarily in Europe and North America, to design public services, among other things. We began to experiment with service design methods in our research projects and found that tools such as customer journey mapping were useful for understanding, connecting with and illustrating the behaviour of individuals and the context of development efforts. I was curious as to why these tools seemed to work so well and wondered how they could be used to complement other disciplines and approaches.

Although design thinking is growing in popularity for designing products and technologies in environments in the 'Global South', service design in developing country contexts has been less explored. Little is known about how service design, its approaches and tools can be used in these contexts to manage complexity and allow local-level influence to shape development efforts. This is problematic because without knowing how service design – an approach derived

from the West – works in these contexts, there is a risk of negative unintended consequences.

This thesis explores how service design can be used to address the complexity associated with designing development interventions and to increase the agency of those involved in the design process. Working in interdisciplinary teams with local partners, I conducted three case studies that examined the design of three development interventions: an improved cook stove in Kenya, an insurance product for small-scale farmers in Uganda and a long-term energy transition plan for a community in rural Kenya. Service design methodology combined with quantitative methods were used in the stove and insurance studies, while we used participatory backcasting to explore possible solutions and alternative futures for the energy transition plan.

We found that, under the right conditions, where there are strong, trust-based collaborations with local actors as well as experience and knowledge of the subject area within the team, service design devices can be useful for managing complexity and to ensure local influence over intervention design. For example, mapping user journeys and the construction of archetypes can be useful devices for understanding how individuals value a service over time and to allow decision makers and programme designers to visualize patterns in the heterogeneity of what people value in a service. These devices can also be used to identify capabilities that are important for the users of an intervention. In other words, these methods can help us to draw out the basic conditions and actions ("beings and doings") that users in a system consider most valuable for living a good life, and to link these factors to design functions in the intervention. In this way, we can ensure that the design work contributes to the long-term development goal linked to the intervention while remaining firmly anchored in the local context. The research also showed that prototyping can give users agency to further develop, challenge or reject the basic assumptions of an intervention. As a means for exploring futures, participatory backcasting can also facilitate community perceptions and understandings of alternative development paths, which can lead to individual and collective action at the grassroots level. Addressing complex social-ecological problems requires transdisciplinary efforts involving researchers from different disciplines, as well as non-academic actors. This thesis shows how service design can be an integrative approach for transdisciplinary research and practice in complex development contexts.

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Fiona Lambe,

Stockholm, January 2023

List of appended papers

I. F. Lambe, Y. Ran, M. Jürisoo, S. Holmlid, C. Muhoza, O. Johnson, M. Osborne, Embracing complexity: A transdisciplinary conceptual framework for understanding behavior change in the context of development-focused interventions, *World Development* 126 (2020) 104703. https://doi.org/10.1016/j.worlddev.2019.104703.

II. F. Lambe, Y. Ran, E. Kwamboka, S. Holmlid, K. Lycke, S. Ringström, J. Annebäck, E. Ghosh, M. O'Conner, R. Bailis, Opening the black pot: A service design-driven approach to understanding the use of cleaner cookstoves in peri-urban Kenya, *Energy Research & Social Science* 70 (2020) 101754. https://doi.org/10.1016/j.erss.2020.101754

III. Osborne, M., Lambe, F., Ran, Y., Dehmel, N., Tabacco, G. A., Balungira, J., Pérez-Viana, B., Widmark, E., Holmlid, S., & Verschoor, A. (2022). Designing development interventions: The application of service design and discrete choice experiments in complex settings. World Development, 158, 105998. https://doi.org/10.1016/j.worlddev.2022.105998

IV. Lambe, F., Osborne, M., Ran, Y., Dehmel, N., & Holmlid, S. (2022). Design devices for human development: A capabilities approach in Kenya and Uganda. *She Ji: The Journal of Design, Economics, and I novation*, *8*(2), 217–243. https://doi.org/10.1016/j.sheji.2022.06.001

V. Lambe, F., Nyambane, A., & Holmlid, S. (2022, June 16). *Backcasting as a design device to support grassroots system change: Insights from a case study on future energy pathways in rural Kenya*. DRS2022: Bilbao. https://doi.org/10.21606/drs.2022.240

Other relevant publications

The publications listed below are not included in this thesis, but they are closely related to the topic and shaped my thinking on the use of service design in interdisciplinary and transdisciplinary development research and research on sustainability transitions. They include peer reviewed articles that resulted from research collaborations immediately prior to and during my PhD education.

Jürisoo, M., **Lambe, F**., & Osborne, M. (2018). Beyond buying: The application of service design methodology to understand adoption of clean cookstoves in Kenya and Zambia. *Energy Research & Social Science*, *39*, 164–176. https://doi.org/10.1016/j.erss.2017.11.023

Dehmel, N., Ran, Y., Osborne, M., Verschoor, A., **Lambe, F**., Balungira, J., Tabacco, G. A., Pérez-Viana, B., Widmark, E., & Holmlid, S. (2021). Combining service design and discrete choice experiments for intervention design: An application to weather index insurance. MethodsX, 8, 101513. https://doi.org/10.1016/j.mex.2021.101513

Bailis, R., Ghosh, E., O'Connor, M., Kwamboka, E., Ran, Y., & **Lambe, F**. (2020). Enhancing clean cooking options in peri-urban Kenya: A pilot study of advanced gasifier pellet stove adoption. *Environmental Research Letters*. https://doi.org/10.1088/1748-9326/ab865a

Ran, Y., Nilsson Lewis, A., Dawkins, E., Grah, R., Vanhuyse, F., Engström, E., & **Lambe, F**. (2022). Information as an enabler of sustainable food choices: A behavioural approach to understanding consumer decision-making. *Sustainable Production and Consumption*, *31*, 642–656. https://doi.org/10.1016/j.spc.2022.03.026

West, S. E., Bowyer, C. J., Apondo, W., Büker, P., Cinderby, S., Gray, C. M., Hahn, M., **Lambe, F**., Loh, M., Medcalf, A., Muhoza, C., Muindi, K., Njoora, T. K., Twigg, M. M., Waelde, C., Walnycki, A., Wainwright, M., Wendler, J., Wilson, M., & Price, H. D. (2021). Using a co-created transdisciplinary approach to explore the complexity of air pollution in informal settlements. *Humanities and Social Sciences Communications*, *8*(1), 285. https://doi.org/10.1057/s41599-021-00969-6

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1 Introduction

Society is facing several novel anthropogenic risks, from climate change to biodiversity loss and deepening inequalities at the global and national levels, as well as traditional threats such as economic and food insecurity, discrimination and conflict (UNDP, 2022). These challenges are interlinked, which means that addressing any one of them will require grappling with complexity and thinking in terms of interconnected social-ecological systems (Berkes et al., 2001; Ostrom, 1990). Approaches that rely on linear thinking are ill-equipped to address complex environmental-development problems, and there are calls for innovative methods that can better meet this challenge (Mohieldin, 2019). It is also acknowledged that tackling complex societal challenges will require transdisciplinary efforts involving broad constellations of actors representing different academic disciplines, as well as non-academic stakeholders (Kates et al., 2001; Lang et al., 2012).

Service design has promise as an approach and practice that can be useful for addressing complex problems when applied in transdisciplinary research collaborations (Gustafsson & Bowen, 2017; Klein, 2007; Patrício et al., 2018). However, there are knowledge gaps regarding how service design methods function when designing development interventions in complex social-ecological contexts, as well as whether or how these tools can support local agency (Nahi, 2016). This thesis is concerned with the design of services that aim to achieve environmental sustainability and social well-being among households living in low and middle-income countries. The work is motivated by the inadequacy of traditional linear intervention design logic in attempts to address poverty in complex social-ecological contexts, and the potential for service design as a human-centred, integrative approach to improve understanding and translate this complexity into feasible development interventions.

The research presented in this thesis is based on three case studies that were conducted in Kenya and Uganda between 2018 and 2020: (*a*) a case study in Gachie, a neighbourhood in peri-urban Nairobi, that focused on the introduction of a clean burning cookstove and fuel, which I will refer to in the thesis as "the cookstove study"; (*b*) a case study in eastern Uganda that looked at the design of an insurance product to protect smallholder farmers from climate shocks, which I will refer to as "the insurance study"; and (*c*) a case study in rural

Machakos County, Kenya that focused on understanding local imaginaries of a long-term energy transition, which I will refer to as "the backcasting study". The cookstove and insurance studies used a service design approach and methods in combination with quantitative methods to support the design of a cookstove service system in Kenya and to inform the design of an insurance product in Uganda. In the backcasting study, participatory backcasting was used as a method to facilitate the co-development of an energy transition roadmap.

Each study was led by a team from the Stockholm Environment Institute (SEI) in Stockholm, where I have been working as a researcher since 2008. SEI is an international environment-development research institute with headquarters in Stockholm, and seven centres globally, including regional centres in Nairobi, Bangkok and Bogota. The three studies were supported by a constellation of collaborators that included SEI colleagues based at the Nairobi office, staff at a Kenyan cookstove company, colleagues at a local research institute in Uganda, Kenyan and Ugandan student research assistants, and community facilitators at each location. There was also international collaboration with the University of East Anglia on the insurance study and a team of Swedish service designers on the cookstove study. Details of each collaboration are provided in chapter 3.
This chapter presents the background to and framing of the research, the aim and research questions and an overview of the five appended papers.

1.1 Poverty and development

Statistically, as of 2022, most of the world's population is living in poverty; two-thirds of people globally are living on less than $10 per day and every tenth person is living on less than $1.90 per day, which is defined as extreme poverty (Castaneda Aguliar et al., 2022). Poverty is often understood as a lack of financial or material assets and measured in terms of how much money is available to a household per day relative to the cost of living in a given country (Ravallion, 2016). A broader understanding of poverty incorporates multiple deprivations beyond income, such as a lack of healthcare and education, lack of access to energy and water, lack of employment, and lack of power and voice (Alkire, 2003). This broadening of the concept of poverty has become known as Multidimensional Poverty (MDP). Approximately 1.3 billion people are multidimensionally poor (UNDP, 2022). Of this number, nearly 85% live in sub-Saharan Africa (556 million) or South Asia (532 million), roughly, 84% (1.1 billion

people) live in rural areas, and 16% (about 209 million) live in urban areas (UNDP, 2022).

Efforts to address poverty through development interventions have been under way since the end of the second world war. Development interventions, projects and programmes have been undertaken by a range of actors from governments to multilateral agencies, such as the World Bank, United Nations agencies and development banks, International and local non-governmental organizations (NGOs), grassroots community organizations and businesses with social innovation objectives. Such activities are often collectively referred to as "development" or "international development", umbrella terms that refer to development practice and research that seek to contribute to poverty alleviation and reduced inequalities (Horner & Hulme, 2019).[1]

1.2 Complexity and the design of development interventions

Although the statistics on the prevalence of global poverty are stark, it should be acknowledged that progress has been made in recent decades in addressing certain development problems. Governments and development organizations have been quite successful at providing development interventions that are standardized, routine and high volume (Andrews et al., 2012; Denizer et al., 2013). This has particularly been the case in situations when development issues are easily defined and specific technological or logistical solutions can be identified, such as, for instance, the eradication of smallpox in the 1980s or the Green Revolution in India in the 1970s (Hopkins, 1988; De Datta et al., 1968). Despite these achievements, more complex problems such as global poverty, climate change, and rising inequality persist (Head, 2008; Head & Alford, 2015; Moser et al., 2012) prompting calls for a greater emphasis on the dynamic, multidimensional, and complex nature of such problems (D. Burns & Worsley, 2015; Reyers et al., 2015; Waddock et al., 2015).

[1] The term "International development" has been criticized for failing to reflect the globalized nature of the world, the fact that poverty and inequality exist everywhere, and the global nature of challenges such as climate change. The term Global Development is increasingly used to better reflect this interconnectedness, and the fact that poverty and inequality exist everywhere, not only in the "Global South". I acknowledge the need for updating the terminology, however in this thesis, to avoid misunderstandings and to properly situate my research, I use the term "development" to refer to the collective efforts of the development community in low and middle income countries.

A central criticism of development research and programming is that it tends to be top-down, or characterized by central planning (Easterly, 2013). This top-down planning applies a linear intervention logic where development practitioners and policymakers first seek to determine the root cause of poverty, and then use causal reasoning to design solutions to address the problem (D. Burns & Worsley, 2015). This mode of intervention design is informed by development research and learning that is situated within the logical positivist paradigm, where experimental approaches are considered the gold standard and the goal is to isolate the causal effect of an intervention on a certain factor by controlling all other variables. While it might be possible to predict responses to interventions in controlled contexts where factors are relatively constant, the picture is much more complex in social situations where there are multiple interacting variables (D. Burns & Worsley, 2015). A linear causal logic is not well-equipped for thinking about and designing development interventions because such problems are typically complex or wicked in nature (Rittel & Webber, 1973). Wicked problems cannot be addressed in a traditional linear manner because the problem definition evolves as new potential solutions are explored and/or implemented, and because problems rarely have a single root cause but rather multiple interacting and interrelated dimensions and elements (Head, 2019).

The wickedness of development problems and processes is rooted in the fact that they often play out in complex adaptive systems where poverty is interconnected with other social-ecological processes (Radosavljevic et al., 2021; Stoett, 2016). In such systems, actors often interact in unstructured and unpredictable ways, resulting in the creation of cross-scale patterns and feedback loops that influence actor interactions (Levin et al., 2013). This means that processes and individual behaviours at a local scale can influence emergent properties at the sub-national, national, regional or global scale, which can, in turn, impact behaviours back at the local scale (Levin, 1999; Lansing, 2003). In other words, change is context dependent and can be triggered by small local interactions that lead to bigger changes over time, which ripple out through the system (Folke et al., 2010; Levin et al., 2013). Furthermore, since development interventions play out over time, and their impacts (both positive and negative) may not be felt for months or even years, it is difficult to predict what the long-term effects might be using a positivist linear intervention logic.

In addition to being ill-equipped to handle systems complexity, top-down, linear models of development have struggled to incorporate the perspectives and needs of people who are supposed to benefit from development interventions (Chambers, 1984; Easterly, 2013). At the individual level, poor people describe their experience of poverty as involving multiple disadvantages in parallel, from poor health and nutrition, to lack of basic services such as energy and water supply, to feeling ashamed and disempowered (Alkire, 2005). Experiences of poverty, and the extent to which someone will benefit from a development intervention, can vary depending on individual circumstances and characteristics, notably gender, age, ethnicity and location. Thus, designing interventions to improve well-being requires a deep understanding of individual situated experiences and context in order to understand what people consider valuable for living a good life (Robeyns, 2005).

Addressing complex social-ecological problems requires the participation and agency of diverse groups of stakeholders, including households and communities (Marshall et al., 2018; Norström et al., 2020). Agency refers to the ability that people have to take action, or to choose which action to take (Mansuri & Rao, 2004). Over the years, various attempts have been made to strengthen local participation and agency in development programming through the use of tools such as participatory rural appraisal in the planning of development interventions (Chambers, 1992). However, these efforts have been extensively critiqued for failing to engage with issues of power and politics, for institutionalizing technocratic and tokenistic approaches to participation in development processes and for putting too much emphasis on the local level (Cooke & Kothari, 2001; Hickey & Mohan, 2005). As a result, the impact of participatory development programmes on poverty reduction and citizen empowerment has been weak (Mansuri & Rao, 2004). Hickey and Mohan (2005, p. 4) argue that participation should be understood as "the exercise of popular agency in development processes", which requires multi-scale approaches that encompass structural and institutional as well as local and individual actors and processes. However, development programme structures are often too inflexible to incorporate citizens' perspectives and needs (Gaventa & McGee, 2013; Mansuri & Rao, 2004).

This systemic understanding of complexity has several major implications for development planning. First, identifying opportunities for creating new

feedbacks or strengthening desirable feedbacks requires an understanding of the drivers of behaviour and decision-making at the micro (local) level, and of how these relate to the larger social-ecological systems in which individuals and households operate (Lambe, Ran, Jürisoo et al., 2020). Transdisciplinary collaboration involving academic and non-academic participants is needed to identify the various important drivers of behaviour and critical system dynamics, including power and politics (Lang et al., 2012; Pohl, 2008). Second, adaptations and feedback loops in the system mean that development practitioners must be innovative, open to experimentation and willing to engage in iterative learning cycles when designing solutions, because the outcomes of interventions will be uncertain and unpredictable (D. Burns & Worsley, 2015; Osborne et al., 2022).

The development research and practitioner communities are under increasing pressure to find creative, scalable approaches that can successfully address this complexity (Mohieldin, 2019). This will require a shift in mindset away from the notion that we can deduce fixed, long-term solutions to complicated problems, and towards an acceptance that designing interventions to address complex problems is of necessity iterative and emergent.

1.3 Design for development: an overview

It is a primary assumption of this thesis that service design has the potential in transdisciplinary research to translate complexity at the local level into the type of information required by actors engaged in the design of development interventions. This assumption hinges on the fact that design methods are epistemologically distinct from the conventional qualitative methods used in development research and that design is often focused on addressing wicked problems in complex settings (Buchanan, 1992; Head & Alford, 2015) in which the information required to produce solutions to a problem depends on the idea for solving it (Rylander, 2009). In practice, this implies that a design process is intentionally iterative and emergent, alternating between problem articulation and proposed solutions, and is characterized by imagination, prototyping, and constant interaction with and empathy for the intended beneficiaries (Conklin, 2006; Lawson, 2010).

The history of design's application in development contexts may be traced back to Victor Papanek and E.F. Schumacher, who studied the appropriate design of products and technologies in contexts where resources are constrained (Papanek, 1972; Schumacher, 1993). Papanek contends that design is crucial for addressing the unmet or under-served needs of marginalized people in resource-limited societies, and that in these contexts, "decent and ethical design interventions" are possible (Papanek, 1972, p 227). This early thinking on design for development focused on technology and product design to achieve social and economic development in disadvantaged societies (Schumacher, 1993). During the past two decades, design methods have been used to develop products, services, and product service systems for so-called Base of the Pyramid (BOP) populations (Jagtap et al., 2013) where poor people are considered as both customers and producers of these products (Prahalad, 2009). The concept of frugal innovation has emerged as a discourse for designing in BOP settings, emphasising the need to do more with less when designing in low-income settings (Radjou et al., 2012; Radjou & Prabhu, 2014). Another strand of the literature on frugal innovation draws attention to bottom up and informal approaches to developing products and services that are often the result of long term experimentation and innovative practices where frugality is required due to resource constraints (Leliveld & Knorringa, 2018).

With the increase in multinational companies targeting relatively poorer and 'new middle class' consumers with frugally designed products and services (Leliveld & Knorringa, 2018), the past 20 years has seen a surge in interest in "design for development", and many universities now offer courses or projects that involve undertaking design research in this field (Donaldson, 2009; Falcioni, 2011; Jagtap et al., 2019a) while several large social innovation and technology companies dedicate entire departments to the topic (e.g., IDEO, Dalberg and Philips). The growth in popularity of design for development has raised concerns that design work carried out in Global South contexts could risk reinforcing colonial power imbalances and alienating local communities (Amatullo et al., 2022; Tlostanova, 2017). For example, Ambole draws attention to the recent proliferation of design thinking agencies in East Africa that are reinforcing a Western design paradigm and decentring local stakeholders from design processes (Ambole, 2020). Given these concerns, there is an urgent need to understand how design tools can be used to support the local agency of marginalized

communities so that useful tools are promoted and less effective or even harmful ones avoided (Jagtap, 2019a).

Despite the wealth of examples of design approaches applied in development contexts, there has been less research and analysis on the role that design tools have played in such cases (Jagtap, 2019a). As with the use of design in other contexts, design for development has tended towards short term, one-off design projects, and design research in these contexts tends not to follow up on the long-term outcomes and impacts of design processes (Donaldson, 2009; Hansen, 2019; Jagtap, 2019a; Jagtap, 2019b). Design work in low- and middle-income contexts has often focused on the development of innovative products and services with end-users either as consumers or beneficiaries, paying less attention to understanding how design approaches can support longer term transformative outcomes, such as participation, agency and empowerment for people and communities (Kraff, 2018; Escobar, 2017; Nahi, 2016).

1.3.1 Why Service Design

The focus of this thesis is service design rather than design more broadly. Prior to my PhD studies, I spent several years applying service design approaches and tools in transdisciplinary research projects at the Stockholm Environment Institute. I found it to be a useful way to conceptualize design work that takes place at and between different levels of a system. For instance, a service design approach was used to inform the design of interventions that aimed to increase the incomes of small-scale mango farmers in Kenya (Lambe, Ran, Jürisoo, et al., 2020), to improve the design of clean cookstove interventions in Kenya and Zambia (Jürisoo et al., 2018; Lambe, Ran, Kwamboka, et al., 2020), and to inform the design of mini-grid services for households and businesses in Tanzania and Zambia (Muhoza & Johnson, 2018; Ogeya et al., 2021). In each of these case studies, the problem in focus was complex. For example, interventions that aim to transform household energy practices require improved technologies and services that can meet the needs of the targeted users in terms of affordability, convenience, cultural compatibility, and so on. In addition, often, for the intervention to be effective, the demands of numerous key stakeholders must be met: the cookstove user, other family members, the service provider, a government agency, international funders, and so on. Accepting this systemic perspective on designing services implies a key role for service design in facilitating the mutual exchange of understandings between micro- and meso-level actors within a

system as part of the process of designing an intervention, which can be empowering for individuals and communities. (Patrício et al., 2018; Sangiorgi, 2011; Vink et al., 2021). Given the inherent complexity described above, however, it is much less clear how designers working in development contexts can bridge the micro- and meso-system levels in practice. A framework is needed that can guide design work that aims to put the needs and perspectives of the beneficiaries of development interventions at its core, while situating this granular knowledge within the wider system of actors, policies and normative goals.

1.3.2 Bridging service design and development

Thus far, I have argued that societies face several complex and persistent social-ecological challenges. Development programming that relies on top-down, linear models has been somewhat successful in meeting some of these challenges, but less so where problems are highly complex or wicked in nature. Development problems often play out in complex adaptive systems, where the impact of an intervention might be difficult to anticipate using linear intervention models. Furthermore, the multidimensionality of poverty as experienced by individuals means that attention must be paid to the situated experiences of the intended beneficiaries of a development intervention. To date, however, traditional approaches to development planning have struggled to incorporate local perspectives and foster participation (Cooke & Kothari, 2001; Kothari, 2019).

Service design appears to have potential as an approach and practice for addressing the challenges outlined above. The human centred and systemic nature of service design suggest that it could be useful for connecting situated, lived experiences at the micro level to development programming and policymaking at the meso level, but there is a lack of research on how his can be achieved in complex social-ecological contexts, and how designers can practically navigate in such contexts. Given the growing interest in applying design approaches to development problems in the Global South, there is an urgent need to understand how these approaches can address some of the challenges related to fostering local agency and participation faced by development research and programming, as well as the challenges and limitations of using service design as part of a transdisciplinary collaboration.

1.4 Aims of the thesis and research questions

Against this backdrop, the aim of this thesis is to explore how service design can be used as an approach to designing development interventions in complex social-ecological contexts. In pursuit of this aim, the thesis addresses three research questions:

RQ1: What are the salient features of service design for capturing and making sense of local contextual information for use by other important actors in the system?

By addressing this research question, the aim is to explore how service design can be used to understand the situated local experiences of individuals and households as the target beneficiaries of development interventions, and to communicate this information to other important actors in the system, including the development planners and policymakers responsible for devising interventions. **Paper 2** addresses RQ1 by applying a service design approach to understand the experiences of users of a clean-burning cookstove and fuel in peri-urban Nairobi. Experience mapping, archetypes and prototyping are used to understand how different types of users value the stove and how this relates to their experience of using the stove over time. **Paper 3** combines a service design approach with discrete choice experiments to understand how farmers experience and respond to risk, and to support the design of an insurance product that can help farmers cope with unexpected shocks related to climate change. P**aper 4** reflects on and synthesizes the findings from the Kenyan and Ugandan studies, using the capabilities approach as an analytical lens to examine how experience mapping, archetypes and prototyping operated as design devices for translating information about individual preferences, values and priorities into the information required by development programme designers in order to address multiple dimensions of poverty.

RQ2: What are the distinguishing features of service design that enable local agency in intervention design?

For this research question, the thesis aims to provide empirical evidence on how service design tools can create the conditions for local (individual, household and community) agency in the design of development interventions. **Papers 3 and 4** address RQ2 by exploring how the use of experience mapping and

prototyping could create space for research participants to input into the design process. **Paper 5** addresses the research question by exploring the use of participatory backcasting as a device for prototyping a future energy transition with a rural community in Kenya.

RQ3: How can service design methods and approaches complement conventional development research methods?

By addressing this research question, the thesis aims to contribute knowledge about how service design can be used alongside more conventional development research methods and approaches, including the effects on project outcomes. **Paper 1** proposes a conceptual framework for combining service design with behaviour change theory and social ecological systems thinking. The framework was developed using the empirical findings from a series of case studies conducted prior to starting my PhD and is the conceptual starting point for the work presented in this thesis. **Papers 2 and 3** apply elements of the conceptual framework in case studies. In **paper 2**, a service design approach was used in combination with quantitative measurement of cookstove usage by households to triangulate the data on user experience and to explain patterns of cookstove usage. **Paper 3** demonstrates the combination of service design and Discrete Choice Experiments (DCE) to understand farmers' strategies for coping with climate uncertainty and to identify the key design attributes of an insurance product that could protect farmers against climate shocks.

Table 1.1 Research questions addressed by each paper

Research Question	Paper 1 (Conceptual framework)	Paper 2 (Cookstoves)	Paper 3 (Insurance)	Paper 4 (Design devices)	Paper 5 (Backcasting)
RQ1		X	X	X	
RQ2				X	X
RQ3	X	X	X		

1.5 Position of service design in the thesis

The interdisciplinary work presented in this thesis is based on applied research conducted at the Stockholm Environment Institute in 2015–2020 as part of a research programme initially funded by the Swedish International Development

Agency (Sida) called the Behaviour and Choice Initiative (BCI). The initiative aimed to generate new knowledge on how individual and household level behaviour affects the outcomes of development interventions, and how this information can be used to support better design of development policies, programmes, interventions and products (Lambe, 2015).

According to the BCI rationale, development policies need genuine local ownership in order to make a lasting impact, which in turn requires an understanding of people's needs and motivations (Lambe, 2015). Although development actors and policymakers are increasingly aware of the importance of local contextual and behavioural factors in the design of effective interventions, this awareness is often not translated into practice in international development planning and policy, largely because this information is difficult to access in conventional approaches and difficult to translate into feasible interventions (Lambe, Ran, Jürisoo, et al., 2020). A central objective of the BCI was to develop and test new methods for understanding individual and household needs, perceptions and behaviours, and for situating this knowledge in the wider system of actors and processes. In the BCI projects, a service design approach was used to shift the focus to the experiences and contexts of the users (beneficiaries) of an intervention. Although I am not trained as a service designer, I had used design tools such as contextual interviews, cultural probes and rapid prototyping as methods for understanding people's perspectives and preferences in previous household energy research projects in India and Kenya (Jürisoo et al., 2018; Lambe & Atteridge, 2012; Lambe & Senyagwa, 2015).

In the initial BCI case studies, which focused on the uptake of advanced biomass burning cookstoves in Kenya and Zambia, the key focus was understanding individual behaviour. Design methods were a means of mapping out and sequencing behavioural drivers over time, and showing how this knowledge could be translated into better interventions and services for the user (Jürisoo et al., 2018). Two of these case studies were conducted together with professional service designers, which gave my colleagues and I the chance to practice using the tools and understand how they could form a valuable part of a research strategy in the BCI. These studies contributed to a conceptual framework that describes how a behaviour change model could be usefully combined with service design to enhance intervention design (**Paper 1**). This framework is the conceptual and

analytical starting point for the thesis. Figure 1.1 depicts the relationship between the BCI and PhD studies, including the timing of the work.

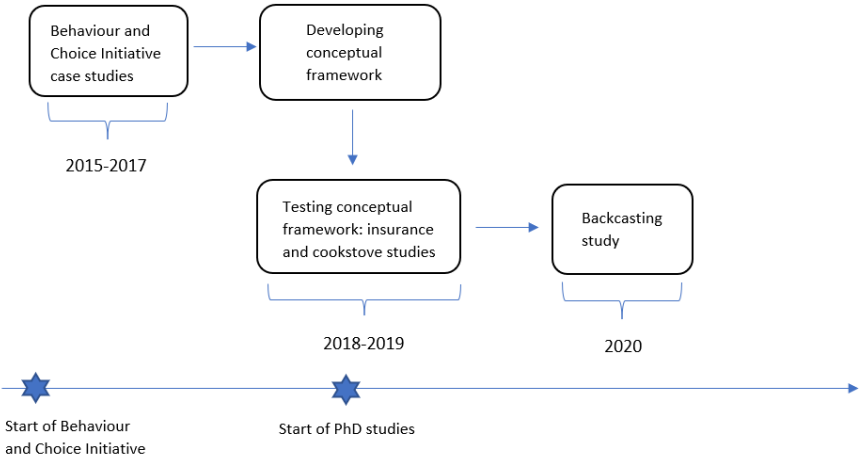


Figure 1.1 Relating the BCI and PhD studies

The value of service design in the early (pre-PhD) case studies was largely instrumental. The design tools and methods were a way to identify and map the drivers of behaviour at the household level, to show how these factors play out over time and how an intervention could be designed to account for these factors (Jürisoo et al., 2018). However, when conducting the case studies in Uganda and Kenya that contribute to this thesis, service design began to play a more fundamental role in the work, and I began to see how design could open up space for communities and households to affect how development-oriented initiatives are conceived and initiated. With this growing realization, I became more curious about the transformative potential of service design to create the conditions for local participation early on in the intervention design process.

In 2020, I had the opportunity to lead a project looking at local conceptions of a long-term energy transition in a community in rural Kenya (the backcasting study). Unlike previous projects I have worked on, there was no specific technology or practice in focus. Instead, the aim was to explore alternative pathways to electrification from the perspective of community members. This project provided a unique opportunity to test a participatory approach to co-designing a transition pathway where there were few donor requirements or parameters to

adhere to, and to further explore some of the questions relating to agency and participation that had emerged during the cookstoves and insurance case studies.

1.6 Overview of appended papers

F. Lambe, Y. Ran, M. Jürisoo, S. Holmlid, C. Muhoza, O. Johnson, M. Osborne, Embracing complexity: A transdisciplinary conceptual framework for understanding behavior change in the context of development-focused interventions, *World Development* 126 (2020) 104703. https://doi.org/10.1016/j.worlddev.2019.104703.

Paper 1 develops a conceptual framework for understanding behaviour change in complex development contexts. The framework is based on four empirical studies that were conducted prior to beginning my PhD, where service design was used to support the design of a development intervention together with external partners. The framework draws on service design as an iterative, human-centred approach used to understand and map individual experiences of new interventions or services, and to pinpoint opportunities for behaviour change to support the implementation of the interventions. Service design is positioned as an integrative approach to designing in complex contexts, which connects a model of behaviour change known as the COM-B model, used to understand individual behaviour and choice, with social-ecological systems thinking, used to account for system complexity.

Author contribution
I led two of the empirical studies that contributed to the framework, as well as the development of the conceptual model and paper writing.

F. Lambe, Y. Ran, E. Kwamboka, S. Holmlid, K. Lycke, S. Ringström, J. Annebäck, E. Ghosh, M. O'Conner, R. Bailis, Opening the black pot: A service design-driven approach to understanding the use of cleaner cookstoves in peri-urban Kenya, *Energy Research & Social Science* 70 (2020) 101754. https://doi.org/10.1016/j.erss.2020.101754

Paper 2 applied aspects of the conceptual framework developed in paper 1 to a case study looking at the adoption of a clean cookstove. A service design approach was applied in combination with quantitative measurement to understand

user experience of a clean cookstove and fuel in two peri-urban areas outside Nairobi. We used user journey mapping, archetype construction, solutions prototyping with users, and service system mapping with representatives from the Kenyan improved cookstove sector. Three different user archetypes were identified, based on the primary motivation for trying new stoves – to save money, convenience, and health and safety. Critical weaknesses in the pilot intervention were identified from the perspective of each archetype The findings of the qualitative analysis were verified by reviewing the corresponding stove use monitoring data.

Author contribution
I led the research design for the service design component, and on data collection tools, as well as one round (of two) of data gathering, and on the data analysis and paper writing. Co-authors at SEI led on the quantitative methods and one round of qualitative data gathering.

Osborne, M., Lambe, F., Ran, Y., Dehmel, N., Tabacco, G. A., Balungira, J., Pérez-Viana, B., Widmark, E., Holmlid, S., & Verschoor, A. (2022). Designing development interventions: The application of service design and discrete choice experiments in complex settings. World Development, 158, 105998. https://doi.org/10.1016/j.worlddev.2022.105998

Paper 3 also applied aspects of the conceptual framework developed in paper 1, this time to a case study involving the design of a climate adaptation service in Uganda. We applied a service design approach in combination with discrete choice experiments (DCEs) to support the design of an innovative insurance product – Weather Index Insurance – to protect small-scale farmers in eastern Uganda from risks associated with climate variability. User journey mapping, archetype construction and solutions prototyping with users were applied during the service design phase.

Author contribution
Research design, data collection (service design phase), analysis and paper writing were shared equally with the joint first author. The last author led on the design and implementation of the DCE.

Lambe, F., Osborne, M., Ran, Y., Dehmel, N., & Holmlid, S. (2022). Design devices for human development: A capabilities approach in Kenya and Uganda. ***She Ji: The Journal of Design, Economics, and Innovation*****, *8*(2), 217–243. https://doi.org/10.1016/j.sheji.2022.06.001**

Paper 4 presents a synthesis of the findings from the cookstoves and insurance case studies to identify the role of service design. In each case study, experience mapping, archetype construction and prototyping were used to understand individual needs, capacities and values, and to translate this information into design features for the interventions. Using Amartya Sen's capabilities approach as an ex-post analytical frame, we examined how these devices supported local participation in the design process.

Author contribution
I led on the reanalysis of findings and paper writing. Co-authors provided critical feedback on various iterations of the paper.

Lambe, F., Nyambane, A., & Holmlid, S. (2022, June 16). ***Backcasting as a design device to support grassroots system change: Insights from a case study on future energy pathways in rural Kenya*****. DRS2022: Bilbao. https://doi.org/10.21606/drs.2022.240**

Paper 5 uses a case study approach to explore the use of participatory backcasting as a design device to facilitate bottom-up community engagement in long term development planning. We used participatory backcasting to facilitate the codesign of a ten-year roadmap detailing a transition to electric cooking with 30 members of a rural Kenyan community. The roadmap articulates a local vision of a long-term development process, including the community's role in that process. Follow-up interviews were conducted with a sample of the workshop participants to understand how they perceived the backcasting process, and whether it had led to any changes at the individual or community level. The findings are discussed in relation to the literature on transformation design.

Author contribution
I led on research design, data collection, analysis and paper writing. Colleagues at SEI Kenya provided support with data collection and training the workshop facilitators.

1.7 Research contributions and target audiences

The work presented in this thesis lies at the intersection between development studies and service design, and therefore makes contributions that should be of interest to both development and service design researchers and practitioners. First, for the development community, the thesis demonstrates the application of a conceptual framework (**Paper 1**), which draws on service design as an approach that can be used to help development researchers and practitioners navigate local level complexity and translate it into valuable services or service components for households and communities. The conceptual framework provides a novel approach to thinking about development research in complex settings where development intervention design is reframed as a collaborative process where service users are active value co-creators rather than passive beneficiaries. Second, for the design research community, the thesis presents new knowledge on how design devices can support the design of development interventions. Specifically, the findings show how prototyping, archetype construction and user journey mapping can operate as devices that can help to ensure that an intervention contributes to the strengthening of local agency (**Papers 2 and 3)** and capabilities **(paper 4**). The findings also demonstrate the value that other disciplines and approaches can bring to design in transdisciplinary research contexts.

Finally, the thesis conceptually bridges service design and the capabilities approach by using design devices to pinpoint the most relevant capabilities sets for a proposed intervention **(paper 4)** and showing how creative methods (participatory backcasting) can support collective capabilities **(paper 5)**. This conceptual linkage can anchor and frame design work in terms of locally defined, long term development goals and help service design practitioners and researchers to navigate normative development objectives while ensuring that their design efforts pay attention to relevant ancillary development interventions and services (**Papers 4 and 5**). Collectively, these contributions provide a conceptual and methodological bridge between the fields of development studies and

service design – two fields that I believe have much to gain from speaking to one another.

1.8 Outline of thesis

This thesis is arranged in two parts: the kappa and the appended papers. In the kappa, this chapter discusses the research context, including the complexity of designing development interventions, and provides an overview of the literature on design in development contexts. The research aims and research questions, and an overview of the appended papers are also presented. Chapter 2 describes the key conceptual and theoretical concepts underpinning the thesis. Chapter 3 outlines the research methodology and provides detailed descriptions of the three case studies and the methods used in each. The strategies used to ensure research quality and ethical considerations are also described. Chapter 4 presents the main findings of each of the five contributing papers. Chapter 5 discusses the key findings in relation to the literature, contextualizing them in the light of the research questions. Some research limitations, implications for policy and practice and research impacts are also discussed. The overall conclusions and suggestions for future research are presented in chapter 6.

2 Conceptual and theoretical framing

This chapter presents the main conceptual and theoretical frames of reference used to situate and analyse the empirical work of the thesis. My work is primarily positioned in relation to service design as a practice for and approach to designing in complex development settings. I also lean on the capabilities approach as a framework for conceptualizing the multidimensionality of poverty and wellbeing. I describe below the key assumptions of these conceptual and theoretical frames, highlight important complementarities and divergences and show how they were used to support the empirical work throughout my PhD journey.

2.1 Service design as an approach and practice

Service design as an approach is a conceptual red thread that runs through this thesis, from underpinning the conceptual framework presented in paper 1 to providing the conceptual and methodological foundation for the empirical studies presented in papers 2, 3 and 4. As described in chapter 1, service design proved a useful approach in the research that led up to my PhD studies for understanding users' perspectives and behaviours, and situating them in relation to a wider system as the first step to designing a development intervention. This section provides background on the field of service design before describing its features as an approach and practice that are relevant for designing in a development context. I also reflect on the contextual origins of service design, and the need to adapt service design methods and approaches when applying them in other sociocultural contexts.

The original focus of service design research and practice was on service interfaces as points of interaction between service providers and service users, which allow designers to draw on their ability to understand human experiences and translate this understanding into improved service experiences (Holmlid, 2007; Sangiorgi, 2009). Over the years, the focus expanded to include service interfaces and experiences (Clatworthy, 2011; Secomandi & Snelders, 2011), service contexts and systems (Morelli, 2002, 2009), and organizational changes (Pinheiro et al., 2012). Thus, service design has had broad application that spans the design of interactions to the design of transformations at the individual, organizational and societal levels (Wetter-Edman, 2014)

In around 2008, there was a notable shift in focus in the service design literature from defining and justifying service design as a field to research on service design, including a focus on the tools and methods of service design practice (Blomkvist et al., 2011). Since I am primarily interested in service design as an approach and practice, I focus on this latter wave of the literature and deal less with service marketing and management literature, although the concepts of value in use and value cocreation are touched on.

As a design *practice,* service design can be seen both as a set of tools and techniques, as well as an approach to service innovation, increasingly used to address challenges within the public sector (Malmberg, 2017), and to address wicked problems in social systems (Banathy, 1996; Jones, 2014). Several defining principles of service design, such as human centredness, co-creation and collaboration with actors operating at different service system levels (Stickdorn & Schneider, 2011), make the practice well suited to the challenges of complexity encountered when working in the development context.

Service design is commonly described as a qualitative approach to understanding people and their contexts, motivations and behaviours, and to co-designing services or systems that better meet their needs (Edvardsson et al., 2012; Manzini, 2015; Pfannstiel & Rasche, 2017). Thus, service design is inherently *human centred*, where service users are understood as experts on their own situations (Ehn, 1993) and services are designed from the perspective of the user. The human centredness of service design can be derived from the prominence given to the concept of value-in-use, that is, that the value of a service is uniquely determined by the beneficiary or user (Vargo et al., 2008; Vargo & Lusch, 2014) who is engaged in both creating and evaluating value over time in an "experiential process of usage" (Grönroos & Voima, 2013, p 138). The centrality of the user is a key reason why service design is well-suited to development contexts, where interventions often fail because the needs, perspectives and contexts of the intended beneficiaries are not well understood or not properly translated into appropriate and desirable services that can meet those needs (Datta & Mullainathan, 2014).

At the same time, service design is concerned with engaging with a range of stakeholders that play a role in the service, beyond the end-users (Akama, 2009; Sangiorgi & Prendiville, 2017). Here, the service logic concept of *cocreation*

directs attention to the various interactions, relationships and processes that play out and connect the service user with the service providers (Vargo & Lusch, 2008), and is a key tenet of service design research. In practice, co-creation entails actors in service systems engaging in a creative process to define problems and explore solutions (Sanders & Stappers, 2008). In complex service systems, co-creation with various key actors allows for a multiplicity of needs to be considered, of both the service provider and the service user (Patrício et al., 2018). This capacity to handle diverse interests while drawing on the creativity and resources of multiple actors within a service system has the potential to support the design of better services and service systems in resource poor contexts where activities are often undertaken in an uncoordinated way by various actors operating in siloes (Jagtap, 2019b, 2022). The importance of co-creation as a service design concept means that collaborative approaches are central to service design practice (Sangiorgi & Prendiville, 2017). Collaborative processes, with their roots in participatory design and co-design, can facilitate the inclusion in the design process of a wide range of contextual factors, such as socio-cultural, political-economic and ecological factors. The dual challenges of understanding individual subjective experiences and managing the complexity of service systems has led to the development of a number of service design methods and tools (Sangiorgi & Prendiville, 2017).

Since experiences are subjective and elusive phenomena, mapping techniques, and observational and ethnographic methods can be used to gain a deeper understanding of user experiences (Holmlid & Evenson, 2008; Meroni & Sangiorgi, 2016; Parker & Heapy, 2006). The practice of making external representations to communicate stakeholders' insights is commonly referred to as *visualization* (Segelström, 2009), considered one of the primary skills of a service designer and a key feature of service design practice (Holmlid, 2007; Kimbell, 2009). Visualization is critical to designing services because services are intangible and abstract, and there are often different actors or processes involved in their conception and construction (Diana et al., 2010). Visualization is used to translate information and support communication between different actors in complex service systems (De Lille et al., 2012). Examples of visualization techniques include customer or user journeys, personas, storyboards, desktop walk throughs and system maps (Segelström, 2013). These tools are often used to help participants in a design process to articulate value(s) and goals, and work as boundary objects between participants and developers, creating opportunities

to prototype and assess the qualities of future value creation (Blomkvist & Segelström, 2014; Foglieni & Holmlid, 2017).

Given that the users of services actively create value in their experiences of services or service systems (Lusch et al., 2008; Prahalad & Ramaswamy, 2004), a key focus of service design is on understanding the user's experience and identifying the "touchpoints" within the service system that most significantly influence the experience (Payne et al., 2008; Teixeira et al., 2012). Mapping techniques, such as customer journey mapping or user journey mapping, are commonly used to identify and visualize these critical touchpoints in an individual's experience of a service or system. In my research, I refer to "experience mapping" as well as user journey mapping because I am interested in a broad understanding of the lived experience of a given service or system, beyond experiences of certain touchpoints from a customer or user perspective. The mapping process involves developing a composite illustration of multiple users' experiences or a new technology, service or practice, from the point of hearing about it, to purchasing/acquiring it and making it part of their daily routines (Jürisoo et al., 2018; Stickdorn & Schneider, 2011).

Archetypes or personas are common classifications based on people's behaviours and needs. These are often developed to inform and "test" interventions in the design process. Since emerging as a customer segmentation tool in marketing, personas have been used in public service innovation to express complex layers of contextual information about people, their values, beliefs and practices, and sometimes including quantitative demographic measures such as age or income (Bason, 2018). To avoid market research connotations and to highlight that this tool enables rich, holistic descriptions based on interactions with real people, my research refers to archetype construction rather than personas. Archetype construction can facilitate an understanding of people's wider lived experience beyond their experiences of specific service touchpoints (Trischler & Scott, 2016).

Useful archetype descriptions might include insights about people's key motivations, the kinds of benefits they might obtain from a service or intervention and information about the types of challenges they face in relation to current services (Bason, 2018). Thus, archetypes can be used as an analytical tool to bridge between qualitative and quantitative insights, and identify patterns in the

information on people's experiences of a current or proposed service. In this way, archetypes support prototyping, allowing the designer to explore whether possible design features would fit archetypes (Bason, 2018).

Prototyping is central to design and has been emphasized in design thinking and transformation design as a strategy to "fail early to succeed sooner," especially when it is conducted in a real-world situation (Brown & Wyatt, 2010; C. Burns et al., 2006). Prototyping is used to explore, evaluate or communicate a future situation using a facsimile version of a to-be-designed artefact (Blomkvist, 2016). In service design, prototyping is used for gathering both insights and feedback on a proposed service from prospective service users and other important stakeholders without having to create the service (Blomkvist & Holmlid, 2011). The practice of quickly testing and evaluating services, or aspects of services, prior to their full development can be useful during the design of development interventions, which typically follow a linear model where learning and evaluation occur after the intervention has been rolled out.

2.1.1 Transferring service design approaches to Global South contexts

Although there is clear potential for service design to enhance development practice and research, key assumptions underpinning design and service design practice have been criticized for being value neutral, for failing to address wider issues of power and structural inequality and for promoting a monolithic view of culture (Duan et al., 2021; Tlostanova, 2017). These critiques draw attention to the contextual origins of service design as an approach that emerged from the Global North, specifically Europe and North America. Many key concepts were framed in the service marketing and management literature, and reflect a capitalist paradigm of modernization and technological progress (Escobar, 2018). From a methodological perspective, design pragmatically focuses on how to transform a situation (Dalsgaard, 2017), which is sometimes referred to as solution-ism (Manzini, 2016), and may not be an appropriate way of operating in all cultures.

Cultural issues are increasingly discussed as significant challenges to service design research and practice (Cipolla & Reynoso, 2017; Duan et al., 2021). Some scholars view western-centric service design models as mismatched with other non-western cultural contexts (Baek et al., 2019; Taoka et al., 2018; Lee

& Lee, 2007). Baek et al. (2019) argue that user-centred design framings often overemphasize visible cultural levels (such as behaviour) and risk neglecting the intangible value of culture. Taoka et al. (2018) suggest that, due to the power hierarchies in Japan, designers involvement in the co-design process restricts the participation and empowerment of non-designers. Cipolla and Reynoso discuss the importance of designers understanding the cultural environment that forms the basis for local sense-making processes (Cipolla & Reynoso, 2017). In the African context, for example, the core ethos of *Ubuntu* (humanness) is a concept that binds individuals to communities, and shapes how people relate to one another and to society (M'Rithaa, 2009).

It is critical that designers from western contexts find ways to ensure that local sense-making and culture are not missed, glossed over or lost in translation (Cipolla & Reynoso, 2017). Scholars and practitioners are increasingly calling for more a more reflexive approach to service design practice in which designers are aware of their potential to impose their own cultural constructs or recreate harmful societal structures (Duan et al., 2021; Tlostanova, 2017), and seek to incorporate indigenous and local approaches to design (Cipolla & Reynoso, 2017; Maré & M'Rithaa, 2018).

2.1.2 Service design and design for transformation

As described in chapter 1, service design played an instrumental role in the work conducted immediately prior to my PhD studies, which was mostly concerned with understanding how best to create value for the users or beneficiaries of a development intervention. During my PhD studies, however, following further exposure to literature on transformation design and systems design, I became interested in the transformative potential of service design beyond a set of tools for the delivery of specific services, to an approach that could be used to instigate structural change. I began to wonder: if the object of service design is value in use, could the object of transformation design be the agency of individuals and communities to drive change processes? Papers 4 and 5 explore what transformation design could mean in the context of development, looking at how the agency and participation of the beneficiaries of development interventions could be strengthened through the design process.

Although the concept of transformation design dates back a long way, often linked to Herbert Simon's characterization of social transformation in *Sciences*

of the Artificial (Simon, 1969), transformation design as a practice emerged strongly in the UK in the 2000s, with the publication of the RED Paper on Transformation Design by the UK Design Council. As Burns et al. (2006) argue, many of today's more complex problems arise because the latent needs and aspirations of 'end-users' – those individuals who will receive the benefit of a given service or system – are not being met by what the service or system is offering, and that this is more likely to be a problem when changes are driven by technical or bureaucratic goals. For Burns et al. (2006), design methods have huge potential for understanding users and the multiple complex problems they face within a wider system that acknowledges the interconnectedness of societal challenges. In the RED paper, Burns et al. argue that power and community empowerment are central to transformation design, which they define as "a human-centred, interdisciplinary process that seeks to create desirable and sustainable changes in behaviour and form – of individuals, systems and organizations – often for socially progressive ends. […]" (C. Burns et al., 2006).

The RED paper inspired a wave of bottom-up public sector design work across Europe characterized by the inclusion and even empowerment of citizens as central actors in the system (Cottam, 2019; Cottam & Leadbeater, 2004; Malmberg et al., 2019; Manzini, 2015). Although the RED paper mostly refers to cases in the UK, its relevance to development-oriented initiatives in low-income settings, where the services or solutions are often developed by large bureaucratic development agencies, disconnected from the needs, behaviours and contexts of the people who are supposed to benefit from them, is clear.

In the ground-breaking work presented in her book, *Radical Help: How We Can Remake the Relationships Between us and Revolutionise the Welfare State*, Hilary Cottam, a co-author of the RED paper, operationalizes the RED paper concepts and demonstrates how design can be used to empower and engage communities, including service users and frontline workers in the creation of new public services (Cottam, 2019). This bottom-up approach uses existing relationships within communities as the building blocks of new services that can enhance wellbeing and capabilities, while reducing the dehumanizing bureaucracy in current welfare systems (Cottam, 2019).

The idea of community as the source and driving force of change is explored by Daniela Sangiorgi who builds on the defining features of transformation design

in the RED paper by making connections with the fields of Community Action Research, and Participatory Design (Sangiorgi, 2011). She argues that Community Action Research, with its focus on consciousness raising at the grassroots level, is a first step towards the bottom-up transformation of a system (Sangiorgi, 2011). As such, community action research does not shy away from issues of power and interests, but instead aims to empower communities to challenge the systems and structures that are failing them, and to put in place the conditions for action. The role of design, therefore, is to facilitate this space making and participation.

Sangiorgi provides a set of principles that unify the transformative practices used in service design, organizational development and community action research, with a particular focus on issues of public service reform and wellbeing. The seven key principles are: (1) active citizens; (2) intervention at community scale; (3) building capacities and project partnerships; (4) redistributing power; (5) designing infrastructures and enabling platforms; (6) enhancing imagination and hope; and (7) evaluating success and impact (Sangiorgi, 2011). Sangiorgi suggests that these principles can be used to reflect on the role of service design in transformative projects, and in particular on the responsibility of the designer in such processes (Sangiorgi, 2011).

The principles of transformative services provide a useful frame of reference for understanding how community empowerment can be supported through design work, and highlight the responsibility of the designer when working with vulnerable communities. The role and responsibility of the designer in managing transformative change processes is also brought into focus. However, these principles are not specifically tailored to the complexity of development contexts described in chapter 1. Core questions remain about how individuals are related to the design process, how transformative services might respond to external normative goals such as sustainable development or poverty reduction, and what this would mean in practice for design work.

2.2 The Capabilities Approach

This thesis aims to contribute new knowledge about how service design can be used as an approach to designing development interventions in complex social-ecological contexts. Until recently, the main approach to understanding and

measuring development and poverty was based on consumption and income. As discussed in chapter 1, I subscribe to a multidimensional understanding of development that goes beyond utilitarian and income- or resource-based measures of welfare. The idea that well-being is based not solely on utility or consumption measured by income, but also on an individual's freedom to lead a life they consider valuable is at the core of the capabilities approach, a framework for well-being and justice introduced by Amartya Sen and Martha Nussbaum (Robeyns, 2005). Given my focus on the multidimensionality of poverty, the capabilities approach is a useful and relevant framework for understanding the role of design in development contexts. This sub-section provides background on the capabilities approach and its key concepts, and an overview of the literature on capabilities and design.

The capabilities approach is a normative framework for the evaluation of human well-being and social arrangements, as well as for the design of social change policies and proposals. (Robeyns, 2005). As a framework, the approach aims to move beyond income-based evaluation models of development, by instead drawing attention to what people are *actually* able to be and to do (their capabilities), and whether a proposed intervention or policy should focus on people's capabilities rather than just their economic interests (Nussbaum, 2013a). Central to the approach is whether policies, practices and institutions are focused on people's capabilities, that is, their opportunities to lead a life they value (Robeyns, 2017). Thus, the capabilities approach is an alternative development framework to income-based measures of well-being that dominated until the 1980s and has become influential in measuring development progress at the macro level, and the basis of the methodology used for the annual UNDP Human Development Report (Sumner & Tribe, 2008). Capabilities are what individuals have the freedom to do or to be. These "beings and doings", or what Sen calls *functionings*, can be understood as "an umbrella term for the resources, activities and attitudes people spontaneously recognize to be important, such as knowledge, a warm friendship, an educated mind, a good job" (Alkire, 2003, Pg 5). Capabilities are the achievement of these functionings and can include, for example, the capability to maintain meaningful relationships and the capability to be educated (Sen, 2001). Capabilities can be combined in that they link internal capabilities, such as physical and mental capabilities, with capabilities realized through training or education, as well as "external conditions", such as the existence and accessibility of relevant structures and institutions that allow

individuals to exercise and expand their capabilities. (Nussbaum, 2013b, pg. 84) At the core of the capabilities approach is an understanding of people as multi-faceted beings who require different types of support to achieve their capabilities, depending on their individual circumstances and conceptualizations of a "good life".

Nussbaum developed Sen's capabilities approach into a theory of social justice focused on what she terms "the narrative context of human lives", in order to ensure the design of policies and interventions that are meaningful to real people (Nussbaum, 2013a). Thus, for Nussbaum, understanding the local context is essential when designing interventions that can enhance capabilities. The capabilities approach is well-suited to revealing these insights because, as Nussbaum says, it "stays close to the ground…it asks and answers a question that real people pose to themselves every day: 'what am I am able to do and to be'" (Nussbaum, 2013a, Pg.106). Sen refused to define a core set of essential capabilities. However, Nussbaum proposes a list of ten central capabilities and argues that a basic threshold level for each is required to secure a life of dignity, and that the role of government is to ensure this threshold level for all citizens (Nussbaum, 2013b).

The overriding focus of the capabilities approach on individual functionings has been critiqued for neglecting the role of communities in expanding individual capabilities, and in generating collective capabilities (Ibrahim, 2017; Stewart, 2005). Sen argues that "societal arrangements, […] are investigated in terms of their contribution to enhancing and guaranteeing the substantive freedoms of individuals" (Sen, 2001, Pg xii-xiii) and rejects the concept of collective capabilities, arguing that that capabilities resulting from collective action depend on individual capabilities (Sen, 2001). Collective capabilities are not simply the sum of individual capabilities, but rather new capabilities that are formed when individuals engage in collective action, which can, in turn alter societal structures. (Evans, 2002).

If achieving individual capabilities is an essential requirement for human development, attaining collective capabilities might be considered essential for transformative development. As the capabilities and development researcher Solava Ibrahim highlights, community level transformation is an incremental process of change that occurs at the local level through the attainment of both individual

and collective capabilities (Ibrahim, 2017). To understand how communities can achieve collective capabilities, Ibrahim developed a conceptual framework known as the 3C model to highlight how external actors, such as NGOs and development partners, can best support grassroots initiatives that aim to achieve social change (Ibrahim, 2017). The framework is based on empirical findings gathered during a decade of research on grassroots community projects in Egypt and West Africa. At its core, the model describes "three Cs", *conscientization*, *conciliation* and *collaboration*, as essential processes for communities to achieve collective capabilities through acts of collective agency (Ibrahim, 2017).

According to the 3C model, the first step in grassroots social innovation is conscientization, which Ibrahim describes as the process of gaining awareness of and reflection on one's situation, as well as knowledge about how the situation might be changed or improved (Ibrahim, 2017). Conscientization might be triggered through the articulation of individuals' values, aspirations and well-being goals, and their ability to achieve these goals. It enables individuals to think critically about their problems and then to actively engage in pursuing solutions to address them (Freire et al., 2021). The capacity to aspire is considered a critical element supporting conscientization, since being able to articulate hopes, dreams and material needs is an important step towards achieving capabilities (Appadurai, 2004).

The second stage of the 3C model is conciliation, the process of community consensus building around key priorities, needs and capacities (Ibrahim, 2017). An important aspect in conciliation is the acknowledgement of differentiated needs and capacities within the community and, based on this acknowledgement, an acceptance that the impacts of community level interventions will be felt differently by community members depending on individual circumstances (Ibrahim, 2017). Crucially, through the conciliation process, community members envisage the potential benefits that can be achieved collectively beyond the benefits that they might experience as individuals. Conciliation takes place through inclusive public discourse, and results in a communal vision that reconciles individual and community well-being objectives (Ibrahim, 2017).

The third and final stage of the 3C model is collaboration with external partners. Collaboration is needed for community level interventions to be sustained in the

longer term. Here, Ibrahim describes how collaboration with civil society actors, such as NGOs, government bodies and donors is essential if the community is to reshape power relations with these actors and redress power imbalances (Ibrahim, 2017). Here, the community vision that results from the first two stages can be a useful reference point when redefining power relations as a clear articulation of community agency (Ibrahim, 2017).

Table 2.1 Nussbaum's 10 sets of capabilities

1. Life The ability to live a full, normal-length human life, and not live a life reduced to worthlessness.
2. Bodily Health The ability to have a good life, which includes (but is not limited to) reproductive health, proper nourishment and shelter.
3. Bodily Integrity The ability to move around freely, in addition to having sovereignty over one's body, which includes being secure against assault, for example, sexual assault, child sexual abuse and domestic violence, *and* the opportunity for sexual satisfaction.
4. Senses, Imagination, and Thought The ability to use one's senses to imagine, think and reason in a "truly human way" informed by an adequate education; the ability to produce self-expressive works and engage in religious rituals without fear of political ramifications; the ability to have pleasurable experiences and avoid unnecessary pain; and, finally, the ability to seek out the meaning of life.
5. Emotions The ability to have attachments to things outside of ourselves; this includes being able to love others, grieve at the loss of loved ones and feel anger when it is justified.
6. Practical Reason The ability to form a conception of the good and critically reflect on it.
7. Affiliation A. The ability to live with and show concern for others, empathize with (and show compassion for) others and the capacity for justice and friendship. Institutions help develop and protect forms of affiliation.

B. The ability to have self-respect and not be humiliated by others, that is, being treated with dignity and equal worth. This entails (at the very least) protections from being discriminated against on the basis of race, sex, sexuality, religion, caste, ethnicity and nationality. In work, this means entering into relationships of mutual recognition.
8. Other Species The ability to have concern for and live with other animals, plants and the environment at large.
9. Play The ability to laugh, play and enjoy recreational activities.
10. Control over One's Environment A. Political: The ability to effectively participate in political life, which includes having the right to free speech and association. B. Material: The ability and capacity to own property not just formally, but materially; that is, as a genuine opportunity, as well as the ability to seek employment on an equal basis with others and freedom from unwarranted search and seizure.

Source: Nussbaum (2013a, p 33-34).

2.2.1 Capabilities and Design

Conversion factors - or the conditions that must be present for a development intervention to expand capabilities (Sen, 2010) - are a crucial component of the capabilities approach. The concept of conversion factors is analogous to that of value-in-use in service research (Vargo et al., 2008). Just as resource provision will only result in capability expansion if the right conditions have been put in place, a designed service or system can only deliver value to end-users if the conditions for their successful use have been intentionally created (Oosterlaken, 2009). There is no inherent value in a service or system; rather, the value is imparted by the user as a co-creator of value, depending on the user's sense of what is important and significant (Kustrak Korper et al., 2022).

Thus, designers must always pay close attention to the values and objectives of individual service users, which requires grappling with a high level of diversity at the local level. Interventions that have a development or wellbeing objective bring an additional layer of complexity. Oosterlaken proposes a "capabilities-

based approach" to designing for society, and particularly for the world's poor (Oosterlaken, 2009) Although Oosterlacken's ideas are generally focused on technology design, they are equally applicable to the design of services and systems. According to Oosterlacken, the capabilities approach can enhance design in in two ways: by fostering a greater awareness of the personal, social, and environmental factors that influence the conversion of resources into capabilities, and by mandating genuine local engagement in design processes (Oosterlaken, 2009, 2012).

Several toolkits have been developed to allow for the practical application of capabilities to the design and evaluation of technologies in development contexts. In her work on designing Information and Communications Technologies for Development (ITC4D), Kleine demonstrates the Choice Model, a framework used to evaluate the development impact of information and communication technology (ICT) services for vulnerable communities by taking the user's own wished-for development outcomes and individual resources (conversion factors) as the starting point (Kleine, 2010). In his work on technology design for well-being, Steen argues that the capabilities approach can help to avoid the prioritization of certain intervention outcomes over others due to biases within the design team or external influences that might restrict individual freedom of choice (Steen, 2016). Steen developed a set of cards that can be used to identify important capabilities for the target group at the outset of an intervention design process (Steen, 2016). Similarly, focusing on the design of medical devices, Aranda-Jan et al. (2016) propose a practical framework to support designers to gain a deeper understanding of the context of use when designing with resource poor communities.

Moving beyond technology and product design, the capabilities approach has inspired the design of public services for marginalized communities. Cottam leans on the capabilities approach in a general sense in her work on co-designing public services with underserved communities in the UK to describe how help can be offered in ways that are empowering, and that build on existing agency as opposed to reducing it (Cottam, 2019). In this way, Cottam reflects Nussbaum's argument that the capabilities approach can shift how we think about the provision of services in resource poor settings by developing a granular understanding of people's lives to see how internal and external structural issues might affect them, and therefore how to address the bigger challenges we face

(Cottam, 2019; Nussbaum, 2013a). Cottam's work provides examples of service design used both to understand people's situated experience and to facilitate the participation of citizens in the design and delivery of public services (Cottam, 2019).

2.3 Synergies and divergences between the theoretical concepts

I have used the theoretical concepts presented above variously to guide and concretize the empirical work conducted in the case studies and as analytical frames to support the interpretation of the findings. The interplay between the theoretical concepts reflects my intellectual journey during the PhD, beginning with a rather instrumental understanding of service design in paper 1, towards a progressively deeper understanding of the transformative potential of design for handling the multidimensionality of wellbeing in complex development contexts as explored in papers 4 and 5.

Service design has potential as a human-centred, integrative and co-creative approach to designing in complex contexts. Depending on how the approach is used, however, there is a risk that that the structural causes of societal problems might be glossed over, and emphasis placed instead on designing tangible "solutions". Without "reflexive self-vigilance" on the part of the designer (Tlostanova, 2017). There is also a risk that design becomes a monolithic and colonial approach that ignores the plurality of local experience (Duan et al., 2021). Transformation design as conceptualized by Burns et al (2006) Cottam (2019) and Sangiorgi (2011) draws attention to the interconnectedness of societal problems, as well as the need for bottom-up community action to be the driving force in the design of services for marginalized communities. Sangiorgi highlights the responsibility of the designer when facilitating design work, particularly when working with vulnerable communities, and suggests principles for facilitating transformative design work with communities (Sangiorgi, 2011). These principles provide a useful guide to facilitating design processes driven by bottom-up community action. However, given that I am interested in approaches and methods that can create space within existing development programme structures for actors at the micro level (individuals and households) to participate in designing development interventions, the ability to accommodate normative development goals within design practice becomes crucial.

The capabilities approach, which views poverty as multidimensional and situated, and has clear social justice objectives, is a normative framework that aligns well with the objectives of both service design and transformation design, and as such can provide a conceptual bridge between design and development. Key concepts such as *functionings*, *beings and doings* and *conversion factors* speak to the importance of understanding local level diversity as a first step in any process to improve people's well-being. They therefore seem to be fully aligned with the human centredness of service design, and with ideas on how value for individuals is understood and created. The concept of collective capabilities and frameworks such as the 3C model could provide a conceptual basis for design work that seeks to support grassroots agency.

3 Methodology

This chapter presents the methods used in the thesis and a description of the three studies. The following sections discuss the overall research approach, positionality and reflexivity. For each study, I provide some brief background and describe the methods of data collection and mode of analysis. The research evaluation criteria and ethical considerations are also discussed.

3.1 Research approach

The research presented in this thesis is exploratory, aiming to understand how service design can be used as an approach to designing development interventions in complex social-ecological contexts. The empirical basis for the thesis is three studies where a service design approach (papers 2 and 3) and participatory backcasting (paper 5) were applied in real world development-oriented projects. Elements of the conceptual framework developed in paper 1 were applied in two of the case studies (papers 2 and 3). See table three for an overview of the methods used in the three case studies,

Situated at the intersection of design and development studies, the work is interdisciplinary. Two of the studies used a mixed methods approach, where quantitative methods were applied alongside a design driven qualitative approach. The decision to combine qualitative and quantitative methods was in part determined by the policy objectives of each study – to provide practical insights about the design of services for development practitioners and other decision makers in each context. As clarified in the introduction, my intention is that this research should be of practical use for the design of development interventions. Although design methods are increasingly applied in development contexts, design and service design approaches on their own are often perceived by policymakers as insufficient for informing decision making, and the use of quantitative approaches can help to validate findings from the design phase of a project (Jagtap, 2019a; Tantia, 2017). There have also been calls from within the design for development research community for the use of quantitative approaches to help evaluate design insights (Jagtap, 2019a). In two of the studies presented here (papers 2 and 3), quantitative approaches were used variously to facilitate pattern recognition, highlight irregularities or peculiarities that might otherwise be missed (Sandelowski et al., 2009) and verify interpretations of the data generated from the service design phase.

I used an abductive approach to reasoning in each of the studies, moving between suggesting solutions as interpretations of alternatives to what currently exists (for example, through the use of prototyping) and gathering data about the context and the problems, all the time gradually refining the solution and the problem articulation as new data and insights were gathered (Alvesson & Kärreman, 2007).

3.2 Ontological perspective and positionality

The research approach taken reflects my worldview and research positionality. I take a social constructivist ontological perspective, which affirms the presence of various, diverse realities. According to this perspective, people negotiate historical, cultural norms and social norms to give situations meaning (Creswell & Poth, 2018). Thus, constructivist research is shaped from the bottom up, that is, from individual perspectives to wider patterns and finally to broad understandings (Cresswell & Plano Clark, 2011).

In social research, paradigms serve as heuristics; they are conceptual and practical "tools" used to solve specific research problems (Abbott, 2004). Following Maxwell, I take the view that philosophical ideas from different paradigms may be combined and used productively to understand the phenomena under study (Maxwell, 2011). My work straddles the interpretivist and pragmatist paradigms. In an interpretive study, the objective is to develop a comprehensive grasp of the subject area, going beyond a comprehension of its fundamental parts (Goldkuhl, 2012). This knowledge results from dialectical interactions between understanding the whole and understanding the parts (Goldkuhl, 2012). This comprehension of subjective meanings in the world can be used as the foundation for a theory (Goldkuhl, 2012). At the same time, my work adopts a strong pragmatist stance, which entails that the social world can only be understood by trying to change it, and that the process of inquiry is more important than finding an absolute answer (Bernstein, 1999). The essence of pragmatic ontology is action and change; humans act in a world that is in a state of ongoing transformation (Bernstein, 1999). As one of the great pragmatists, John Dewey, contends, knowledge and understanding rely on our interactions with the world and how we reflect on these interactions (Hickman, 1998). The goal of pragmatism is to generate actionable information that can be used to make meaningful changes in practice (Goldkuhl, 2012).

I have a background in political science and development studies, rather than design, but the work presented here is situated at the hinterland between design and development research. My vantage point as an interdisciplinary researcher has allowed me to observe the weaknesses and biases that come with each, and to spot synergies and complementarities between them. At the same time, I carry disciplinary baggage as a development researcher, working for many years within a field that is methodologically conservative and dominated by a logical positivist worldview that tends to favour a linear, deductive logic when answering research questions. I have brought elements of this logic to my PhD research, both consciously and subconsciously, and it has influenced the design and implementation of the studies. For example, as mentioned above, two of the studies employed qualitative and quantitative methods, initially as ways to triangulate the findings generated during the design phases. It should also be noted that much of my research to date has focused on energy access and household energy transitions, and significant prior research was conducted in East Africa, and Kenya in particular. This background becomes important later on when reflecting on how service design can be used in development contexts, and the other types of knowledge that are needed if service design is to add value.

I consider myself to be a pragmatic idealist. My career path for the past 18 years as a development practitioner and researcher reflects a deep conviction that I have a moral obligation to try to improve conditions for disadvantaged people and communities. This conviction implies that there is a role for development practice and research in improving conditions in society, a position that is contested by some who view development as an entirely neo-colonial project that has largely failed to deliver benefits to low income countries and should be dismantled (Escobar, 2017; Kothari, 2019). I fully acknowledge the colonial heritage of the current development architecture, and that many of these structures need to be radically changed, but I do not submit to the nihilistic view that efforts should not be made to improve the situation for disadvantaged and disempowered people. It is my view that a balanced approach is needed, between criticality and pragmatism, so that the design work conducted in the context of development is reflective but not paralysed by reflexivity.

Finally, I must acknowledge my position as a white, European academic, conducting research in marginalized communities in East Africa. This is a privileged position that has implications for power dynamics in the planning and

implementation of research in Kenya and Uganda. I attempted to address this through continuous personal reflection and soliciting feedback from trusted colleagues and collaborators, particularly collaborators who come from the areas where the research was conducted.

Table 3.1 Overview of methods and roles of collaborators in the empirical studies

Study (Paper)	**Methods**	**Role of collaborators**	**Duration and timing**
(2) Cookstoves	Baseline survey, semi-structured interviews with stove users, quantitative stove use monitoring, prototyping workshop with stove users (including 5 focus group discussions), user journey mapping, workshop with stakeholders from the Kenyan clean cookstove sector	Burn (Kenyan cookstove manufacturer): supported recruitment of interviewees; provided access to data on fuel sales	Nov. 2018 to May 2019
		Research assistants: translation, note taking, data interpretation and support with data analysis, facilitation of focus groups	
		Service designers: support with workshop design and methods; design of workshop materials; workshop facilitation; support with data analysis	
		SEI Africa: led the semi-structured interviews; led the set-up of workshops; support with data analysis	
		SEI US: designed and led the stove use monitoring	
(3) Insurance	Open-ended interviews with farmers, participant observation, in-field focus group discussions, validation workshop, discrete choice experiment	Lab-in-the-field Uganda: participant recruitment, research assistants provided support with translation and note taking; interpretation of meaning, in-field data analysis; workshop facilitation	Feb. to Aug. 2018
		University of East Anglia: led on the discrete choice experiment; support with preliminary data gathering	
(5) Backcasting	Participatory backcasting in a workshop setting; semi-structured interviews.	Community mobilizer: recruitment of participants; recruitment of research assistants; engagement with district chief; focus group facilitation, translation during follow-up interviews; support with in-field analysis	Feb. 2020 to Sep. 2020
		Research assistants: facilitation of breakout sessions; support with in-field analysis.	
		SEI Africa: input into research design; support with in-field analysis; paper writing	

3.3 Paper 2

A heavy reliance on fuelwood, charcoal, and kerosene causes 4 million premature deaths annually from exposure to smoke, particularly in low- and middle-income nations (Institute for Health Metrics and Evaluation (IHME), 2019). Over 90% of Kenyan households depend on solid fuels or kerosene for part or all of their cooking needs (EED & Stockholm Environment Institute, 2019). This heavy reliance on biomass for cooking takes a toll on Kenyans' health with approximately 5% of premature deaths in Kenya attributable to exposure to household air pollution (IHME, 2019). For decades, governments, development organisations, and social entrepreneurs have worked to transition communities to cleaner cooking fuels, such as Liquefied Petroleum Gas (LPG) and electricity, but progress has been slow (Lewis & Pattanayak, 2012; Quinn et al., 2018). Transitioning from traditional biomass for cooking to cleaner alternatives is highly complex. Households that adopt clean cookstoves typically continue to use their older stoves alongside the newly acquired technology, and frequently discontinue using the new technology over time (Ruiz-Mercado & Masera, 2015). Surprisingly little is known about how user and sociocultural influences on cooking practices affect the adoption and sustained use of clean cookstoves (Jagadish & Dwivedi, 2018; Jürisoo et al., 2018). There is also a gap in our understanding of how individuals value energy interventions such as clean cookstoves and how these values translate over time into technology adoption (Furszyfer Del Rio et al., 2020; Hirmer & Cruickshank, 2014).

Background to the case study

This case study involved a team of researchers from SEI (two from HQ, two from the Africa centre and two from the US centre); a team of two Swedish service designers from a small Stockholm-based design consultancy, Expedition Mondial; Burn, a Nairobi-based cookstove manufacturer that has been operating in Kenya for 10 years, which was seeking to introduce a clean-burning, forced-air pellet stove called the Mimi Moto (MM) to the Kenyan market; and four Kenyan student research assistants recruited from Jomo Kenyatta University, with whom we had collaborated on two previous SEI projects. The study was funded by the Climate and Clean Air Coalition (CCAC), part of the United Nations Environment Programme (UNEP) and co-financed by the Swedish International Development Agency (Sida). The “we” I refer to in the description of the study is the team involved in gathering and analysing the data, which

comprised SEI colleagues, the Swedish service designers and the student research assistants.

We conducted this study between November 2018 and May 2019. In addition to understanding the overall user experience of the MM stove and fuel, Burn, the cookstove manufacturer, was interested in testing an innovative pay-as-you-go model, whereby stove users acquire credits when they purchase the biomass pellet fuel which are used to activate the stove using a digital card reader attached to the stove. The purpose of the study was to understand the value users placed on the MM stove—attributes that may be used to improve the design of a clean cooking system from the users' perspective. Two research questions were posed: How do various stove users value the MM stove? How do their perceptions of the stove's value correlate with their use of the new cooking system over time?

Methods of data collection

The case study involved 30 households located in Gachie, a peri-urban neighbourhood on the outskirts of Nairobi. We combined quantitative data from Stove Use Monitors (SUMs) with qualitative data gathered using a service design approach. I led the qualitative research phase and a colleague at SEI US led the SUMS monitoring. SUMS are small digital thermal monitors that can be attached to the exterior of a cookstove to monitor stove use over time. SUMS were used to monitor the actual use of cookstoves in each of the 30 participating households. See Bailis et al. (2020) for a detailed description of the SUMs methodology and data collection. Comparing observations of the same phenomenon via two distinct methodological lenses allows for a more robust evaluation of stove use patterns and user experience (Lambe, Ran, Kwamboka, et al., 2020). The qualitative and quantitative components of the study were conducted in parallel between November 2018 and May 2019. The study began with a structured baseline survey of the 30 households to gather basic demographic data and establish baseline stove and fuel use patterns.

For the service design phase, we conducted 30 interviews with cookstove users, all women from low- and middle-income households in Gachie. In-home interviews were conducted in Kiswahili by SEI colleagues based at SEI in Nairobi and the student research assistants. Two research assistants used pen and paper

to record interview responses. At the time the interviews were conducted, households had been using MM stoves for approximately ten days. The purpose of the interviews was to map in detail the cookstove users' experience with the MM stove, from hearing about it for the first time to purchasing it and beginning to use it. The interview guide was structured to capture the experiences and perceptions of these stages of the user journey.

We conducted five focus group discussions (FGDs) during a prototyping workshop that was attended by 25 of the same users. During the initial FGD, we asked participants to reflect on their motivations for purchasing the MM and to rank the importance of each. During the second FGD, participants created individual user journey maps using pre-prepared templates and added sticky notes to describe their positive and negative experiences and thoughts throughout the journey. During the third FGD, participants merged their user journey maps in groups based on their primary motivation factor and discussed where their experiences converged and diverged (Figure 3.1). Again, sticky notes were collected using a pre-designed template printed on a poster. In the fourth FGD, working in the same groups, the participants discussed the key challenges they faced with the MM service system and began to prototype solutions to these, first discussing ideas in a group and then sketching solutions on to a pre-prepared template using pen and paper. Each group then presented its ideas to the other participants and received feedback. In the fifth and final FGD, participants refined their prototypes to include details about which key actors would be required to implement their proposed solutions. All the additional information was recorded on the prototyping templates. Our team of local research assistants facilitated all the FGDs in Kiswahili and took notes during each session. Feedback from the FGD sessions was later translated into English.



Figure 3.1 Participants merging their user journeys during the third FGD. Photo credit: Fiona Lambe

We then held a stakeholder workshop in February 2018 with 25 meso-level cookstove sector stakeholders, such as representatives of the private sector, non-governmental organizations, the research community and government agencies, to discuss the users' experiences and the prototypes they had developed. We presented the user journey maps for each archetype group, highlighting the areas where each archetype encountered the most difficulty interacting with the MM service system. We also mapped where each representative was situated within the cookstove sector in relationship to the user; that is, whether they were actively involved in promoting or implementing clean cookstove or fuel interventions and, if so, which part of the user journey they were most involved in. The sector was mapped using a pre-prepared poster, and participants used sticky notes to indicate their location. Using the sector map and the prototypes created by the cookstove users as a jumping off point, the workshop participants then discussed in groups their capacity to implement the proposed service system changes. The workshop was conducted in English and the research team took notes on the discussions.

Mode of analysis

The field notes on the interviews and focus groups were coded using open coding (Strauss, 1987). Open coding involves line-by-line analysis of data for "empirical indicators" such as any reported interviewee behaviours, experiences, opinions or other observations (Lambe, Ran, Kwamboka, et al., 2020). The research team manually performed the open coding during field analysis and immediately after the FGDs and the empirical indicators were structured using the interview and focus group guides (Lambe, Ran, Kwamboka, et al., 2020). The data were disassembled into subsets that were categorized in accordance with the research questions. Coding was iterative; for instance, when analyzing interview data, two lead researchers identified key factors that influenced the purchase of the MM stove. These were then reviewed and revised after consultation with a third researcher who had participated in data collection. During the subsequent workshop with users, one of the motivating factors was eliminated as irrelevant and another was revised based on participant feedback. The local research assistants played a key role in data analysis, helping the team to understand and interpret cultural nuances.

3.4 Paper 3

In Uganda, like many parts of East Africa, climate-induced drought and pests are key threats to farmers' livelihoods. Although insurance would seem to be an obvious measure for reducing risk, traditional insurance schemes are often too expensive for low-income rural farmers. Weather Index Insurance (WII) has emerged as a financial innovation that could protect rural livelihoods from the severe impacts of climate change. WII works by using localized weather data and measured variables, such as the lack of rainfall, to validate individual farmer claims, which, once verified, trigger a payment. This means that insurance companies have no need to send claims adjusters to assess damage, which reduces the cost of the insurance package (Platteau et al., 2017). Despite its promise as an intervention that could alleviate poverty and prevent food insecurity in low-income countries, WII schemes have not been widely taken up (Platteau et al., 2017). Studies of WII interventions show that design deficiencies often related to local, contextual factors contribute to limited uptake of these products (Dercon, 2007; Nshakira-Rukundo et al., 2021).

Background to the case study

In 2018, my colleague at SEI HQ and I were invited by a team of development economists from the University of East Anglia (UEA) to conduct participatory research in Bwikhonge, in the Bulambuli district of eastern Uganda to understand the viability of Weather Index Insurance as a means of reducing risks associated with climate shocks. This team had previously studied the risk taking and risk sharing behaviour of farmers in Bugisu, eastern Uganda and, in close collaboration with local partners including Ugandan insurance companies, had used their findings to propose measures for reducing farmers' risk, while boosting agricultural productivity (McSherry, 2017; Verschoor et al., 2016). Their primary recommendation was to bundle WII with high-quality agricultural inputs and credit, and to provide this package to local risk-sharing groups, especially village savings and loans associations (VSLAs) (Verschoor, D'Exelle, Balungira, et al., 2016). However, the team of economists was concerned that the recommendation might not properly represent the local context because it was derived from data collected using lab-in-the-field experiments and was therefore based on behavior observed in artificial settings where hypotheses are tested about particular factors in isolation while keeping all other relevant factors, which typically determine behavior in the real world, at bay (Dehmel et al., 2021). In complex social-ecological settings, these other factors interact with the factors being tested, and may strengthen or negate the effect of the factor being studied. The team from the UEA was interested in methods that could be used alongside their field experiments to capture this complexity and contribute a more nuanced understanding of the factors that determine risk for smallholder farmers in Bwikhonge – and how they would interplay with the provision of WII. It was agreed that the conceptual framework described in Paper 1would serve well as the foundation for the methodological approach taken in the study.

The collaborators in the project were: my colleague and I from SEI HQ, who both have a background in development studies and experience using service design approaches and tools in SEI projects; three colleagues at UEA, one of whom travelled to Bwikhonge two months before fieldwork commenced to work on site selection, participant recruitment and compilation of relevant data on the farmers in the area; and a local research organization, the Field Lab Uganda, a local non-profit company that specializes in lab-in-the field experiments.

The study combined a service design approach with a discrete choice experiment (DCE). Each method operated as an independent body of research but was conducted in a way that allowed the triangulation of findings. The service design work led by SEI was conducted during phase one in February 2018. The discrete choice experiment led by the UEA was conducted in phase two in July 2018. Local research assistants from the Field Lab Uganda were involved in data collection for both phases. We also worked with a team of local research assistants that regularly supports Field Lab Uganda research projects. All the team members live in the area and are well known in the communities. We provided a one-day introduction to service design for the research assistants and a UEA colleague prior to commencing the fieldwork. The fieldwork assistants played several important roles, such as acting as translators during interviews, correcting misinterpretations and misunderstandings during interviews and focus groups, and supporting the analysis and interpretation of findings, in particular as we moved from understanding the context to prototyping with farmers. The "we" referred to in the following description of methods comprised my colleague and I from SEI HQ, the UEA colleague and the team of research assistants from the Field Lab Uganda.

Methods of data collection

For the service design phase, we conducted 20 participant observations, 48 key-informant interviews, 14 field workshops and two larger workshops – a solutions prototyping and a testing and validation workshop, each with 40 local farmers and stakeholders. Research participants were randomly sampled household heads and spouses taken from population lists that had been generated by the UEA team during previous rounds of fieldwork. The order of fieldwork activities broadly followed the model outlined in paper 1: problem co-definition, actor-centred mapping, experience-based problem definition, rapid prototyping and design, and testing of prototypes (Lambe, Ran, Jürisoo, et al., 2020). However, there was a degree of iteration of methods as new insights were gathered (for more detail, see Dehmel et al., (2021).

Participant observation is a qualitative research method where the researcher observes the members of a group or community being researched while participating in their activities. Participant observation is often a starting point in a

design process and is useful for building a relationship with research participants, for affording access to the "backstage culture" of everyday life (De Munck & Sobo, 1998, p 43) and for facilitating the development of new research questions or hypotheses (DeWalt & DeWalt, 2011). It also provides "richly detailed descriptions of behaviours, intentions, situations and events as understood by one's informants" (De Munck & Sobo, 1998, p 43). In our case study, participant observation involved accompanying farmers for approximately 30–40 minutes as they went about their daily activities, such as checking their crops for pests or checking on livestock. Observations were captured as notes using pen and paper, and as photographs using digital cameras. The participant observations directly preceded the key informant interviews, where we had an opportunity to ask follow-up questions about our observations.

Semi-structured interviews were conducted in respondents' homes and/or farms or places of work by four research teams of one research lead and one assistant, working in parallel. All interviews were conducted in a mixture of the local language, Lugisu, and English. Where needed, interviews were translated on the spot into English by the research assistants. Interview responses were captured using pen and paper. Various activities were conducted as part of the interviews, such as detailed mapping of the annual farming, income and risk cycles, and of risk response journeys, including a mapping of key actors and financial flows. The focus shifted as the interviews progressed and new insights were gathered, and the interview guide was adjusted according to the new lines of inquiry.

The initial 12 interviews lasted 1–2 hours and focused on mapping the annual agricultural cycle. Instead of asking leading questions regarding farming practices, we allowed the respondent to guide us through their daily lives and context (Dehmel et al., 2021). We used trigger materials such as picture cards depicting everyday items and tasks for the interviewees to use to support the mapping of activities over a time cycle of a day, a month and a year (Figure 3.2). Eight further interviews each lasting 1–2 hours, were conducted to understand how farmers respond to risk. We mapped the individual households' risk response journeys by inquiring about the coping strategies they used the last time they were affected by a drought or pests. From there, we began to identify a general pattern in the risk response mechanisms of different households. To facilitate the development of our hypotheses, we shifted to conducting shorter, more targeted interviews (n = 23) in 30-minute sessions that, in addition to mapping risk

responses, also explored households' agricultural and economic diversity, formal institutional and informal social networks (Dehmel et al., 2021). To better understand the role of these various support mechanisms, we conducted five semi-structured interviews of 1–2 hours with key informants along market value chains who had been identified by the respondents as trusted sources of financial support and advice, such as cotton agents (intermediaries between the community and the cotton development organization) and store owners (Dehmel et al., 2021).

In-field focus group discussions were conducted with household heads and spouses, sometimes in groups divided by gender to encourage open participation, in various meeting spaces such as churches and barns. Each FGD included between four and six participants and lasted between one and two hours. Seven FGDs were used to map typical daily activities and annual cycles of agricultural production, weather patterns and risks, and market value chains for various commodities. Similar to the interviews, participants guided researchers through the mapping of daily, monthly, and annual cycles using trigger materials (see Figure 3.1) The FGDs were facilitated by lead researchers who posed questions along the main lines of inquiry. We also conducted additional FGDs (n = 4, of 4–6 participants in sessions of 1–2 hours) with members of village savings and loans associations (VSLA), burial groups (groups that form to save collectively for the cost of funerals), and merry-go-rounds (informal savings groups that make payouts to members on a rotating basis) and non-members to understand their reasons for not participating (Dehmel et al., 2021).



Figure 3.2 Mapping the annual farming cycle during in-field focus group discussion
Photo credit: Fiona Lambe

At the end of the qualitative fieldwork phase, we convened a validation workshop in Bwikhonge to which we invited 40 representatives from local community organizations, including VSLAs, farmer groups, representatives of various churches, mosques, and schools, and local government. The workshop began with the presentation and validation of research findings, followed by a discussion on the viability of the various prototypes. We ensured close to equal representation of men and women. The participants were divided into four working groups in which the results of stages two through four were presented and debated. Research assistants led the conversations in the local language and took notes on the discussions. The feedback from each group was reported in a plenary.

During the workshop, we used storytelling and sketching to describe and visualize the three farmer archetypes that emerged during the qualitative fieldwork. After presenting each archetype, we invited stakeholders to comment and make modifications to bring them closer to community reality. Once they were

satisfied with the archetype descriptions, participants then worked in groups to map out the risk response paths they imagined each archetype would take in the event of a sudden shock.

Early sketches of ideas for prototype insurance products had emerged during the initial 12 interviews with farmers. These prototypes were further refined through manual coding and analysis of the interviews, FGDs and participatory observations. In our study, prototypes took the form of hypothetical models of insurance products or features of an insurance model, for example pay out models. We identified four potential insurance designs to propose to farmers as prototypes for WII interventions. These included an expensive "full coverage" option that would cover all losses, a less expensive option that would cover the cost of inputs (seed and fertilizer), and an inexpensive "basic coverage" option that would serve as a safety net in the event of a natural disaster. Given that some farmers reported being unable to afford even inexpensive insurance, we also included a prototyped savings scheme as an alternative to insurance.

During the second half of the stakeholder workshop, the participants discussed the prototype models in relation to the three archetypes introduced during the first part of the workshop. Again, working in groups, the participants were asked to match the appropriate insurance package with each archetype, with the option of not selecting any if none were considered appropriate.

Discrete Choice Experiments (DCEs) are a quantitative technique used to systematically elicit individual preferences of a product, service, or program (Mangham et al., 2009). DCEs derive from the economic theory of demand, specifically the work of Lancaster, who argued that the demand for commodities is driven by the demand for specific combinations of their qualities (Lancaster, 1966). In the context of a face-to-face questionnaire, potential consumers are repeatedly presented with a set of hypothetical alternatives to a product, service, or program and asked to indicate which they would prefer to purchase or participate in (Greiner et al., 2014; Raes et al., 2017; Vorlaufer et al., 2017). On the basis of utility theory (Hauber et al., 2016; McFadden, 1974) users' preferences are assumed to be determined by the design attributes that define and distinguish interventions. DCEs can be used to rigorously 'test' user support for a proposed future policy or intervention in a reasonably inexpensive and quick manner prior to intervention (Osborne et al., 2022). The results can tell a policymaker whether

a particular program or service is valued and provide a precise measurement of how the policy's many components are valued in relation to one another (Hensher et al., 2005; Landmann et al., 2018).

For the insurance study, 196 farmers were randomly selected from 10 villages in the district of Bwikhonge. Face-to-face interviews lasting around 45 minutes were used to conduct discrete choice experiments with individual responders. Each respondent engaged in a 1.5-hour session prior to the interview to ensure that they had a complete and accurate understanding of the insurance product and its various features. For a detailed description of the experiment's design see Osborne et al. (2022) and Dehmel et al. (Dehmel et al., 2021).

Mode of analysis

We used thematic analysis (Clarke & Braun, 2017) to analyse the participant interview and in-field workshop data, initially clustering data according to the interview guide/focus group discussion guide structure and then iteratively refining the themes through several rounds of manual coding. Four researchers led the coding process and most of the analysis was conducted in Uganda, which allowed our local research assistants to participate actively in the analysis, and to correct misunderstandings over how the team interpreted the data. For a detailed description of the data collection and analysis, including DCE analysis see Dehmel et al. (2021).

3.5 Applying elements of the conceptual framework

As described above, we applied elements of the conceptual framework described in paper 1 in two of the case studies (the cookstove study and the insurance study). For the stages of the conceptual framework that were applied in each study and an overview of the specific methods used at each stage, see Table 3.2.

Table 3.2 Stages of the conceptual framework applied in each study and overview of methods used at each stage

Stages in the SD conceptual framework	**Methods applied**	
	Kenya	**Uganda**
Problem co-definition	Interviews	Household survey, Focus Group Discussions
Actor-centred mapping	User Journey Mapping (during interviews); System mapping (during stakeholder workshop)	Interviews (Farming cycle mapping, risk response mapping)
Experience-based problem definition	Archetype development, quantitative stove use monitoring (SUMS)	Archetype development
Rapid prototyping	Prototyping workshop with users; refinement of prototypes in stakeholder workshop	Prototype development (in interviews); prototype refinement in workshop with farmers
Testing of prototypes		Discrete choice experiment

3.6 Paper 4

The purpose of the study presented in paper 4 is to answer two research questions regarding how design tools can be used to create transformative development interventions: How can design facilitate meaningful participation of the intended beneficiaries of development interventions in their design? And how can design devices help align development interventions with local capabilities? Paper 4 is a conceptual paper and did not involve the gathering of empirical data. Instead, the findings presented in paper 2 and paper 3 were reanalysed using a capabilities approach lens. Reanalysis in qualitative research is often used to reveal fundamental patterns and general themes in the data and for advancing theory (Wästerfors et al., 2014). For details on the data collection methods used to gather the empirical data, see sections 3.3.2 and 3.4.2.

Mode of analysis

Two steps of deductive thematic coding were used to analyse the findings of each study. In the first step, I reviewed the qualitative findings from each study

and coded them using functionings and conversion factors for the various archetypes identified in the original analysis. In the second step, I used Nussbaum's Capabilities Lists (Nussbaum, 2013a) to identify sets of capabilities relevant to the archetypes in each study. At each step of the analysis, I checked the emerging themes with two of the co-authors, one of whom was involved in gathering the original data.

3.7 Paper 5

Recent research indicates that in both urban and rural parts of sub-Saharan Africa, cooking with electricity, which was previously thought to be too expensive for rural African homes, is swiftly becoming a cost-effective and viable alternative (Couture et al., 2016; Loughborough University, 2019). Although the economic and technological case for a shift to cooking with electricity is growing, little research has been conducted to explore local imaginaries of large-scale transitions such as rural electrification schemes, particularly the imaginaries of energy consumers themselves (Simmet, 2018).

Background to the case study

The case study presented in paper 5 was commissioned in 2020 by a Dutch NGO interested in researching local perspectives on a long-term energy transition in rural Kenya from biomass to electricity-based cooking. The primary objective of this study was to investigate the sociocultural implications of a large-scale energy transition at the micro level, focusing on how households and communities envisage such a transition and how they conceptualize their role in the system transformation. Unlike the case studies on cookstoves and insurance, this study provided an opportunity to engage with a community early on in a transformation process, prior to the implementation of a specific energy access programme or technology. The case study addressed two research questions: How might a participatory design approach serve to foster dialogue with a community over a future energy transition?; and What parts of a transformational change might such an approach support?

Machakos County was chosen as the study location due to its proximity to Nairobi, where the majority of the research team is located, and because cooking methods there are comparable to those in other regions of the country, which

would allow the findings to be generalized. Only households lacking an electrical connection were chosen. Despite being only 10 kilometres from an urban centre, the villages have a rural character due to their level of infrastructure and access to services, and primary means of subsistence. Each village is connected to the main route into Machakos by unpaved roads, there are few healthcare facilities in the area and the majority of households rely on subsistence agriculture.

My collaborators on the study were two colleagues from SEI: one from SEI Africa, who is Kenyan, is based in Nairobi and has a background in energy access; and one from SEI US, with a background in biomass energy and energy transitions in developing countries. I led the research design with input from my SEI colleagues and was supported in the fieldwork by my colleagues from SEI Africa. For the data collection, we worked with a community mobilizer who comes from the villages where the research was conducted and is known to the communities, and a team of student research assistants from Machakos University, several of whom live in the communities where the research took place. As with the other case studies, the research assistants played a pivotal role during fieldwork. They facilitated the breakout discussions throughout, translated feedback from participants from Kamba to English, and supported the in-field analysis of the workshop findings, correcting misinterpretations of the data and suggesting tweaks to the backcasting methodology based on their understanding of the local context. Crucially, they led on developing the narrative that was used at the start of the workshop to trigger the participants to imagine life in the community in the year 2030. My colleague from SEI Africa and I provided a one-day training session on backcasting and the workshop methodology for the research assistants and the community mobilizer prior to running the workshops, as well as explicit written instructions for use as needed during the workshops. The "we" referred to in the description of methods below includes me, my colleague from SEI Africa, the community mobilizer, and the four research assistants.

Methods of data collection

Two phases of empirical work contributed to the case study: a workshop and an interview study. Participants in the workshop were recruited with the aid of the Kitulu area chief and the community mobilizer. The group comprised 21 women

and 9 men. More women than men were included since women are more often responsible for cooking and acquiring fuel. Participants ranged in age from 18 to 78 years old. The workshop was held in the church in Kitulu village, which is a common location for discussing community issues. Backcasting was selected as the workshop method because it has proved to be an effective technique for stimulating creative thought about ideal futures and has been used to study a wide range of sustainability issues (Dreborg, 1996). Backcasting is a type of future planning in which participants discuss a desired future and then work backwards to discover the steps or actions required to achieve the desired result (Dreborg, 1996; McPhearson et al., 2016; Wiek & Iwaniec, 2014). The strategy incorporates visioning, which involves the construction of a future-oriented image (McPhearson et al., 2016).

We used participatory goal-oriented backcasting, where the emphasis is placed on identifying future perceptions and ideas as goal-achieving (Wangel, 2011). Our goal was to replace all forms of cooking with cooking by electricity by 2030.The workshop took place over the course of two days. The participants worked in groups of six and a research assistant was appointed to each group to guide it through the sessions and take detailed notes on the conversations.
The first part of the workshop focused on defining the future objective of 100% electric cooking. A large, printed timeline depicting three time intervals between 2020 and 2030 was hung on the wall.

During the introduction to the workshop, we drew the participants' attention to the year 2030 on the timeline. A brief pre-written narrative was delivered aloud to establish the concept of 100% electric cooking by 2030. The participants were then asked to consider in their groups what 2030 would look like. Participants were provided with materials such as coloured pens and paper to illustrate their anticipated futures. Facilitators used guiding questions to assist their groups in describing potential futures in detail and consider their implications, such as the impact on family interactions, technical elements and the type of food that would be cooked. The second part of the workshop was dedicated to backcasting. In the same groups, the participants began to consider what would be needed to realize their 2030 vision.

Open-ended, semi-structured interviews were conducted with 10 workshop participants six months after the workshop. The community mobilizer facilitated

the follow-up interviews in person and I participated via mobile phone, since travel to Machakos was not possible due to Covid-19 travel restrictions. Interviews were conducted in Kamba and translated into English by the community mobilizer. All the interviews were recorded and transcribed in English.

Mode of analysis

The data, which consisted of the field notes taken by each facilitator, the notes taken by the research team documenting our observations and transcriptions of both rounds of interviews, were manually coded using open structured coding (Strauss, 1987). For the workshop data, coding was conducted manually by the research team (three individuals) in two rounds: one directly following the workshop and the other a week later. Interview data were analysed by the lead author. The interview and workshop guides were used to structure the empirical indicators.

3.8 Research evaluation

Numerous principles and strategies have been proposed for ensuring quality and rigour in qualitative studies. The two central criteria used to assess research conducted within the interpretive paradigm are validity and transferability. Within the pragmatist paradigm, quality of research is typically assessed based on its usefulness for addressing a specific problem in a given context. Below I detail the strategies used to strengthen the validity, transferability and usefulness of the research conducted.

3.8.1 Research Validity

A study's validity refers to its credibility or reliability and to a deliberate effort to establish trust in an accurate interpretation of the data's meaning (Carboni, 1995). A key aspect of establishing credibility is the extent to which the research accurately reflects the experience of the participants or the context (Lincoln & Guba, 1986). Importantly, the reader must be reassured that the interpretations are reliable and reveal truths external to the investigators' experiences (Thorne et al., 1997). The strategies I used to ensure validity in my studies were method and data triangulation, member checking and critical reflection.

Triangulation emphasizes the use of multiple methods and theoretical constructs (Guba, 1990, p 23), thereby enhancing the study's rigour, breadth, and depth (Denzin & Lincoln, 2008, p. 4). In all three studies we used multiple methods and sources of data to build our understanding of the phenomena under study. In the Kenya cookstove study we used cookstove monitors to quantitatively measure cookstove usage in households, alongside qualitative methods such as semi-structured interviews and FGDs. This combination of methods allowed us to triangulate the experiences and behaviours described in the interviews, and to establish patterns and similarities emerging from the data. Similarly, in Uganda, we applied qualitative methods such as semi-structured interviews, participant observation and in-field workshops alongside discrete choice experiments to develop a comprehensive understanding of the key design parameters of an insurance product for small-scale farmers. For the backcasting study we used FGDs, observations and follow-up interviews with a sample of workshop participants six months after the workshop in order to triangulate the data.

We also used investigator triangulation in each of the three studies, which refers to the involvement of multiple researchers in the collection and analysis of data. In the Kenya cookstove and Uganda insurance studies, the interview data were gathered by at least two teams working in parallel. Each team included local research assistants as well as researchers from Sweden. The teams would meet periodically to present a summary of preliminary findings and to agree on whether the interview guides should be revised to probe the themes that were emerging. This in-field analysis was a critically important mechanism for ensuring the validity of the insights emerging, and to avoid meanings being misinterpreted due to some team members being unfamiliar with the local context, although I acknowledge that misinterpretation can still happen. In the Kenya backcasting study, in-field analysis was conducted together with the local team to share and consolidate insights from the first day of the workshop and to adjust the guiding questions that would be used during the second day.

Member checking, also known as informant feedback or respondent validation, is a method of determining the reliability of qualitative research findings (Lincoln & Guba, 1985). Research participants are provided with data or draft reports to verify the accuracy in the light of their experiences. Member checking was used in different ways in all three studies. In the Uganda study on insurance and the Kenya study on cookstoves, member checking was conducted in a

workshop setting midway through the data collection period to share the preliminary findings, including the experience mapping of each case. In the backcasting study in Kenya, a draft report of the findings was shared with a selection of the research participants approximately six weeks after the workshop, and a local facilitator convened a meeting to discuss the report's contents with these community members and to gather verbal feedback on the findings. Community members also had the opportunity to provide written comments to the research team, which they did by sending their comments to the community mobilizer.
Continuous reflection on the research questions, methods and interpretation of data is essential when aiming for objectivity in research findings. I worked with interdisciplinary research teams in each of the three studies. This required us to clearly articulate our positions and assumptions throughout the research process. For example, in the insurance study, although it seemed as though the development economists from UEA understood what we were trying to achieve by using service design, I discovered during the process of writing paper 3 that some colleagues at UEA had not fully understood the epistemology of service design, and why it is distinct as an approach from other qualitative approaches.

As described above, in-field analysis was a feature of all three studies, which involved code checking by other researchers and by local research assistants. In addition, at least two researchers were involved in coding the data for the final analysis in each of the studies.

I also sought feedback on research design, methodological decisions and data analysis from my supervisors and engaged in debriefing sessions with them following each round of fieldwork. This provided an opportunity to gather advice from peers not directly involved in the research and to establish an audit trail documenting why certain methodological and analytical decisions were made (Lincoln & Guba, 1985). Finally, all of the appended papers have undergone double blind peer review with external reviewers.

3.8.2 Research Transferability

Generalizability is a key criterion for assessing the quality of quantitative research, and it is demonstrated by showing that data have been collected from a sample that is representative of the population for which generalization is sought (Guba & Lincoln, 1982). However, the epistemological nature of qualitative research, in particular the subjectivity of the researcher, means that generalization

of this kind is not possible (Guba & Lincoln, 1982) Instead, qualitative researchers aim for transferability of the findings; that is, the extent to which findings generated in one context might be transferrable to a second, similar context (Guba & Lincoln, 1982, 1994).

Strategies for enhancing transferability included the use of purposive sampling and thick descriptions. Purposive sampling is widely used in qualitative research for the identification and selection of information-rich cases to maximize the use of limited resources (Patton, 2002). This entails identifying and selecting individuals or groups with specialized knowledge or experience regarding a phenomenon of interest (Creswell & Plano Clark, 2011). In the cookstoves case, we selected a sample of households that had opted to purchase a clean burning pellet stove and we sought to interview women as primary stove users. In the insurance study we selected small-scale farmers who had recently experienced the negative effects of climate variability, and in the backcasting study we sought rural participants who were not connected to the electricity grid. For each study, I provide thick descriptions of study background, local context, methods, data collection and analysis procedures to give the reader enough information to assess whether the findings can be transferred to other similar contexts (Creswell & Poth, 2018; Lincoln & Guba, 1985).

3.8.3 The generation of actionable knowledge

Research located within the pragmatist paradigm is concerned with generating knowledge that is practically relevant for research participants in a given context (Cresswell & Plano Clark, 2011; Dewey, 1938). Understanding "what works" for addressing a specific challenge is therefore a central criterion for evaluating pragmatist inquiry (Kelly & Cordeiro, 2020, P. 4). Each of the three studies involved engaging the users of various services and service systems in a process of improving these interventions. Through the use of interviews, observations and rapid prototyping with users, we gathered insights about the "problem" in focus, as well as practical actions for improving the situation that are anchored in the users' experience. By engaging with other key stakeholders beyond the users, we aimed to further develop the prototyped measures for addressing problems to ensure that they are practically feasible to implement. As researchers we do not have control over how the knowledge generated is applied once the research project has been finalized, or the means for conducting long-term evaluations of the work conducted. However, we have observed early signs that the

knowledge generated during some of the studies has been applied by development practitioners and individual research respondents (for a discussion on implications for policy and practice and impacts of the research see chapter 5).

3.9 Ethical considerations

We did not gather any sensitive data in the course of the three studies. Aside from the DCE in the insurance study, no personal data was gathered. The SEI ethical practice guidelines were adhered to in all three studies, requiring research ethics protocols to be developed in English and the local language. Protocols contained the following main elements:

- *A project information form* providing background to the research, with information about SEI and Linköping University and an overview of all the planned research activities. The forms also explained how data would be managed and anonymized. Information was provided on expected project outcomes, compensation to participants such as per diem and reimbursed travel costs for attending workshops, and the possible negative aspects for participants, such as the time requirement for participation in interviews and workshops. It was made clear that participants were free to withdraw from the study at any time, and to request that their data be deleted.
- *A description of measures taken to ensure pseudonymity*. In all three studies, pseudonymity was ensured by removing identifying elements, such as name and address, from the data set, assigning codes to each participant and ensuring that the key for the coding was not shared but kept by the project lead on a secure server. Participants' identities were not revealed in research publications.
- *Informed consent*, both written and verbal, was obtained from the research participants in all three studies. The consent forms were designed to allow participants to opt in or out of having their image recorded.
- *A data management plan outlined protocols for handling data*: This set out who would have access to data, how and for how long data would be stored, as well as stipulations that data would not be used for research beyond these studies.
- *A dissemination plan for the findings*: For each study, a plan was developed for how to share the results with research participants. For the insurance and cookstove studies, workshops were held to present and

discuss the findings with stakeholders including the research participants. For the backcasting study, the draft report was shared with the workshop participants during a series of face-to-face meetings. Feedback on the report was gathered, shared with the research team and incorporated into the final report.

For the Uganda study, which entailed a field experiment that required the gathering and processing of personal data, local ethical approval was obtained from the Uganda National Council for Science and Technology (research registration number SS448ES).

4 Findings

This chapter presents the main findings of each of the five contributing papers.

4.1 Paper 1

Paper 1 presents a conceptual framework to support the design of development focused interventions that combines service design as an approach with a theory of behaviour change and social ecological systems thinking. The framework was developed using three case studies that were conducted prior to starting my PhD studies. The case studies focused on understanding and situating individual behaviours in relation to a wider system and proposed intervention. Two of the case studies looked at interventions to enhance energy access: a clean cookstove project in Kenya and a mini-grid electricity programme in Zambia. The third study focused on technologies to reduce post-harvest losses among small scale mango farmers in Kenya. The conceptual framework proposes a six-stage intervention process that is iterative and adaptable to a given context. The six stages as presented in Lambe et al., (2020) are:

Problem co-definition: A formal and thorough effort to gather evidence to support the initial underlying assumptions behind an intervention. This is done in close collaboration with key actors and stakeholders, such as the intervention's funder(s), target beneficiaries and sector experts.
E*xperience-based problem diagnosis*: Verify the problem identified in stage 1 from the perspective of the beneficiary (for example, individual, household, farmer) to identify the underlying causes. User journey mapping is conducted and key behavioural drivers at the individual level are identified in different phases of the journey. This stage also merges iterative intervention design with the *actual* journey of the target actors through an intervention, which allows pivotal 'behavioural moments' to be identified, highlighting, for example, where an intervention could be derailed due to a disconnect with users' needs or motivations, or the emergence of previously undetected opportunities to support the intervention.
System mapping: Create a detailed map of the entire system, from the socio-economic to the ecological, structural and institutional dimensions of the setting, using the target beneficiary's experience or 'journey' as the starting point. The

system map resulting from stage 2 is 'verified' with stakeholders in a workshop setting to ensure accuracy.

*Rapid prototypin*g: Use the insights gathered in the previous three stages to develop rapid prototypes of an intervention that can then be piloted in a subset of the target population to verify the insights gathered so far. Rapid prototyping can be used in parallel with *experience-based problem diagnosis* to generate rough ideas about the design of the intervention, which can be quickly tested during interviews with users.

Design and testing: Test a fully designed intervention in a subset of the population and make changes based on the results.

Upscaling: Scale-up the intervention beyond the initial population, possibly in a new location. This stage entails returning to stage 1 and following the cycle again in order to verify assumptions about a new location and new actors.

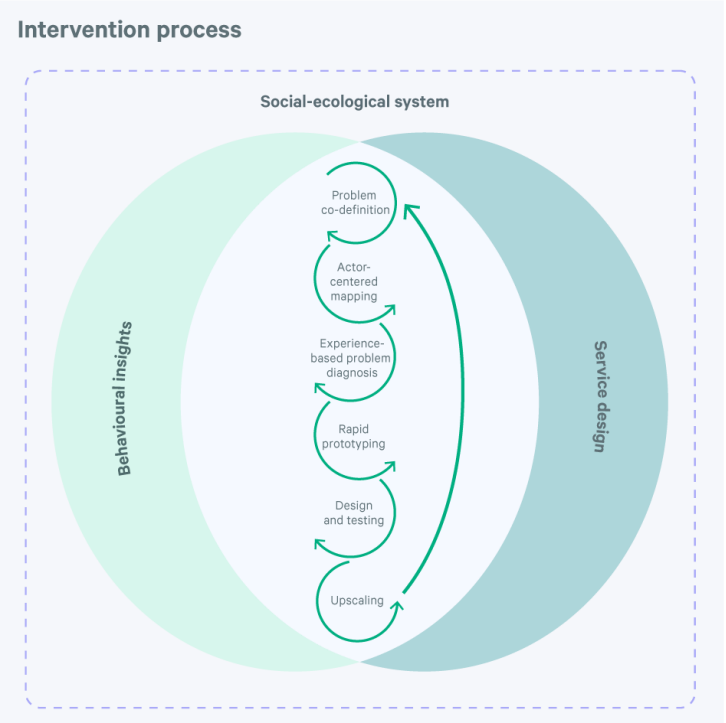


Figure 4.1 Conceptual framework for behaviour-based intervention design
Source: Lambe, Ran, Jüriscoo et al. (2020), reproduced with permission.

The paper presents a summary of the key insights gathered in the process of developing and testing the framework in the three case studies. The paper finds that in the context of development-focused interventions, behaviour change occurs and is reinforced in separate but highly interconnected phases of the actor journey (Lambe, Ran, Jürisoo et al., 2020). In interventions where the aim is to introduce a new technology, a significant shift in behaviour is typically required on the part of the user for the intervention to be successful, and this change in behaviour may need to stay in place for some time. In addition, different behaviours are often required at different phases of the user journey (Lambe, Ran, Jürisoo et al., 2020). Service design can be used to handle this complexity by, for example, mapping user journeys to visualize the multiple phases in focus and to show where high effort behaviour is required, and thus where users might need additional support as they transition to a new behaviour or routine. The paper also finds that the conceptual framework can help to identify user archetypes, which are a useful for categorizing users based on their experiences with an intervention over time, and to identify the type of support each archetype will need at different points in the user journey.

Finally, the paper demonstrates the importance of zooming out and situating an intervention or change process within the broader societal context or social system, including the social ecological context (Lambe, Ran, Jürisoo et al., 2020). For example, in the mini-grid case study, the business model studied had failed to recognize that the community where the mini-grid was being introduced is highly dependent on seasonal incomes, and the need to diversify incomes to be able to afford to pay for electricity from the mini-grid year-round (Muhoza & Johnson, 2018). The systems mapping component of the conceptual framework situates the lived experiences of users within complex social-ecological systems and emphasizes the connections between users and other important actors and processes at various societal levels. (Lambe, Ran, Jürisoo et al., 2020).

4.2 Paper 2

This paper analyses how users experience and value an advanced biomass cookstove and fuel and, based on this understanding, whether distinct types of cookstove users can be identified. Service design methods such as interviews with users, user journey mapping, rapid prototyping with users and system

mapping were combined with quantitative stove use monitoring in a case study that involved 30 users of the MimiMoto (MM) clean burning biomass pellet cookstove in peri-urban Nairobi, Kenya. The findings were discussed in relation to Boztepe's value in experience framework (Boztepe, 2007), focusing on how value is assigned by different types of users at various stages of the user journey.

Identification of experience-based archetypes

Based on the interview data, we identified several factors that influenced the purchase of the MM: cost savings, convenience, health, aesthetic appeal and income generation potential. The majority of respondents indicated that a combination of these factors influenced their decision. These factors were discussed during a workshop with the same users to give them the chance to modify or completely remove motivating factors that did not make sense. The participants then ranked the importance of the motivating factors. The most frequently cited primary motivation was financial savings (n = 15), followed by convenience (n = 6) and health and safety (n = 4). None of the respondents cited the MM's aesthetic appeal or potential to generate income as the primary reason for their purchase.

Working in their archetype groups, the participants then generated user journeys based on their primary motivating factor (for an example of a user journey, see figure 4.2). We compiled descriptions for each archetype – saving money, convenience, and health and safety – based on the user journey maps and group discussions. The descriptions were dynamic, capturing fluctuations in user satisfaction with the new stove over a four-month time span as well as the various actions users took to mitigate issues throughout the user journey. During the awareness phase, members of the money saving group reported feeling very satisfied with the stove, expressing appreciation of the information received on the stove's benefits and being drawn to the model of payment in instalments. This group encountered problems more frequently during the use phase, related to inconsistent fuel availability and a lack of technical support. The data from the Stove Use Monitors (SUMs) confirmed the difficulties reported during the use phase. In weeks 8–10, approximately 2 months into the six-month study period, when fuel availability was the most unstable, there was a significant decline in stove use.

This group remained motivated to save money throughout the user journey, and respondents described going to great lengths to achieve fuel savings by experimenting with the stove, such as testing different power settings for different dishes and adjusting fuel consumption to optimize fuel consumption. These reports are supported by the SUMS data, which show that this group's increased use of the MM stove was accompanied by a decrease in LPG consumption over the course of the study.

Those who cited convenience as their primary motivation reported being attracted to the MM stove from the moment they heard the sales pitch as they perceived it to be a relatively low-risk investment that would "make life easier". All respondents in this group emphasized the importance of hearing about the stove from a reliable source, such as a friend or neighbour. They noted that the stove cooks quickly and is portable, allowing them to cook in multiple rooms of the house and bring it when visiting relatives "upcountry." Several respondents were also pleased that the stove was clean and did not produce soot, allowing them to spend less time cleaning pots and kitchen walls. One respondent summarized the multiple convenient aspects:

> The MM is portable so you can use it to cook inside the house, which was not the case with the charcoal jiko which produces a lot of smoke. Also, the MM produces comfortable heat that warms the house when it is cold.

The convenience archetype group expressed widespread dissatisfaction with fuel availability during the use phase: their preferred fuel was often unavailable, it was not always possible to buy smaller packages, and overall fuel availability was unpredictable, or the vendor frequently absent. The frustration this caused is expressed by one respondent:

> I felt discouraged because I had gotten used to using the MM stove. Also, I live far from where the MM fuel vendors are found so I feel like my time and energy has been wasted.

As in the case of the saving money group, the SUMs data demonstrates a sharp decline in MM stove use as fuel supply decreased. However, in contrast to the saving money group, the convenience group’s MM usage did not reach zero during this period.

Members of the health and safety archetype group reported having relatives with chronic health issues that are exacerbated by exposure to charcoal smoke. This group was unhappy that kerosene is required to ignite the stove because it produces smoke. In addition, each respondent in this group reported initial difficulties with activating the stove, lighting it, and determining how much fuel to use for different dishes. The disappointment experienced by these users is linked to their expectations of a clean stove should function:

> I expected that the stove would work similarly to the LPG cooker where after finishing to cook you just switch it off, but when you turn off the fan when you have finished cooking while there is still some fuel left, it produces smoke.

The users in this group did not find the manual helpful for resolving these issues, preferring to contact the MM vendor for personal assistance.

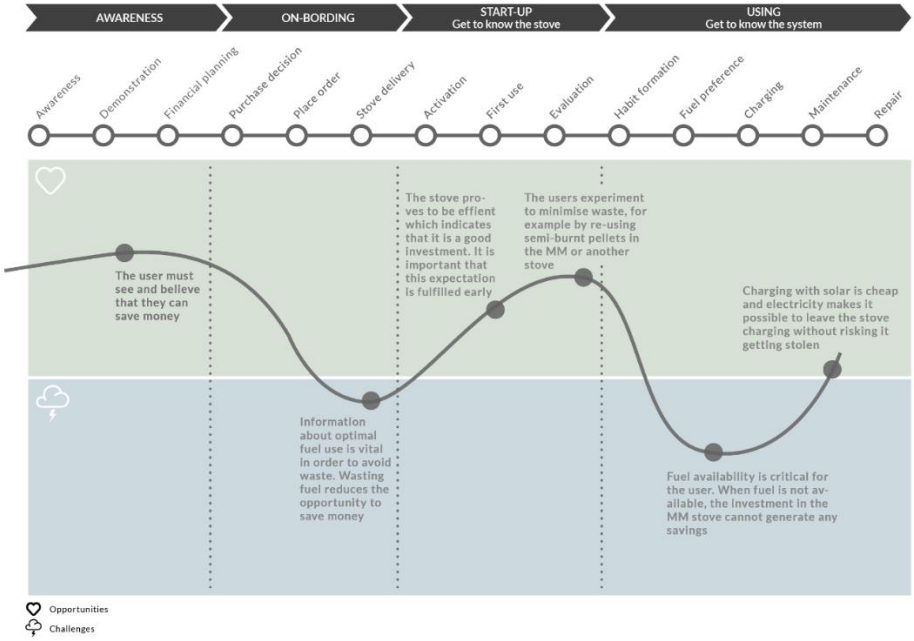


Figure 4.2 User journey for the Mimi Moto stove for the saving money archetype group. *Source*. Lambe, Ran Kwamboka et al. (2020), reproduced with permission.

Outcomes from solution prototyping for each archetype group

During the workshop, each archetype group devised solutions for what they deemed to be the most urgent concerns with the MM cookstove and fuel system. To overcome the fuel supply issues, the saving money group proposed having additional fuels sales vendors in the neighbourhood, and several respondents suggested they could be involved as vendors as it would provide a business opportunity. Similarly, this group suggested that technical support and repairs services could be provided by stove users themselves who could receive training from Burn. Technical support and advice could be provided from a central location or through multiple service points in the neighbourhood. The convenience group proposed a similar solution for overcoming fuel supply issues, where stove users would be employed as fuel vendors, and play a key role gathering real-time feedback from stove users on the functioning of the fuel supply system. The health and safety group developed several ideas for reducing the smoke produced when lighting the stove, including coating the processed fuel with firelighter and including firelighters with each bag of fuel sold.

The user journeys and ideas generated by the three archetype groups were presented during a workshop with representatives from the Kenyan clean cookstove sector and formed the basis for discussion on how to address key challenges faced by the three archetype groups[2]. During this stakeholder workshop, we mapped key stakeholders, including private-sector stove and fuel sellers, user and community representatives, academia, finance providers, government and development partners in Kenya in terms of their proximity to the user along the user journey map. This mapping exercise showed that most stakeholders focus their efforts on the early stages of the user journey – specifically, *awareness* and *on-boarding*. Only two workshop attendees placed themselves near users during the *start-up* and *using* phases, which is when most users reported experiencing problems with the MM stove and fuels (see figure 4.3).

[2] Due to limited space, the stakeholder workshop is not reported on in paper 2. Instead these findings are reported in paper 4.

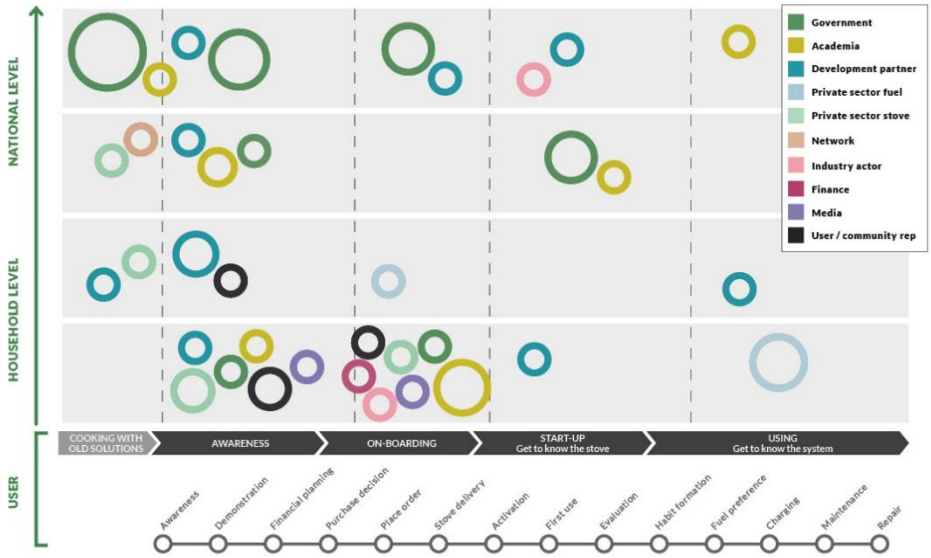


Figure 4.3 Map of Kenyan clean cookstove stakeholders in relation to the MM user journey. The size of the circles represents the influence of the actor type of the actor.

The stakeholders also identified several opportunities for improving the MM stove and pellet fuel system, based on the prototypes generated by the user archetype groups. Due to time constraints, the workshop focused on opportunities pertaining to the two largest archetype groups, *saving money* and *convenience*. Building on the prototypes for improving the fuel supply infrastructure and for technical support and repairs services, Participants at the workshop suggested that the government should mandate the Clean Cooking Association of Kenya, an industry network supporting the cookstove industry, to establish "support hubs" for advanced biomass cookstoves. These hubs could provide stove users with training for providing peer-to-peer support services.

Summary of findings

Three important insights emerge from the empirical findings presented in the second paper. First, mapping the user journeys of various archetype groups allowed us to identify the value in experience of a seemingly heterogeneous group

of stove users, while also shedding light on the type of support required by specific archetypes at different stages of the user journey. The user journey maps provide rich detail on what the different user archetypes expect, feel, experience and need at different points along the journey, resulting in a more detailed and comprehensive understanding of what works for cookstove users and why than is typically available from conventional evaluation techniques. The user journey maps also provided visually engaging material that could be used as a shared canvas between the researchers and the research participants to validate insights and correct flawed assumptions about the users' experiences, as well as a way to communicate complex information about the users' experience to a policy audience. Mapping the stakeholders' roles in the system in relation to the users' experience was a powerful way to highlight the lack of coordination amongst stakeholders working within the Kenyan clean cookstove sector.

Second, there appears to be a symbiotic relationship between user journey mapping and archetype construction. The user journey maps informed by the interviews were the starting point for identifying user archetypes; and the archetypes allowed for more meaningful and actionable user journey maps.

Third, the concept of value-in-use led to new insights about how users value improved cookstoves. We found that for the cookstove users in Gachie, value is assigned throughout the user's journey. Based on the various expectations held by each archetype group, which were sometimes influenced by the type of information they had received, our findings show that users had already assigned value to the stove before they began using it. In addition, the results suggest that value derived from multiple sources can facilitate the adoption of cookstoves. The convenience archetype had the highest correlation between qualitative and quantitative usage patterns. The SUMS data for this group corresponded to the user journey generated during the service design phase, where high levels of MM usage were reported throughout the entire user journey. It is notable that this was the only group that continued to use the MM stove during the fuel shortage. In general, the convenience group was optimistic about the stove's multiple benefits, and the fact that its members valued a wide variety of MM features in addition to fuel efficiency may explain why they were less disappointed when things went wrong than the other archetype groups.

Finally, the case study demonstrated the importance of combining qualitative and quantitative methods for understanding the complex interaction between the user, technology, and service provider in the clean cooking system. Often, quantitative measurement of the "fuel stack" is more objective than self-reported cookstove usage, which can be susceptible to confirmation bias. (Jewitt et al., 2020). However, quantitative stove use data alone cannot explain the observed stove usage patterns; the service design methods provided an in-depth understanding of the underlying reasons for the observed patterns. In a workshop we convened to share the study findings with Kenyan policy makers, a representative of the energy ministry expressed strong appreciation for the mixed methods approach, saying that despite the richness of qualitative insights, quantitative data is often a requirement for developing policy mechanisms.

4.3 Paper 3

The objective of paper 3 was to explore how service design could be used alongside discrete choice analysis experiments to inform the design of a Weather Index Insurance (WII) product for smallholder farmers in Uganda. As part of the service design phase, we used archetypes to capture and articulate the farmers' multiple vulnerabilities and risk-coping strategies. In addition, we used rapid prototyping to identify key design features of the insurance product. Discrete choice experiments (DCE) were used to evaluate the promising Weather Index Insurance design features that emerged from the service design phase in a manner that complemented and triangulated with the qualitative phase.

Findings from the service design phase

The service design phase generated important insights regarding the farming cycle in Bwikhonge and the diverse risk-related coping mechanisms used by various types of farmers in the area. Our mapping of the risk management strategies used throughout the annual farming cycle found substantial differences in how farmers cope with risk, even among households with a similar socio-economic profile. We grouped 11 types of farmer risk-response behaviour into four major categories: social networks, social institutions, agricultural diversity and economic diversity. These categories became the building blocks for farmer archetypes, which we used as models to represent observed patterns (Vaillancourt et

al., 2014) and as a means of representing and communicating the multiple complex characteristics that determine how farmers cope with risk.

The three archetypes that emerged from the interviews were named Andrew, Betty and Charles. These archetypes were fictional characters created to represent the key characteristics of the various farmer types in the region based on their risk response capacity and behaviours. Andrew represented a high capacity for responding to shock, Betty represented a low capacity for shock tolerance and Charles represented a moderate capacity for shock tolerance. Together with local colleagues, we created narrative life stories for the three characters, including their histories in the community and how their risk-coping behaviours align with the four categories discussed above. The three archetypes were used during the stakeholder validation workshop to communicate and validate our insights with community stakeholders, and to further develop and refine ideas for insurance products by having the workshop participants map out the likely risk response journey for each archetype (see Table 4.1). We sketched each archetype on a large poster, eliciting feedback from the workshop participants about their key characteristics (see figure 4.4)

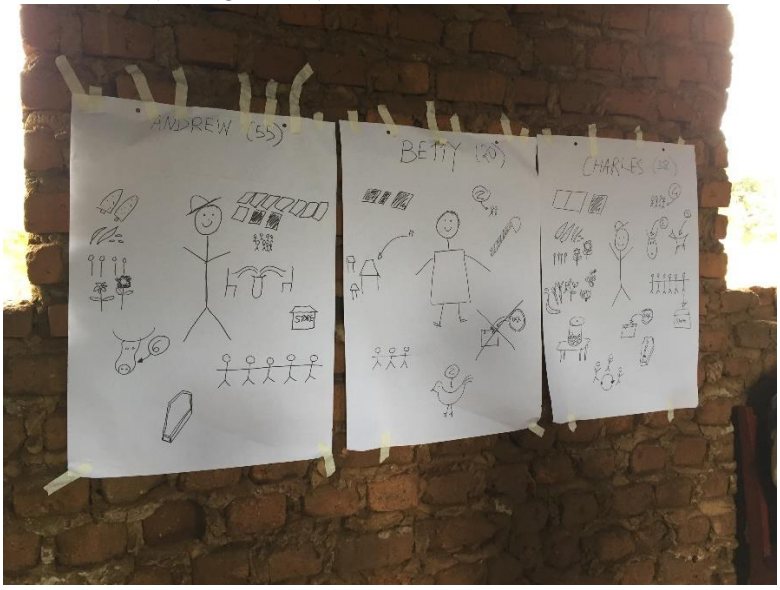


Figure 4.4 Sketches used to communicate the farmer archetypes Andrew, Betty and Charles. Photo credit: Fiona Lambe

The value of the archetypes was their ability to capture and communicate multiple layers of information about various types of farmers in relation to their risk response. By packaging and layering complex information in this manner, research participants were able to comment on, further develop and critique it. During the validation workshop, for instance, multiple participants shared that they recognized the Betty character. As one participant put it: "I know too many Bettys in this community; we need to do more to help our Bettys".

Table 4.1 Risk response pathways for archetypes Andrew, Betty and Charles

Andrew	**Betty**	**Charles**
Seek advice (family & friends)	Seek advice (family & friends)	Seek advice (family & friends)
Take a loan (Bank/VSLA)	Sell off live-stock/chickens	Take a loan (savings group)
Sell land	Seek casual work	Ask for loan from family
Sell livestock	Ask for loan from friends/family	Diversify crops
	Sell crop reserves	Diversify income sources

Source: Lambe, Osborne et al. (2022), reproduced with permission.

Prototyping

The process of prototyping insurance products with farmers began during the interviews, where we tested out various ideas on how the insurance product might work. We found that asking farmers to think through scenarios and then tell us whether these would be feasible or desirable provided richer insights, because it allowed us to swiftly eliminate design features that were originally thought useful, gave us an opportunity to ask why certain scenarios would not work and opened up space for the farmers to suggest alternative scenarios or product features. For instance, a previous round of data collection (prior to our study) suggested that village-level savings and loan groups would be well-positioned as local actors to process insurance pay-outs to farmers. During the interviews and field workshops, however, this feature was rejected as a component

of a prototyped insurance product due to the farmers' lack of trust in the Village Savings and Loan Association (VSLA). Similarly, by testing out scenarios during interviews, we learned that some farmers with a very low risk tolerance (Bettys) would struggle to pay even a low insurance premium, and that it would make more sense to develop a savings scheme as an alternative to an insurance product for these farmers. This insight was confirmed during the validation workshop, where we further developed the prototypes by having the stakeholders work in groups to determine which insurance product prototypes would be most suitable for each of the farmer archetypes. There was broad agreement among the workshop participants that insurance was not an appropriate intervention for Bettys, who were most in need of a savings scheme and enhanced social embeddedness. This finding is important, as it challenges a core assumption of the project – that insurance can be a useful approach to mitigating the risk of vulnerable farmers in these communities.

Table 4.2 Summary of key findings from the service design phase

1.	Insurance was seen as an attractive option for some, but not all, farmers
2.	There is significant heterogeneity in how farmers respond to risk, with financial capacity being just one source of risk response capacity. The key categories of risk response capacity identified were: social network; social institutions; agricultural diversity and economic diversity.
3.	Farmers in this area can be considered to have either strong, medium, or weak capacity to respond to shocks. Insurance does not appear to be an appropriate option for farmers with weak risk response capacity.
4.	Trust in the formal VSLAs is generally low, and most farmers responded negatively to the idea of a VSLA being involved in handling insurance pay outs.
5.	Trust in other community organizations (e.g., agricultural input stores and alternative finance organizations) is high.

Source: Osborne et al. (2022), reproduced with permission.

Findings from the Discrete Choice Experiment phase

From the DCE, we learned that farmers have a clear preference for being insured as individuals rather than as part of a group. In addition, they displayed a preference for bundling insurance with pesticides, seeds and credit over unbundled products, and for insurance pay-outs to be made via mobile money rather than village-based agents. The experiment also revealed that farmers would prefer an insurance package that covers the entire growing season, as opposed to individual growth phases. This preference, along with the bundling

preference, demonstrates the significance of having comprehensive risk protection throughout the farming cycle (Osborne et al., 2022).

Insights on combining service design and a Discrete Choice Experiment

In addition to the triangulation of findings, each methodology identified important aspects of risk management and protection aspects that the other did not (for a summary of contributions from each method see Figure 4.3). Unlike the service design phase, the DCE found strong evidence that bundling WII with inputs and credit would be likely to increase its uptake (Awondo et al., 2017; Ward & Makhija, 2018). In contrast, the service design phase produced nuanced insights that a DCE methodology would be unlikely to reveal. For example, service design found that farmers with few assets and weak social ties require risk protections other than insurance. The service design phase also demonstrated that risk coping is complex and multidimensional; the attempt to approximate this in the DCE analysis was unsuccessful, which indicates a limitation of the quantitative method when used alone. Lastly, by mapping the farming cycle and the farmers' risk coping mechanisms, the service design phase demonstrated that WII adoption is influenced by the annual liquidity cycle of the study area. The DCE missed this because it was unable to separate this factor from competing factors.

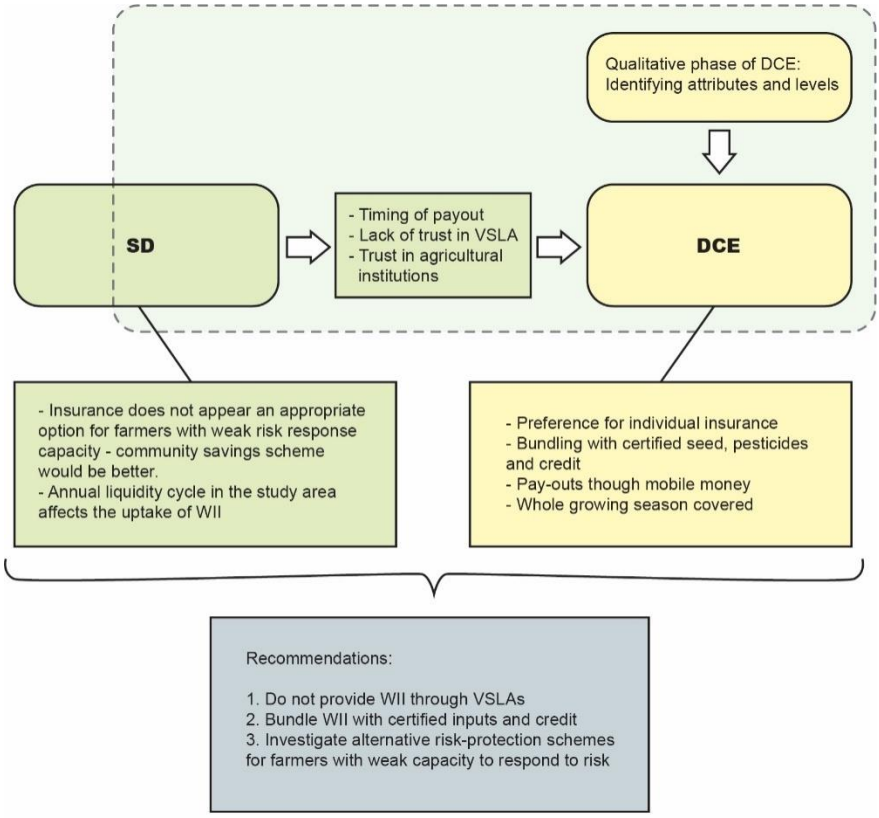


Figure 4.5 The discrete and collective contributions of the service design and discrete choice experiment research phases
Source: Osborne et al. (2022), reproduced with permission.

Since service design requires future intervention users to participate in both problem definition and solution design (Lambe, Ran, Jürisoo, et al., 2020), the approach enables local actors to challenge the underlying assumptions of a proposed intervention and to critique proposed design features. We demonstrated, for instance, how prototyping insurance products enabled community members to reject the assumption that insurance was suitable for all farmers in the region and to propose an alternative form of assistance for the most vulnerable farmers in their community. Such early involvement of users in the design of the intervention also helps reduce the risk of unintended negative consequences of interventions (Escobar, 2017)

4.4 Paper 4

The purpose of paper 4 was to understand how design devices, such as experience mapping, archetype construction and prototyping, can be used to guide design work that aims for environmental sustainability and social well-being in low-income communities. We re-analysed the findings in papers 2 and 3 using a capabilities approach lens, specifically Nussbaum's capability lists (Nussbaum, 2013a). We then related the findings to Arnstein's ladder of citizen participation (Arnstein, 1969) to describe how the design devices supported participation and agency in each case.

Applying the Capabilities lens to the Kenyan cookstove study

In the case study on cookstoves, we examined how different MM stove users experienced the clean cookstove over time and how these experiences translated into value for the user. Based on these findings, we identified three primary archetype groups and the service system requirements for each to achieve value. When we applied the capabilities approach to the data, we found that each archetype group had distinct functionings, conversion factors and service system requirements (Table 4.3). For the saving money archetype, the beings and doings associated with using the stove included being able to expand employment opportunities by cooking food for sale, being able to plan for the future (including saving for a rainy day), and being able to provide education for children.

To realize their beings and doings in this context, women must be able to use the stove effectively, have access to fuel and have the freedom to choose which stove and fuel to purchase and use. Understanding these individual conversion factors helps us make sense of the requirements for the service system that were highlighted by the users during the prototyping workshop. Although the functionings and conversion factors varied across archetype groups, all three sets of functionings contain beings and doings that are associated with Nussbaum's "control over material environment" capability set.

Table 4.3. A capabilities analysis of clean cookstove user archetypes

Values-Based Archetype Group	Nussbaum's Capabilities Set	Functionings (Beings and Doings) Supported MM Stove Use	Conversion Factors	Service System Needs
Saving Money	- Control over environment (material) - Senses, imagination, and thought	- Being able to pursue employment opportunities - Being able to plan for the future - Being able to provide an education for their children	- Ability to use efficiently - Fuel availability and affordability - Functioning stove, inc. spare parts if needed - Women's decision making autonomy in the home	- Physical demonstration - User manual - Employ additional fuel vendors - Ensure local technicians and spare parts
Convenience	- Control over environment (material) - Play - Affiliation	- Being in control of one's time (using the time spent cooking and cleaning pots for other purposes, e.g., socializing, relaxing) - Mobility: being able to take the stove when travelling	- Fuel availability - Ability to use the stove effectively - Flexibility to use other types of fuel	- Fuel made available in various quantities - Expand types of fuel that can be used - Support and advice on using the stove to cook different meals
Health	- Control over environment (material) - Bodily health	- Having a clean and healthy living environment - Having a healthy household - Feeling safe	- Ability to use the stove properly and safely	- Solution to lighting the stove without using kerosene or producing smoke

Source: Lambe, Osborne et al. (2022), reproduced with permission.

Applying the Capabilities lens to the Ugandan insurance study

In the case study on insurance in Uganda, we used farmer archetypes to capture and understand the multiple factors determining how various farmers respond to risk. We identified three main farmer archetypes: Andrew, with a high capacity for shock, Betty, with a low capacity for shock, and Charles, with a relatively high capacity for shock.

Table 4.4 shows the relevant risk response functionings available to each archetype group. Beyond his financial safety net, Andrew's strong capacity to cope with risk is supported by the variety of available functionings. Given that he could afford to pay the premium required to insure all of his land, the insurance product prototype deemed most suitable for Andrew in the event of a climate shock was the more expensive, full coverage insurance (a conversion factor).
Betty's risk response-relevant capabilities include the ability to earn a stable income, pay for her children's education and purchase property. Due to her lack of land ownership and inability to pay an insurance premium, Betty is unlikely to benefit from insurance in the event of an emergency. A savings plan that allows her to set aside money would be useful so she can one day purchase her own land and provide a financial safety net in case of an emergency. Ideally, the savings plan would also increase Betty's involvement in the community and support her to establish a strong social network, a crucial function for a person's ability to withstand sudden disasters.

Charles represents a relatively prosperous farmer archetype since he owns land and is already socially integrated into the community. His status is reflected in his available risk response functionings. Although this archetype is able to avoid disaster by relying on small loans from family and friends, they struggle to pay for farming inputs such as pesticides and fertilizer that could increase their yield and protect them from sudden pest infestations. The most appropriate insurance product for Charles was one that covered the cost of essential farming inputs.

Table 4.4. A capabilities analysis of risk response by archetype

Risk Response Archetype Groups	Nussbaum's Capabilities Set	Available Functionings Relevant for Responding to Risk	Selected Insurance Product	Conversion Factors
Strong Capacity (Andrew)	- Control over environment (material) - Affiliation - Emotion	- Having a financial safety net - Being able to own land - Being part of multiple local institutions - Being socially connected	- Full coverage insurance	- Being able to afford the insurance premium
Weak Capacity (Betty)	- Control over environment (material) - Education - Affiliation	- Being able to earn a steady income - Being able to pay for children's education	- Savings scheme	- Ability to save regularly - Better connections to the community
Relatively Good Capacity (Charles)	- Control over environment (material) - Affiliation - Emotion	- Being part of a social network - Being able to own land - Being respected and established in the community	- Insurance product that covers the cost of key farming inputs	- Ability to pay for the insurance product (savings scheme)

Source: Lambe, Osborne et al. (2022), reproduced with permission.

Capturing multidimensionality

Archetypes as a design tool enabled the identification of relevant collections of beings and doings for the design of the interventions in question, as opposed to viewing them as separate needs. Using this method, we were able to develop a multidimensional understanding of people and the various financial and social resources available to them for mitigating the effects of unexpected challenges. Understanding the farmers in terms of their capabilities enabled the identification of the conversion factors that, in terms of capabilities, are required for an intervention to benefit the various farmer archetypes. In both case studies, the

concept of conversion factors was useful for relating the proposed service or service system to people's beings and doings as well as for demonstrating how the designed intervention could enhance people's individual capabilities.

Mapping the experiences of cookstove users and farmers over time resulted in a more nuanced understanding of community members' needs and capabilities. By mapping the experiences of various user archetypes in Kenya, we discovered that needs evolve over time and that different users require different types of assistance with the MM stove to maximize their capabilities.

In order to visualize the complexity of the farming system in Uganda and to begin prototyping interventions with farmers, we mapped risks and capacities to respond to risk onto the annual farming cycle. Since we took a comprehensive approach to mapping the annual farming cycle, which included non-farming factors such as people's personal histories and social ties within the community, the mapping captured the broader context, beyond farming activities and financial resources. This broader perspective allowed us to recognize that the capacity to mitigate disaster depends on farmers' level of social embeddedness as well as their power within the community, a crucial insight for the development of the insurance product.

Facilitating participation: connecting the micro and meso levels of the system

In Kenya, developing user archetypes and archetype user journeys enabled us to communicate the experiences of diverse user groups to meso-level stakeholders in the Kenyan cookstove industry. Due to cultural norms and social hierarchies in Kenya, we were unable to invite cookstove users to the Nairobi workshop. Instead, the prototyped solutions, which had been co-created by groups of cookstove users, served as the basis for the workshop with sector stakeholders and served as a constant reminder of whose needs and priorities were in focus.
In Uganda, farmer archetypes were used as a shared canvas to facilitate discussion with community representatives about different risk-coping capacities and how to strengthen them. The prototyped interventions appeared to stimulate further exploration of farmers' needs and risk response journeys, and were used primarily to identify important features of an insurance product and associate those features with archetypes. Importantly, the use of co-designed prototypes enabled community members to reject one of the project's fundamental

assumptions, namely that insurance is an effective risk reduction intervention for all types of farmers in the region, and thereby exercise agency in the design process.

4.5 Paper 5

The fifth paper explored whether – and, if so, how – participatory design can support grassroots transformational change by facilitating community engagement around the challenge of energy access. We used participatory backcasting to facilitate the co-design of a 10-year transition roadmap to electric cooking with 30 members of a rural community in Kenya. Here, backcasting was positioned as a method for prototyping in the context of a co-design process. The methodology was applied during a two-day workshop in Machakos County and follow-up interviews with 10 of the workshop participants were conducted six months later. The findings were discussed in relation to the literature on transformative service research and collective capabilities.

Visioning as a tool for connecting energy to the wider system
From the visioning exercises conducted during the first part of the workshop, we found that the participants were thinking about much more than energy technologies in their homes; their discussions covered numerous facets of development. For instance, participants discussed how access to electricity would alter gender roles and household relationships since relieving women of the burden of cooking with traditional biomass would increase their opportunities for education and income generation, but also encourage men to assume more responsibilities at home, as electricity in the home signifies a modern home in which men and women share the workload.

Employment in the region was frequently mentioned. Electricity in the village would create many business opportunities and attract investment to the area. In addition, if implemented properly, the electricity extension programme would itself generate employment opportunities.

Alongside the positive visions of 2030, concerns were raised, for instance, that if everyone had access to electricity, there would be less sharing of fuel between households and less cooperation, such as with gathering fuel. Participants also expressed concern about possible electrical accidents. Interestingly, despite the

fact that the majority of cooking in 2030 would be done with electricity, people could not imagine abandoning traditional cooking altogether; participants envisaged cooking with electricity alongside charcoal and wood, where the latter would be required as a back-up or to cook traditional food for special occasions. This finding is supported by the literature on household energy transitions, which suggests that households are more likely to stack fuels and stoves, rather than replace one fuel with another (Ruiz-Mercado & Masera, 2015). However, deep attachment to cooking practices is often overlooked in development planning that assumes a technocentric transition pathway where traditional technologies are eventually replaced with "modern" ones.

From visioning the future to articulation of a change process

The main output of the second part of the workshop was a transition road map that outlined the major changes required at each time interval, as well as which actors – community, government or civil society – would be responsible for what (see Figure 4.4). It is interesting to note the emphasis placed on civic engagement and civic education. The participants expressed a need to learn about what they were entitled to and how to exert pressure on the authorities to obtain their entitlements. For instance, one participant mentioned that there is a government programme to which villages can apply to obtain financial assistance with grid connection. Most of the other participants were unaware of this. That 2022 was an election year in Kenya was also discussed, as this could be an opportunity for the community to put pressure on the local government to deliver the services to which they were entitled. In addition, participants recognized that some community members were in a better position than others to assert their rights or make personal investments to gain access to electricity. It was suggested, for instance, that a community fund should be established to assist those households which required assistance.

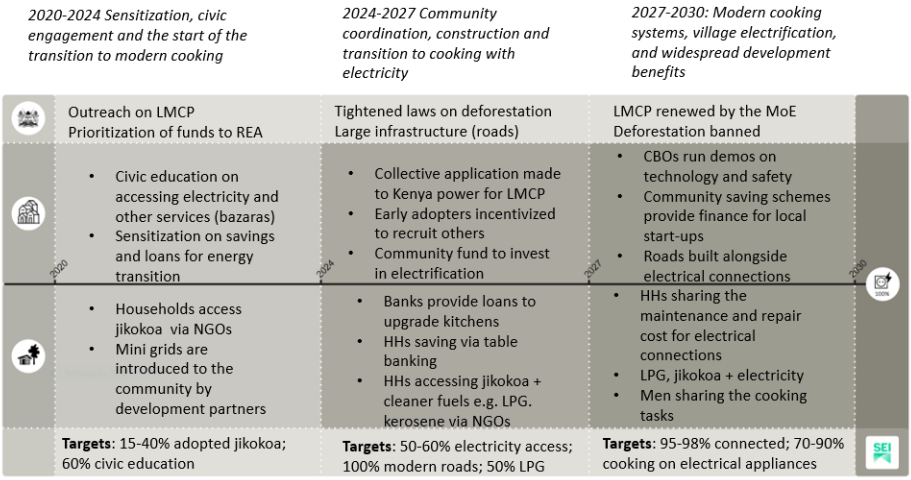


Figure 4.6 Summary of actions and targets for meeting the goal of 100% cooking with electricity by 2030. *Source*: Lambe, Nyambane et al. (2022), reproduced with permission.

Reflecting on the road map, several workshop participants mentioned that it was a useful tool for visualizing and framing the community's desired changes, and that it could be used in future as a reference tool when interacting with external actors. As one participant explained:

> NGOs sometimes come here and ask what we would like them to do for us. This is a difficult question to answer. There are so many things needed here. But now we have a plan, we can use it to remind ourselves of where we would like to go, and what help we should be asking for.

Findings from the follow-up interviews

In follow-up interviews six months after the workshop, many participants recalled having moments of realization. Five participants, for instance, stated that they were unaware of the government's Last Mile Connectivity Project (LMCP) prior to the workshop, and that this information was "eye-opening for them". When asked about the workshop's impact, a number of interviewees mentioned making changes to their homes, such as purchasing solar lamps and saving money to purchase kitchen utensils. Several of the interviewees explained that they viewed these minor improvements as steps towards a future they had envisaged during the workshop. All the interviewees either stated that they had

begun saving through a local savings and loans club or that their savings club intended to start a household energy-specific savings scheme. Six respondents reported that a group in the area had been formed to collect the contact information of those interested in connecting to the grid. This group had then approached the Machakos County Member of Parliament with the list of names and asked that they be considered for the LMCP.

Facilitating conscientization and conciliation

The backcasting process appears to have made the future tangible to the workshop attendees, allowing an in-depth discussion on how the energy system transformation could take place in their community. The speculative aspect of the backcasting methodology influenced how the participants engaged with the questions. Several local facilitators noted that they felt as though the participants had transported themselves mentally into the future. We observed that, unprompted by the facilitators, some participants wrote the future date rather than the actual date in their notebooks (Figure 4.5).



Figure 4.7 Workshop participant writes the date as '18/02/2030'
Photo credit: Fiona Lambe

According to the follow-up interviews, the process of collectively imagining an alternative future appears to have resulted in a moment of conscientization or realization for a number of the participants. By discussing and sketching out their imagined future in some detail, participants realized that this alternative future was not only possible, but something they are entitled to.

There is some evidence from the interviews that the moments of conscientization led to household and community-level changes. Throughout the workshop, civic engagement was emphasized as a prerequisite for achieving the 2030 objective. It was striking to hear interviewees describe how, shortly after the workshop, they approached their local representative to request funding to which they now knew they were entitled. Here we can see some early signs of conscientization as the community begins, in small ways, to push for change. In addition, there were early indications that the process provided space for conciliation through reflection on existing resources and acknowledgement of differing circumstances within the community.

Participatory methods are frequently criticized within the field of development studies for glossing over important differences between community members (Mansuri & Rao, 2004). In the backcasting workshop, participants acknowledged that the energy transition would have varying effects on them, and that a nuanced understanding would be required to ensure that no household was left behind. Rather than generating a single technical path, the process enabled a variety of solutions with roles for various actors in the system.

4.6 Summary of main findings

Table 4.5 provides a summary of the key findings from each of the five papers.

Table 4.5 Summary of key findings

Paper 1 Conceptual framework	- Service design can be used to map and visualize complex behaviour change at the household level and situate this in relation to a wider system of actors and processes. - Mapping provides a way to zoom out and situate an intervention or change process within the wider social-ecological context.
Paper 2 Cookstoves	- Together, archetypes and user journey mapping can be used to make sense of and communicate the heterogeneity of user needs and behaviours. - Value is assigned throughout the user's journey (and before the stove and fuel are in use). - Combining service design and quantitative measurement of stove usage allowed triangulation, enhancing the credibility of the findings.
Paper 3 Insurance	- Archetypes were used to capture and communicate multiple layers of information about various types of farmers in relation to their risk response. - Prototyping provided opportunities for participants to challenge (and reject) underlying assumptions of the intervention, and to suggest alternative models. - Each methodology was able to identify aspects of risk management that the other was not.
Paper 4 Capabilities	- Experience mapping allowed for the identification of capability sets related to the proposed intervention. - Archetypes allowed us to package and communicate collections of beings and doings for policymakers. - Prototypes supported the participation and agency of the research participants by ensuring that the messiness of the context was captured and reflected in the material communicated to programme designers and policymakers.
Paper 5 Backcasting	- Participatory backcasting facilitated envisaging, which opened up space for creative thinking on future development pathways. - Envisaging triggered conscientization and conciliation, which in turn led to individual and collective action.

5 Discussion

The thesis aims to explore how service design can be used as an approach to designing development interventions in complex social-ecological contexts. In addressing this aim through three empirical studies and two conceptual papers, the thesis makes contributions that should be of interest to both design and development researchers and practitioners. This chapter discusses and contextualizes the main findings in the light of the research questions and presents some research limitations. Some impacts of the work are presented and the main contributions and implications for development policymakers and practitioners are outlined.

5.1 Addressing Research Question 1

What are the salient features of service design for capturing and making sense of local contextual information for other important actors in the system?

Systems complexity means that linear models for designing interventions are often ineffective when designing interventions in complex social-ecological systems (D. Burns & Worsley, 2015). Since local level behaviours can lead to unintended consequences elsewhere in the system, innovative approaches are needed to understand and incorporate local contextual factors into the design of interventions. Research question 1 focuses on how service design can be used to identify the local contextual information required in the design of a given intervention, and to package or translate this information for key actors responsible for delivering the intervention, be they policymakers, development programmers or the private sector.

5.1.1 Devices for understanding how services are valued over time

Papers 1 and 2 show how service design devices for mapping and visualizing people's needs, behaviours, values and experiences can help to make sense of local level complexity and heterogeneity without compromising the richness of the data. This is important since development interventions often fail because of a lack of understanding of the situated experiences of individuals and households (D. Burns & Worsley, 2015; Datta & Mullainathan, 2014).

As a device, user experience mapping performs several key functions. On the one hand, it allows for the mapping of experiences over time, including mapping expectations and perceptions prior to the introduction of an intervention, and after – when an intervention is disused or ceases to function. Mapping experiences over time can help to narrate the lived experience of the user and to see the full or "synchronic" picture of the service system (Diana et al., 2009, p. 67). Viewing these experiences as part of an ongoing continuum can help development researchers and practitioners to zoom out and understand how their efforts relate to the wider social-ecological system. The approach taken to experience mapping in the case studies was broad and zoomed out, involving a process of understanding people's daily lives as a starting point before detailing their experiences of a particular service or system. Taking this wide view early on is important because it allows identification of key contextual information that might otherwise be missed by applying a more linear, technocentric logic to designing interventions, where there is a tendency to quickly zoom in on a solution to a particular problem (Ramalingam, 2013). These findings are in line with previous studies that demonstrate the crucial role that visualization plays in service design as a way of understanding complexity in the experiences of users and of communicating design ideas with diverse sets of actors (Diana et al., 2009, 2010; Trischler & Scott, 2016).

The user experience maps played a critical role in the cookstoves and insurance studies as shared canvases that research participants and researchers could use to communicate key information and to cross-check understandings within the research team. The mapping exercises also allowed communication of uncomfortable realities. In the cookstove study, for example, the user journey was the starting point of mapping the wider cookstove and fuel service system in Kenya together with representatives from the cookstove sector. The exercise of mapping where the various cookstove sector actors were situated in relation to the users resulted in a visualization that depicted critical gaps in service provision at certain points of the user journey and duplication of efforts at others. Importantly, the resulting sector map was sufficiently zoomed out to allow stakeholders to understand their position in relation to others while still being connected to the user's experience of adopting a clean cookstove, which served as an anchor throughout the discussions. In this way the user journey map created a tangible connection between the micro and meso levels of the system.

When combined with archetype construction, user journey mapping became a powerful device for identifying the value in experience of heterogeneous groups. Paper 2 demonstrates how archetypes can be used to gain a deeper understanding of how value in an intervention is assigned by different types of cookstove users, and how this value in use evolves over time. This finding addresses a call for further research on how the process of mapping user journeys can be used to understand and represent complex behaviours and values in relation to a service (Diana et al., 2009). The finding is also an innovation from a development studies perspective, since there is a tendency in development studies to measure the success of interventions in a linear and static manner, using one dimensional metrics such as money or time saved to evaluate impact (Adams, 2016; Ramalingam, 2013) The idea that value in experience is dynamic and shifts over time, and that it is possible to articulate how different people assign value provides opportunities for intervention design processes that are better positioned to anticipate feedback loops in the system.

5.1.2 From understanding user needs to strengthening capabilities: devices for capturing complexity

Describing user archetypes in detail, giving them names and back stories, and mapping out their experiences over time allowed us to capture and communicate multiple layers of information about the micro context to meso-level stakeholders. In Uganda, using storytelling and live sketching to describe and discuss a set of farmer archetypes allowed us to validate findings, but was also a mechanism for supporting self-reflection in the community and a way to discuss differences between community members, which was important in the design of an insurance product. The archetypes were constructed from multiple strands of information about people living in the community. These covered social, economic, political and historical aspects, and as such were complex representations of society and social processes. This function of archetyping is in line with the reflections of Blyth et al. (2011) on how designers can reveal the construction of social problems through visual representations that include aspects that are often overlooked. In this sense, the design process can reveal aspects of a problem that were previously unnoticed, bringing differences to light rather than erasing them. (Ehn, 1993). In the Uganda case, the process of constructing archetypes led to the realization that a particularly vulnerable segment of society had not been considered in previous studies conducted to inform the design of an insurance product.

Paper 4 frames this ability of archetypes to capture multiple layers of information as a device for identifying relevant collections of beings and doings in the design of the interventions in question. The multidimensionality of poverty implies a need for a deep engagement with the situated experiences of community members to ensure that the outcome of a designed intervention is aligned with what they view as attributes of a good life (Oosterlaken, 2009). The capabilities approach can be a useful framework for designers working in development contexts to ensure that the capabilities of the people they are designing for are embedded in their designs (Kleine, 2010; Oosterlaken, 2012; Steen, 2016).

Aranda-Jan et al. (2016) developed a framework that used the capabilities approach to incorporate contextual factors into the design of technologies for low-income communities. Similarly, Steen presents a framework and practical tool in his work on design for well-being to assist designers in applying the capabilities approach while designing with marginalized communities (Steen, 2016). The framework provides a set of important questions that can be used early on in the design process to identify the most salient capabilities for the targeted users and other key stakeholders (Steen, 2016). In line with Steen and Aranda-Jan, this thesis suggests that a capabilities framing could help guide development-focused design work with marginalised communities. Paper 4 proposes that the capabilities in focus should be drawn from experience mapping conducted early on in the design process, rather than from the normative goals and criteria established by an external actor (such as the priorities of a development donor) and highlights the importance of identifying collections of capabilities that are relevant to a proposed intervention. Furthermore, paper 4 shows how the concept of conversion factors – the personal, social and environmental factors that determine whether an individual's capabilities will be strengthened by an intervention – can be identified through archetype construction and prototyping (Lambe et al., 2022). Since value is assigned over time by the user of a service, and capabilities may be strengthened (or weakened) by using the service, it is reasonable to assume that that the relative importance of different functionings and conversion factors will also shift, requiring a more dynamic and adaptive approach when designing for capabilities.

Due to the inherent systemic complexity encountered when addressing wicked problems such as poverty, there is a need for design researchers and practitioners

to relate their work to long-term normative frameworks, such as the Sustainable Development Goals or a country's long-term development plan. It might be difficult for designers without a foundation in these contexts or conceptual understanding of the sectors in which they work to identify the development dimensions relevant to their design work (Sangiorgi, 2011). By anchoring design work in people's essential beings and doings, a capabilities framing can help designers navigate between micro-level needs and behaviours, and normative development goals.

A capabilities framing requires an integrated approach to designing development interventions. From a capabilities standpoint, development is about overcoming multiple constraints on individual freedoms. Using a capabilities lens could help designers avoid becoming siloed in their efforts to address development problems, thereby increasing the likelihood that interventions target the underlying causes of poverty as opposed to simply providing short-term technological solutions. Here the thesis addresses calls for design to take an integrated approach to developing solutions for and with marginalized communities(Jagtap, 2019b, 2022). In our studies, we observed that user-generated prototypes frequently addressed issues beyond the intended scope of the interventions. In Kenya, for instance, improvement of the service system, which included fuel distribution and technical support, necessitated the training and employment of community members, which required a significantly higher level of societal engagement than the cookstove intervention had enabled. Social isolation emerged as a key factor in determining how farmers cope with disasters in Uganda, and whether insurance could be a useful intervention to reduce risk. Moreover, in both studies, the ability to own property, to be employed, to be treated with respect and to have relationships with other members of the community were identified as essential functionings. These functionings are closely connected to Martha Nussbaum's capability sets "having influence over one's environment" and "affiliation," which could be considered overarching design principles for the interventions in question (Lambe et al., 2022; Nussbaum, 2013a, p 34).

Here I should point out that I am not arguing for the application of universal design principles. Rather, I am suggesting that the capabilities approach can provide a framing for connecting design to the factors that people consider most valuable for leading a good life, and for aligning it with overarching

development goals. Of course, one might question whether the capabilities approach can be considered universally appropriate as a framework for guiding design work, and whether suggesting a single set of norms for people everywhere is imperialist. However, as Martha Nussbaum argues, the capabilities approach was developed in a bottom-up manner by a group of researchers from more than 80 countries, representing both western and non-western cultural traditions (Nussbaum, 2013a). At the core of the approach is a concern for human dignity, and Nussbaum argues that: "the capabilities approach asks a question that people everywhere pose to themselves everyday 'what am I able to do and to be?', which allows it to bypass accusations of imperialism" (Nussbaum, 2013a, p.106).

5.2 Addressing Research Question 2

What are the distinguishing features of service design that enable local agency in intervention design?

It is widely assumed by the development community that local participation in development programming leads to improved socio-economic outcomes and citizen empowerment. Development practitioners and policymakers argue that participation enhances aid effectiveness because it generates information on citizen needs and helps to secure local support for project implementation. Development researchers and practitioners also believe that giving people a voice in development programmes is valuable for its own sake, as a way of empowering citizens (Mansuri & Rao, 2004).

To date, however, the development community has struggled to ensure effective local participation in development programming (Cooke & Kothari, 2001; Easterly, 2007). The impact of participatory development programmes on poverty reduction and citizen empowerment has been weak, programme designers have overlooked issues of power and politics, and programme structures are often too inflexible to incorporate citizens' needs (Gaventa & McGee, 2013; Mansuri & Rao, 2004). Although design approaches have been used in development contexts for decades, their ability to enhance local agency is not well understood. Research question 2 focused on how service design devices can help to create the conditions for local (individual, household and community) agency – that is, the ability to take action or to choose which action to take – in the design of development interventions.

5.2.1 Prototyping as a device for shifting power

Papers 2 and 3 found that prototyping provided opportunities for participants to critique proposed design features and to challenge the underlying assumptions of proposed interventions, indicating a shift in power. In Uganda, prototyping afforded opportunities for farmers to challenge the core assumption that insurance would be a feasible intervention for all farmers in the community. The use of scenarios during interviews seemed to provide a space for farmers to provide honest feedback on the various design features that were proposed, including rejecting ideas they felt would simply not work and suggesting alternative scenarios. By decentring our roles as researchers and designers, the prototypes functioned as devices for shifting power to the research participants (Kimbell, 2012). If agency is understood as the ability to take action or to choose which action to take, then it could be argued that prototyping in the Uganda case created the conditions for the farmers to exert agency on the design of the insurance product.

In the cookstove study, prototyping allowed the users to articulate solutions to specific challenges they identified within the cookstove and fuel system. The prototyped solutions called for the engagement of specific actors within the clean cooking sector in Kenya, and were presented during a stakeholder workshop where they formed the material basis for the discussions with stakeholders, and for further prototyping of the users' solutions. Since the cookstove users were unable to participate in the stakeholder workshop for social hierarchy reasons, their prototyped solutions, along with their archetype journeys provided a way to centre the users' voices in the discussion. In paper 4 we suggest that prototyping facilitated agency among the users in the cookstove study by allowing them to articulate the changes they would like to see in the clean cookstove and fuel system and provided a way to represent their views in the stakeholder workshop. On reflection, however, I acknowledge that being excluded from the stakeholder workshop meant that the users were prevented from participating directly in a co-design process with other stakeholders, which eroded their agency in this part of the design process.

5.2.2 Backcasting as a device for supporting collective capabilities

Paper 5 showed how participatory backcasting can combine creative visioning with structured deliberation and can enhance local participation in a planning process. The visioning element of the participatory backcasting approach in

Kenya provided workshop participants with an opportunity to include a wide array of development considerations far beyond energy for cooking, connecting a future energy service to a wider set of changes that they considered important in their community. One reason the approach worked so well was that no limitations were placed on what the participants could include as part of their desired future; the only factor imposed by the researchers was the quantitative goal of 100% cooking with electricity by 2030. Similarly, the participants were free to propose any actions they felt necessary to achieve their 2030 vision, and to include any actor they felt would need to be involved to achieve their envisaged future. The result was a holistic depiction of a development transition. Rather than generating a single technical pathway, the process facilitated discussion of a wide range of possible solutions with roles for various actors operating at different levels of the system, which would seem appropriate given the complexity of the system in question (Metcalf, 2014; van der Bijl-Brouwer, 2017).

Allowing people the freedom to imagine a desired future and to propose how it might be achieved could be seen as entering the realm of science fiction or encouraging utopian dreaming, which could be viewed as irresponsible as it might risk raising participants' expectations about unrealistic futures. However, the structured and systematic process used to backcast from the envisaged futures to the present day required that the participants connect their visions to concrete actions or processes. Without limiting the space for ideas and imagination, the process requires logical connection of activities and events through time, and careful articulation of causal connections between events and processes. The resulting timeline – in our case, the energy transition roadmap – captures in detail both an imagined, desired future and the concrete steps needed to get there, according to the research participants.

The backcasting process seems to have led to individual and collective action following the workshop, indicating the emergence of collective capabilities. Capabilities scholar Solava Ibrahim argues that transformational development in communities requires collective capabilities. According to Ibrahim's 3C model of grassroots community development, community members initially experience a moment of conscientization, which is the process of developing consciousness and reflecting about one's situation, as well as understanding how the situation could be changed or improved (Ibrahim, 2017). Conscientization can be triggered by articulating individual beliefs, aspirations and well-being

objectives, and discussing how to achieve these objectives. Conscientization encourages individuals to reflect critically on their problems and then actively explore solutions (Freire et al., 2021). The capacity to aspire is regarded as a crucial component of conscientization, as the ability to verbalize ambitions, dreams and material demands is a critical step in acquiring capabilities (Appadurai, 2004). The combination of visioning and structured discussions that contributed to the backcasting appears to have produced a moment of conscientization, which was later shown to have led to individual and collective activities aimed at achieving the community's envisaged futures.

Paper 5 also shows that backcasting as a device facilitated deliberation among the workshop participants about the existing capacities of individuals within the community, as well as the resources needed and identification of key actors within the community and beyond that would need to be involved for the desired change to happen. The backcasting process provided space for participants to discuss the difficult aspects of a development transition, such as, that there are differences between community members in terms of existing resources and capabilities, and that the pathway to 100% cooking with electricity would therefore not be the same for everyone. These discussions touched on the social justice aspects of the transition, highlighting that not all households in the community would be able to enact the changes required at each stage of the transition timeline, and that there would be a need for mechanisms within the community, such as savings funds, to support less well-off households to participate and benefit from the transition. The government's responsibility was also discussed with reference to an existing funding mechanism to which households are entitled access to cover the costs of connecting to the grid. Ibrahim refers to this form of community-level discussion in the context of development planning as conciliation, or the process of forging community consensus around important priorities, needs and capacities (Ibrahim, 2017). Through the conciliation process, community members visualize the possible benefits that can be realized collectively, in addition to any individual gains they might experience (Ibrahim, 2017).

For Ibrahim, conciliation and conscientization are two key steps towards building the collective capabilities needed for grassroots-led transformative development (Ibrahim, 2017). The notion of raising consciousness is also present in Sangiorgi's framework for transformative services (Sangiorgi, 2011). For

Sangiorgi, consciousness raising is part and parcel of being an active citizen, where participation is seen as a right and involves questioning societal structures (Sangiorgi, 2011). Sangiorgi's principle of enhancing imagination and hope is linked to collective imagination of an alternative future and could be seen as a critical step in the design process supporting conscientization.

Facilitating a participatory backcasting process places significant responsibility on the design researcher to manage the various ideas and contestations that might emerge, and to try to ensure that certain ideas are not excluded (Sangiorgi, 2011). Given how the workshop was set up, with local facilitators running the breakout sessions and managing the documentation of discussions, there is a risk that certain ideas may have been elevated and others sidelined, which calls into question whether we can claim that the backcasting process facilitated the agency of all participants. In fact, the process of gathering feedback on the draft workshop report revealed that some ideas had been omitted from the workshop proceedings. The decision to hand over facilitation and control of the breakout sessions involved a trade-off in terms of being able to see and hear all the interactions as researchers, and the research participants feeling comfortable as part of a process facilitated by people they know, and in their own language.

Both Sangiorgi and Ibrahim assume long-term engagement with communities in their conceptualizations of community-driven change processes. According to Sangiorgi, a transformative design process can support the conciliation phase by establishing an enabling platform, or space for this deliberation that endures beyond the lifespan of the project. For Ibrahim, the enabling platform is an agreement, among community members that emerges from the conciliation process on how they should engage with external actors to secure their development objectives (Ibrahim, 2017). This implies a long-term process of negotiation at the community level. In the limited timeframe of the backcasting study, beyond conducting follow-up interviews, we were unable to explore whether or how such an enabling platform could emerge in the community.

5.2.3 Service design and the capabilities approach: implications for theory

Previous studies have linked design to the capabilities approach in various ways. As discussed, several scholars have explored how the capabilities approach can support the design of technologies and products for low-income communities.

To my knowledge, however, this thesis represents the first attempt to link the capabilities approach with the field of service design in the context of development intervention design, and here some theoretical contributions can be discussed.

First, the thesis articulates the conceptual congruity between conversion factors and value-in-use, and shows how this conceptual linkage can be useful for aligning service design work with individual capabilities. Understanding the value-in-use of a particular service for different types of users through the application of design devices can help to identify the conversion factors that must be in place for different users to benefit from that service (and thus strengthen their capabilities). As a concept, value-in-use can therefore help to orient the design process towards the most relevant factors for strengthening capabilities.

Second, the findings suggest that design devices such as backcasting used in the context of service or system design processes, can support grassroots development at the community level. Creative methods for facilitating imagination and hope (Sangiorgi, 2011) can be used to support conscientization and conciliation as necessary stages in grassroots community development. Thus, the thesis demonstrates how transformative service design can support the 3C model of grassroots development and collective capabilities (Ibrahim, 2017).

5.3 Addressing Research Question 3

How can service design methods and approaches complement conventional development research methods?

The starting point of this thesis was a critique of linear intervention logic for addressing complex development problems. The cookstove and insurance case studies explored how service design could be used as an alternative approach to intervention design in standard project and programme structures, where a solution to a development problem had been determined using a linear causal logic. Conducting exploratory service design work in these settings entailed a transdisciplinary research approach, which combined different types of knowledge from various stakeholders, both academic and non-academic. Research question 3 sought to understand this transdisciplinary set-up and the role that service design played alongside conventional development research methods and approaches.

5.3.1 The role of service design in transdisciplinary development research

In practice, the transdisciplinary work carried out for the cookstove and insurance studies entailed a balancing act where a service design epistemology was central to addressing the complexity that characterized each of the development problems, but where methods more commonly associated with development research and practice were used in parallel. This balance was necessary partly for pragmatic reasons, since design approaches are not yet well established within development research and programming, which means that decision makers, be they government or private sector, tend to require quantitative data to underpin decisions. For example, when we presented the findings of the cookstove study to a Kenyan policy audience, an official from the Kenyan energy ministry expressed appreciation for the mixed methods approach, reflecting that qualitative data is often needed to understand whether technical solutions work in the real world, but it is difficult to base policy decisions solely on qualitative insights. The mixed methods approach was also needed to provide a comprehensive and detailed understanding of the context required for designing an intervention.

The service design process model presented in paper 1 represents a stepwise, albeit iterative process for designing interventions, where one of the later stages involves quantitative testing of the insights emerging from the preceding qualitative steps. In the insurance study, however, rather than the service design insights feeding into the discrete choice experiment, as per the model, we found that both methodologies operated in parallel, each providing unique understandings about the farmers' risk coping mechanisms, needs and preferences that the other methodology could not, indicating a high level of complementarity. This highlights the role that service design can play in a transdisciplinary research set-up, where various ways of knowing contributed to a comprehensive understanding of a context (West et al., 2021). These findings challenge more simplistic framings of service design as detached methods that can be applied easily by anyone in any situation (Akama & Prendiville, 2013).

Each of the studies presented in this thesis was an example of a transdisciplinary collaboration, involving researchers from various academic disciplines, as well as (in one case) professional service designers, and non-academic collaborators such as research assistants, community mobilizers and the private sector. In each case, several members of the team brought knowledge and experience of the

sector or problem area in focus. For example, I have worked for almost 20 years on the topic of energy access and household energy transitions, and much of this work was conducted in the East African context. This knowledge and contextual experience meant that I could quickly relate the insights we gathered during the cookstove study to previous research conducted on the topic. Similarly, the teams from the University of East Anglia and the Field Lab, Uganda, with which we collaborated in Uganda, brought decades of experience of conducting research on innovations for building resilience among smallholder farmers, which was useful for contextualizing the work conducted in our case study. Crucial to the success of the transdisciplinary collaborations was an attitude of openness to experimentation and a willingness to engage in an open-ended process of enquiry on the part of all collaborators, which can be seen as keys aspects of a design attitude (Michlewski, 2008).

The methodological complementarity also hinged on having someone with dual expertise on the team who can see both sides of the coin and play a facilitator role, conveying important insights and knowledge to team members who either lacked thematic expertise or were unfamiliar with service design. This role reflects another key aspect of design attitude: the ability to consolidate multidimensional meanings, including reconciling different objectives (Michlewski, 2008). In recent years, service design has been framed as an integrative approach to cross-disciplinary and collaborative service innovation (Yu & Sangiorgi, 2018) and the role of service designers in complex transdisciplinary projects typically understood as facilitative, bringing together multiple perspectives into a new service solution (Sangiorgi et al., 2017). However, I believe that in the case studies presented in this thesis, this facilitative role would have been difficult for a service designer without thematic knowledge or prior experience of working in these contexts.

5.3.2 Reframing development intervention design as value co-creation

Although value co-creation was not a core focus of this thesis, the findings suggest that framing development interventions as services where value is co-created with service users could radically shift how interventions are conceptualized by development actors and could open up new avenues of research on the design of public services in development contexts. In Kenya, we studied how the value of a clean cookstove and fuel was created in use by different types of

users, and how this shifted over time depending on how well different users' expectations were met. This implies that the users of the service are active creators of value in the system, rather than passive beneficiaries, which changes the type of user engagement needed to design a service or service system from consultation to co-creation. Within development research, this is a fundamentally different way of viewing the people on the receiving end of development interventions, which demands a new, more reciprocal intervention design logic – one which implies an obligation or responsibility on the part of the service provider to deliver a service that will address the needs of the service user.

Shifting to a value co-creation logic would also mean a more holistic or systemic approach to designing services, since users' experiences of a given service will probably be affected by other adjacent systems and services (Yu & Sangiorgi, 2018). This is critically important in development programming where inflexible project designs and pre-set donor goals often lead to a siloed approach to development intervention design, in which services or sectors are viewed in isolation and where there is an urgent need for an integrated approach to service delivery (Jagtap, 2019b).

5.3.3 Reflecting on the use of service design in Global South contexts

How can the service design approaches and practices that emerged in the context of service management and business innovation in the Global North be usefully and ethically applied to development contexts (Cipolla & Reynoso, 2017) in the Global South? I have discussed the usefulness of service design devices above, with reference to how these devices might support and guide design work with marginalized communities in complex social-ecological contexts. However, methods and tools do not implement themselves in a vacuum; rather, they require embodiment (Light, 2010; Light & Akama, 2012). It is important to consider the role that the various collaborators played in shaping the outcomes of the studies presented here, and what these different roles meant for the use of service design in Kenya and Uganda.

In all three case studies, the research assistants played a key role in interpreting meaning in the gathered data, including highlighting important cultural idiosyncrasies that might otherwise have led to misunderstandings and incorrect inferences. For example, in Kenya the research assistants helped us to understand the

complexities of intra-household decision making, how the level of autonomy that women have to make financial decisions can differ and how to interpret interviewee responses about decision making. In Uganda, the research assistants educated us about the role of various social structures in the community and their political nature, which was critical for contextualizing farmers' responses to early prototypes of insurance products.

Playing this interpretive role required the research assistants to challenge the assertions of non-local team members at different points in the research process, which can be difficult due to power asymmetries between team members from the Global South and the Global North, and due to social hierarchies in both Uganda and Kenya which can prevent younger people and women from questioning or challenging the viewpoints of older generations and of men. In each of the case studies, we provided training for the research assistants on service design methods or participatory backcasting, where a key focus was on collaborative approaches and the importance of questioning emerging insights and testing assumptions as a central part of the research process. The team of research assistants with which we collaborated on the cookstove study had participated in a course on service design co-organized by SEI and a Swedish service design consultancy in 2016, and had collaborated with us on two service design projects prior to the cookstove study. This training and experience of working with the same team gave the research assistants a deep understanding of the methods and, more importantly, cemented a trust-based relationship in the team, where the research assistants felt comfortable challenging and questioning the insights that emerged during the research process. In addition, having an established relationship or prior association with the communities where the research was carried out allowed us, as outsiders, to quickly build trust with the research participants and become attuned (Light & Akama, 2012) to their realities within the relatively short case study time frames.

Design has been critiqued for promoting western-centric values such as solutionism and scientism (Lee & Lee, 2007; Manzini, 2016; Tlostanova, 2017), and for rigidly applying methods in a formulaic and distanced manner that can ignore how the design process is entangled with particular dynamic cultural contexts (Akama & Prendiville, 2013). Through the course of my PhD studies, I have come to realize that it is impossible to separate service design from the culture and context where it is applied, and that when applying a service design

approach, often the most important insights emerge from messy entanglements (Akama & Prendiville, 2013). The service design process model presented in paper 1 reflects a somewhat simplistic framing of service design as a methodological toolkit that can be picked up and applied by development researchers grappling with complex questions. The process of applying elements of the conceptual framework in the design of real-world interventions has led to a more nuanced understanding of the role and potential of service design in these settings, and an acknowledgement that service design on its own will never be sufficient for designing development interventions.

5.4 Limitations

The research presented in this thesis should be viewed in the light of various limitations, many of which have been touched on implicitly in the discussion. Each of the appended papers provides an overview of the conceptual, methodological and contextual limitations. Here I summarize some more cross-cutting limitations.

Limits on research funding meant that the case studies included in this thesis were relatively short in terms of the length of time spent in the field. There is therefore a risk that I missed or misinterpreted important contextual factors in each of the settings. However, I believe that this risk is low, given the measures taken to ensure research quality outlined in chapter 3. First, I collaborated closely with my colleagues at SEI in Nairobi for the six-month duration of the study – on the research design and on the in-field analysis of the qualitative interviews. Second, working closely with well-established local partners and spending time training research assistants on the methods and approaches used helped us to avoid sociocultural misinterpretations. I acknowledge that conducting one longitudinal study of one case would probably have provided more in-depth insights, but I see it as a strength that this thesis includes several studies looking at different types of development interventions in different contexts, as I feel that this diversity is needed to be able to draw conclusions about how service design functions as an approach and practice in these settings.

Since I am not based in Kenya or Uganda, I do not speak Kiswahili, Kamba or Lugisu, and I was coming and going during the research, I will certainly have missed out on cultural nuances. As described elsewhere in the discussion and in

chapter 3, I attempted to mitigate being an outsider by collaborating with insiders, but there remains a risk that some cultural aspects were lost in translation. As is the case in most qualitative research, the findings and contributions in this thesis were to some degree influenced by my subjectivity and role in each of the studies. Throughout my PhD studies and in the drafting of this kappa I have attempted to reflect on and be transparent about the various personal, interpersonal, methodological and contextual factors that brought this research into being and influenced its outcomes. The measures taken to ensure research quality, including statements on research reflexivity and positionality are detailed in chapter 3. The mixed methods approach taken in two of the studies was selected in part to enhance research objectivity.

In each case study, the users – or prospective users – of the development interventions were purposefully selected as the central stakeholders. This focus on the micro level was required because of how complexity in development intervention design is problematized in this thesis: as systems complexity where local level heterogeneity is typically not well accounted for and where multidimensional poverty requires an understanding of individual situated experience. Although other actors were included, there were fewer interviews with policymakers, development programme designers and private sector actors. This imbalance could be seen as a weakness given the important roles of these other stakeholders in delivering interventions. As discussed above, it would have been valuable to have been able to engage directly with the insurance providers in Uganda, to understand their needs and requirements.

Finally, like most fieldwork planned for 2020, my research was disrupted by the COVID-19 pandemic. I was unable to return to Kenya in late 2020 to conduct follow-up interviews with research participants in the backcasting study, and instead conducted the interviews by telephone with the support of the community mobilizer in Machakos. Without being able to make eye contact and observe body language, it was difficult to build trust and rapport with the research participants and this certainly reduced the richness of the interviews.

5.5 Contributions and implications for policy and practice

In addition to the research contributions discussed above, this thesis also makes contributions to policy and practice. Paper 1 provides a framework for

combining service design and development methods with approaches to addressing challenges in complex social ecological settings. This framework can be a useful guide for development practitioners who are new to service design, and provides a set of practical stages to guide the application of service design alongside other non-design approaches.

Paper 2 shows how the design devices experience mapping and archetype construction can be used to map how different types of service users assign value to a service over time, which will help service providers to align their intervention with users' needs and expectations, and to plan their resource allocation accordingly. The ability to better tailor interventions is valuable given the resource constraints in developing country contexts. Papers 2 and 3 demonstrate how prototyping can be a low-cost way to quickly evaluate service features early on in a design process, as well as an approach for facilitating local engagement in intervention design, thereby overcoming some of the key challenges development practitioners face in ensuring local participation. Together, these contributions have the potential to enhance aid effectiveness and to address the call for innovative approaches to tackling complex development problems. These contributions highlight the value of developing service design capabilities within development organizations, and the need for future research on how best to develop and embed such capabilities.

Paper 4 demonstrates how prototyping, archetype construction and experience mapping can be used as design devices for connecting intervention design to individual capabilities, thereby anchoring design work to normative development goals. This is an important contribution for design practice as it provides a framing to help designers ensure that their work addresses the underlying causes of development problems, and to navigate between competing development objectives.

Paper 5 demonstrates how participatory backcasting could be used as a method for engaging a local community in a long-term planning process and, through the visioning component of the methodology, broadening the scope beyond a single sector or technology. Finding ways to connect sector-specific and societal policy goals might create new avenues for addressing complex development challenges and new opportunities for the private and public sectors to collaborate (Trotter & Brophy, 2022). Asking service users about their broader needs

beyond a single sector or specific technology opens up ways to connect solutions and to engage the work of a diverse set of actors (Trotter & Brophy, 2022).

Finally, in reflecting on the research that contributed to this thesis, the kappa provides practical insights on how design practitioners from the Global North can apply design approaches in Global South settings in a reflexive way that pays attention to culture and context. The kappa demonstrates the importance of team composition to ensuring thematic knowledge, local knowledge and design expertise.

5.6 Impact of the work

Building on the methodological and conceptual contributions outlined thus far, several impacts for policy and practice can be discussed. Here I present evidence of the impact of the PhD work, some of which is captured in the appended papers, but most of which was noted several months or even years after the studies were completed. Attributing impact is not easy since impacts are often the result of a complex interplay of factors over time. However, given my pragmatic stance both personally and ontologically, I believe that it is important to attempt to articulate what the research led to, albeit in a reflexive way. Here, I describe some impacts and signs of impact on communities where we worked, for development programmers we engaged with and for the research community within and beyond SEI.

The findings from the Uganda study informed the design of a Weather Index Insurance product for smallholder farmers that was trialled in several regions of Uganda. The design has since been adopted by 11 insurance companies in Uganda, which have collectively sold the product to approximately 300 000 farmers. Unfortunately, we did not have the resources to follow the subsequent trials or upscaling of the intervention, or to conduct follow-up work with the farmers now using the product. Future work is needed to study the institutional and organizational arrangements that allowed for the insurance product to be further developed and scaled-up.

As discussed, the backcasting process used in Kenya led to changes at the individual and community levels. At the end of the workshop, several participants mentioned that the roadmap that had been co-created was useful because it

captured their vision of a local development pathway, as well as the resources needed to achieve this vision. They told us that they often found it difficult to engage with external actors like NGOs as a collective. Futures studies approaches, such as backcasting and scenario building, are increasingly being used to address complex environment-development challenges, but they are typically applied at the "expert level", engaging government representatives and sector experts who are responsible for designing programmes or policies (Dreborg, 1996; Wangel, 2011). The backcasting study in Kenya demonstrated that such approaches can be used to involve citizens who are on the receiving end of policies and programmes.

Within development policy and practice, scientific publications are often important for conveying the credibility and legitimacy of new approaches. Three of the papers that contribute to this thesis have been published in high-ranking development journals, and we are beginning to see some indication that these ideas are of interest to policymakers in the development space. Insights from the Kenyan studies were presented during a series of workshops for Sida energy experts in 2020–2021. Sida colleagues were interested in how we had combined qualitative and quantitative findings to generate practical guidance on designing clean cooking interventions. The findings from the Ugandan study were presented at a roundtable on climate adaptation for UN agencies in Rome, and at the UN Finance for Development meeting in New York in 2019. At both meetings, there were interesting discussions about the differences between service design and participatory rural appraisal, a similar approach from the field of development studies, and on how service design could be used to facilitate citizen engagement in local development processes and to enhance aid effectiveness.

There are some early indications that the conceptual framework developed during my PhD studies is inspiring similar research on sustainability transitions more broadly. In the past three years, I have been invited to provide training and give talks on the methodological approach used in Kenya and Uganda by colleagues at SEI. This has led to the approach being used to investigate sustainability transitions on food purchasing behaviour in Sweden (Ran et al., 2022), recycling behaviour in Bangkok (Pottinger-Glass et al., 2022) and sustainable mobility in Bogota (Hernandez et al., 2021), and to inform the design of a tool to reduce air travel across the institute (Savvidou et al., 2020). All these cases

involved a complex behaviour change process at the individual level, and the design of interventions, policies or tools to support this transition.

6. Conclusions

This chapter summarizes the main contributions of the thesis and outlines several avenues for future research. Some reflections on the PhD journey are also presented.

6.1 Revisiting the research aim

The aim of this thesis was to explore how service design can be used as an approach to designing development interventions in complex social-ecological contexts. This aim was motivated by the need for innovative responses to complex development challenges (D. Burns & Worsley, 2015). Development programming that depends on top-down, linear intervention logic has been effective in tackling certain development challenges, but less so in situations where problems are highly complex or wicked. Development problems often emerge in complex adaptive systems, where it may be difficult to predict the impact of an intervention using linear intervention models. The multidimensionality of poverty means that attention must be paid to the situated experiences of the intended beneficiaries of a development intervention, but traditional approaches to development planning have struggled to incorporate local perspectives and to foster participation (Cooke & Kothari, 2001; Mansuri & Rao, 2004).

Service design has emerged as a promising approach for addressing complex systemic challenges. The human-centred and systemic nature of service design suggests that it could be useful for connecting situated, lived experiences at the local or micro level to development programming and policymaking at the meso level. However, there is a lack of research on how this can be achieved in complex social-ecological contexts and how designers can practically navigate in such contexts. Given the growing interest in applying design approaches to development problems in the Global South, there is an urgent need to understand how service design can address some of the challenges associated with fostering local agency and participation in development research and programming, as well as the challenges and limitations of using service design as part of transdisciplinary research collaboration. In pursuit of the aim, three research questions were devised:

RQ1: What are the salient features of service design for capturing and making sense of local contextual information for use by other important actors in the system?

RQ2: What are the distinguishing features of service design that enable local agency in intervention design?

RQ3: How can service design methods and approaches complement conventional development research methods?

These research questions were addressed through empirical work conducted in three case studies that examined the design of real-world development interventions. The findings, presented in three empirical and two conceptual papers, and in this kappa, have contributed new knowledge to address the research aim.

6.2 Summary of the thesis

This thesis makes three main contributions. First, it positions service design as an integrative approach for conducting transdisciplinary development research in complex social-ecological contexts in the Global South. The conceptual framework developed in paper 1 provides an empirically grounded process for designing development interventions that combines behavioural science, service design and social ecological systems thinking. For development researchers and practitioners, the conceptual framework provides a novel approach to thinking about development research in complex settings and reframes development intervention design as a collaborative process where service users are active value co-creators rather than passive beneficiaries. Applying the framework to the cookstove and insurance studies generated insights about the role of service design alongside other disciplines, and other types of knowledge when designing development interventions. These insights are valuable for service design researchers and practitioners seeking to apply service design tools and approaches to development contexts.

Second, through the empirical findings from the three case studies, the research demonstrates how design devices, applied as part of a service design approach, can be used to design development interventions in complex settings. The devices can be used to understand how services are valued by diverse types of

users (through archetype construction), and to communicate this understanding to programme designers and policy makers (through experience mapping and archetype construction) operating beyond the micro level. The devices can also support agency by decentring the designers (through prototyping) and facilitating the moments of conscientization and conciliation (using participatory backcasting) required for grassroots community development. Thus, the thesis addresses a key gap in the design literature on understanding how design tools and methods work in development settings (Jagtap, 2019a) and how agency can be supported through their application (Nahi, 2016). This knowledge is valuable for both design and development researchers, as well as practitioners interested in how service design can be used in the design of development interventions in complex social ecological settings.

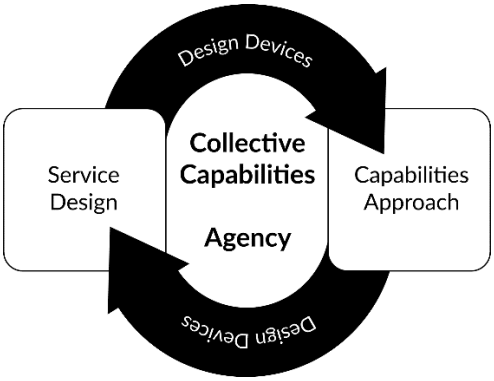


Figure 6.1 Conceptual link between service design and the capabilities approach

Third, the thesis conceptually bridges service design and the capabilities approach by using design devices to pinpoint the most relevant capabilities sets for a proposed intervention and showing how creative methods (for example, participatory backcasting) can support agency and collective capabilities (see figure 6.1). This conceptual linkage can anchor design work in the local context and help service design practitioners and researchers to navigate normative development objectives while ensuring that their design efforts pay attention to relevant ancillary development interventions and services. The capabilities approach has emerged as a universal framework, so anchoring service design research and practice in the capabilities approach can help to avoid imposing western ideals in non-western settings.

Taken together, these contributions are valuable for service design researchers and practitioners interested in how service design can be used alongside other disciplines to support long-term development objectives. The research presented in this thesis demonstrates a pragmatic approach to innovating how interventions are designed within existing institutional and normative structures, thereby addressing increasingly urgent calls for innovative approaches to designing development interventions (Heeks et al., 2014; Leliveld & Knorringa, 2018). For the development community, the contributions represent a fundamentally different approach to designing interventions, in which complexity and non-linearity are embraced.

6.3 Future research directions

Service design is positioned in this thesis as an approach that is capable of integrating different types of knowledge into transdisciplinary collaborations. In the empirical work that underpins the thesis, however, emphasis is given to actors at the micro-level, predominantly individuals and households. Future research on service design in transdisciplinary development research should aim to involve key actors operating at other system levels, including development programmers and local policymakers. Here, a key focus could be on the role of design devices in facilitating agonistic spaces (Mouffe, 2009) to explore how conflicts and tensions between diverse stakeholders can catalyse change.

Given the complexity of designing development interventions and the growing global problem of eroding public trust in public institutions, a fruitful area for future research would be on design devices for engaging with citizens on long-term development planning. In this thesis, backcasting is framed as a design device for prototyping alternative futures with citizens, and for co-designing transition pathways to achieve desired futures. Backcasting was shown to have potential as a device for facilitating citizen engagement in long-term development planning, and in short-term collective action. However, as discussed above, the backcasting study was relatively short and did not allow for the development of an enabling platform to support collective action. Future research should explore backcasting as a community action research method, focused on how service design could facilitate the implementation and testing of the proposed actions that emerged from the process, and allowing sufficient time to study the dynamics of grassroots collective action.

Further work is also needed develop and refine the conceptual linkage established in this thesis between service design and the capabilities approach. Here, future research could take an ex-ante approach to exploring how design devices can be used as a part of a service design approach to identifying capabilities sets early on in the intervention design process. Future work could also explore how service design can support collective capabilities at the community level. For example, research could apply Ibrahim's 3C model as an analytical frame for assessing the role of service design in supporting conscientization, conciliation and collaboration as key elements of grassroots transformative development (Ibrahim, 2017).

Nussbaum's capabilities sets provide the foundation for a theory of social justice that could be used to evaluate policies and programmes aimed at enhancing well-being (Nussbaum, 2013a). Given that service provision in low-income settings is increasingly framed as a social justice issue (Devereux, 2016; Hassim, 2008; Uche Ordu, 2022), linking service design and the capabilities approach could help service design practitioners and researchers to align their work with social justice goals.

The capabilities approach is typically associated with development in low-income countries. However, I believe that the conceptual bridge between the capabilities approach and service design is just as relevant in high-income countries as in low- or middle-income settings. As Nussbaum points out, "all nations are developing nations, in that they contain problems of human development and struggles for a fully adequate quality of life and for minimal justice. All are currently failing at the aim of ensuring dignity and opportunity for each person. For all, then, the Capabilities Approach supplies insight – all have room for improvement" (Nussbaum, 2013a, p 16). At its core, the capabilities approach addresses what people in all societies are concerned with: having the freedom to live a life they consider valuable. Grounding the design of public services and systems in an understanding of individual capabilities will help to ensure that the designs enhance the quality of life of the users of those services and systems. Future research could look at how a capabilities framing could support service and system design work beyond development contexts, for example, in the context of complex sustainability transitions in the Global North.

6.4 Personal reflections on the PhD journey

I have been warned by colleagues that a year from now I will not be able to read this kappa without cringing and wishing I had done things differently. Indeed, even now when I read paper 1, I am struck by the optimistic and somewhat unreflective tone used to frame service design and the instrumental way in which the service design process is described as a toolkit. The paper reflects my position at the time as a development researcher who had recently discovered service design and recognized its potential as an innovative approach to my work. This understanding could be considered naive but without the belief that we were "on to something" that could make a difference for development research and practice, I would not have pursued a PhD in service design. In that sense, my position as a researcher not trained in design afforded me this naivety, which helped fuel my curiosity about the role and potential of service design in the contexts where I work.

During the past four years, my position at the boundary between two academic fields has provided a unique vantage point for analysing the complementarities and divergences between these disciplines in the context of development research and practice. Through my studies, I have shifted from understanding service design as a set of tools and methods that can be usefully applied by the development community, to seeing how service design as an approach and a set of devices can enhance transdisciplinary development research. My pragmatic stance as a researcher working within the established structures of development research and practice has guided and framed my understanding and use of service design in the empirical studies that contribute to this thesis. This has meant using service design to carve out new space, in theory and in practice, within the given structures, for meaningful engagement with the people who are supposed to be at the heart of development interventions – the users and future users of these services.

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